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Assessing the Effectiveness of Large Language Model Support for Generating GDPR ROPA Documentation

Master Thesis at Ulm University

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2026

Version May 21, 2026

Satz: PDF-LaTeX 2_ε

Contents

1 Introduction

1.1 Context

The General Data Protection Regulation (GDPR) requires organisations to maintain a Record of Processing Activities (ROPA), a formal document under Article 30 GDPR that records how personal data is collected, processed, stored, and shared. The content of a ROPA is rigidly specified (controller details, purposes, lawful bases, categories of data and recipients, retention periods, and technical and organisational measures), yet its substance is project-specific prose that has to be written from a blank page and kept current as processing activities change. Maintaining a ROPA is therefore a sustained cost rather than a one-off task; it requires legal and data-protection expertise that small and medium-sized organisations rarely have, and falls disproportionately on those least equipped to bear it [29].

ROPAgen, the tool evaluated in this thesis, is a web-based application developed at the Institute of Databases and Information Systems (DBIS) at the University of Ulm to address exactly this gap. It uses Large Language Models (LLMs) to assist novice users in drafting GDPR-compliant ROPA documentation, offering three interaction modes, each with a different level of LLM support:

Form mode, in which the user fills structured fields directly, with LLM suggestions available on demand.

Ask mode, in which the user works through a structured form while a chatbot answers questions and explains each section.

Chat mode, in which the user describes their requirements conversationally, and the LLM autonomously drafts and populates the ROPA. The three modes are described in detail in Section 3.1; by comparing them, the thesis evaluates how different levels of LLM support affect user performance and trust.

Although the application provides a functional framework for assisted ROPA generation, its effectiveness, usability, and impact on documentation quality have not yet been systematically evaluated. To ensure that such a system can be confidently integrated into practical compliance workflows, both qualitative and quantitative assessments are needed. These evaluations determine whether users can efficiently and correctly create ROPA documents, how the varying levels of LLM support influence performance and trust, and whether the system produces outputs aligned with GDPR requirements.

1.2 Research Problem and Research Questions

The goal of this thesis is to conduct a comprehensive evaluation of the existing LLM-supported ROPA generation application. The evaluation asks how the three interaction modes shape the experience of novice users, how they shape the documents those users produce, and whether the two are aligned. Each overarching research question is broken into three more specific sub-questions tied to the instruments used to answer them.

RQ1 (User experience). *How do varying degrees of LLM support influence novice users during ROPA document creation?*

RQ1.1 How do novice users perceive the *usability* of the three interaction modes (Form, Ask, Chat)?

RQ1.2 How does *cognitive load* differ across the three modes for users with no prior GDPR knowledge?

RQ1.3 How do novice users subjectively evaluate each mode: in their perception of the LLM's support, in their confidence about the documents they produce, and in the mode they ultimately prefer?

RQ2 (Document quality). *How do varying degrees of LLM support influence the quality of ROPA documents produced by novice users?*

RQ2.1 How does document quality differ across the three modes when measured by reference-based NLP similarity to an expert-authored ROPA (BLEU, ROUGE, METEOR, BERTScore, SBERT)?

RQ2.2 As a supplementary, reference-free signal, what additional GDPR-specific quality properties (completeness, scenario faithfulness, legal correctness, hallucination) does an LLM-as-a-judge surface across the three modes?

RQ2.3 Is there a discrepancy between user confidence (RQ1.3) and the document quality actually achieved (RQ2.1, RQ2.2)?

Going into the study, the implicit prediction was that more LLM support would translate into a better user experience and a more preferred mode: the ordering Chat $\succ$ Form $\succ$ Ask was anticipated, with Chat (the free-form conversational mode where the LLM does the most work) expected to be the overall favourite among novice users. The evaluation that follows tests this prediction against the experience and outputs of 73 novice participants.

1.3 Thesis Structure

The remainder of this thesis is organised into five further chapters and three appendices.

Chapter 2 (Background) defines the GDPR and the Article 30 ROPA obligation and locates this thesis in the related work on LLM-supported legal-document generation. **Chapter 3** (Methodology) introduces ROPAgen and its three interaction modes, then describes the user study: design, instruments, sample sizes, and the analytical approach through which the tool is evaluated. **Chapter 4** (Evaluation and Results) reports the empirical findings from the four user-study instruments and from the document-quality analyses, descriptively first and then inferentially. **Chapter 5** (Discussion) interprets the central findings (the completeness–hallucination tradeoff and the confidence–quality gap) against the related work and names the study's limitations. **Chapter 6** (Conclusion) answers RQ1 and RQ2 in plain English and draws the design implications for LLM-assisted compliance tooling that follow. **Appendices A–C** contain the ROPAgen backend prompts and judge configuration, the verbatim study materials, and the full inferential statistics referenced from the Evaluation chapter.

2 Background

This chapter establishes the legal and methodological context for the thesis. The first two sections introduce the GDPR and the Article 30 obligation to maintain a Record of Processing Activities (ROPA), explaining why a ROPA is both legally required and practically demanding to produce. The third section surveys related work along four lenses — GDPR-specific compliance automation, structured legal-document generation, evaluation methodologies, and user-facing legal systems — and the chapter closes by identifying the gap that this thesis addresses.

2.1 General Data Protection Regulation (GDPR)

The General Data Protection Regulation (GDPR), in force since 25 May 2018, is the European Union's framework for personal-data protection [9]. Its scope is extraterritorial: any organisation that processes data on individuals within the EU is subject to it, regardless of its own location. The regulation rests on privacy as a fundamental right under the 1950 European Convention on Human Rights and modernises the 1995 Data Protection Directive in response to the growth of cloud services, social media, and large-scale data breaches. [9, 8]

Under the GDPR, personal data is any information relating to an identifiable individual, ranging from names and email addresses to biometric data, location information, and web cookies. The regulation distinguishes between the data subject (the individual), the data controller (the entity determining the purposes and means of processing), and the data processor (a third party handling data on behalf of a controller). All processing must adhere to seven principles: lawfulness, fairness and transparency; purpose limitation; data minimisation; accuracy; storage limitation; integrity and confidentiality; and accountability. Accountability shifts the burden of proof to the organisation, which must produce documented evidence of compliance. [9]

The regulation mandates that processing is only legal if justified by one of six specific bases, most notably through consent, which must be freely given, specific, and unambiguous. Other justifications include contractual necessity, legal obligations, or a legitimate interest that does not override the subject's rights. Furthermore, the GDPR introduces the concept of data protection by design and by default, requiring that privacy considerations be integrated into the initial development phases of any product or service. Large-scale processors or public authorities are often required to appoint a Data Protection Officer (DPO) to oversee these internal processes and serve as a liaison with regulatory bodies [9].

The GDPR grants individuals a defined set of privacy rights, including the rights to be informed, of access, of rectification, and of erasure (often termed the "right to be forgotten"). Data subjects also hold the right to data portability and to protections against automated decision-making and profiling. Compliance is enforced through substantial administrative fines under Article 83 GDPR, which has placed data protection among the core legal and operational concerns of organisations that handle personal data. [9]

2.2 Records of Processing Activities under Article 30 GDPR

Article 30 of the General Data Protection Regulation (GDPR) establishes the obligation for controllers and, where applicable, processors to maintain a Record of Processing Activities (ROPA). The ROPA functions as a core accountability instrument within the GDPR framework, operationalizing the principle that organizations must not only comply with data protection requirements but also be able to demonstrate such compliance to supervisory authorities. Its primary purpose is to provide a structured, transparent, and up-to-date overview of how personal data are processed within an organization, thereby enabling effective oversight, risk assessment, and regulatory control [8, 29].

From a regulatory perspective, the ROPA serves several interrelated objectives.

First, it enhances organizational awareness of data processing operations by requiring a systematic mapping of purposes, data flows, and safeguards. Second, it supports data protection by design and by default, as organizations must explicitly consider data categories, retention periods, and security measures. Third, it enables supervisory authorities to efficiently assess compliance, particularly in the context of audits, investigations, or following personal data breaches. As such, the ROPA is not merely a formal record, but a living compliance document that reflects actual processing practices [8].

Article 30 distinguishes between the obligations of data controllers and those of data processors. Controllers are required to document processing activities carried out under their responsibility, while processors must maintain records of processing performed on behalf of controllers. These records must be maintained in writing, including in electronic form, and must be made available to the competent supervisory authority upon request. Although the GDPR provides a limited exemption for organizations employing fewer than 250 persons, this exemption does not apply where processing is non-occasional, involves special categories of personal data, concerns criminal convictions, or is likely to result in risks to the rights and freedoms of data subjects. Consequently, in practice, many small and medium-sized organizations remain subject to the ROPA requirement [8].

The content of a ROPA is explicitly defined by Article 30 GDPR and is intended to be concise, structured, and easily comprehensible. Excessive or irrelevant information should be avoided, as clarity and usability are essential for both internal governance and external supervision. At a minimum, a ROPA should include the following elements.

- Name and contact details of the data controller (and, where applicable, joint controllers, representatives, and the data protection officer).
- Purposes of the processing activities.
- Categories of data subjects.
- Categories of personal data processed.
- Categories of recipients, including recipients in third countries or international organizations.

- Details of transfers of personal data to third countries or international organizations, including safeguards where applicable.
- Time limits for the erasure of different categories of personal data.
- A general description of the technical and organizational security measures implemented pursuant to Article 32 GDPR.

Where an organization acts as a data processor, the ROPA must additionally reflect the processing carried out on behalf of each controller, including the identity of those controllers, the categories of processing involved, relevant international data transfers, and applicable security measures. In both cases, the ROPA should be sufficiently detailed to provide a faithful representation of processing activities, while remaining accessible to non-technical stakeholders [8].

The GDPR does not prescribe a fixed update cycle for ROPAs; instead, it requires that records remain accurate and up to date. Consequently, any substantive change in processing purposes, data categories, recipients, or safeguards should trigger a review and update of the ROPA. Many organizations delegate the responsibility for maintaining the ROPA to a data protection officer, where one is appointed, to ensure consistency, avoid duplication of effort, and reduce the risk of omissions [8, 24].

Finally, Article 30 emphasizes the external relevance of ROPA. Organizations must be prepared to present their records to supervisory authorities upon request, and such requests may be accompanied by demands for supplementary documentation, including privacy notices, consent records, or contractual arrangements. In this sense, ROPA operates as a foundational compliance artifact, anchoring broader data protection governance and demonstrating an organization's commitment to accountability under the GDPR [8, 24].

2.3 Related Work

Prior work relevant to this thesis can be grouped along four lenses that mirror the design choices of the present study. The first concerns *domain*: research that targets GDPR compliance specifically and the automated production of regulatory artifacts such as ROPA. The second concerns *task*: architectures for generating structured, template-bound legal documents in adjacent jurisdictions. The third concerns *evaluation*: how the legal-AI literature measures the quality of generated documents, increasingly through a combination of surface-overlap metrics, semantic-similarity metrics, and either expert or LLM-based judgment. The fourth concerns the *user*: how legal AI is presented to non-expert audiences and whether it is accessible to people without prior legal training. Reviewing the literature through these four lenses makes clear what has been studied, what has not, and where the present thesis sits.

2.3.1 GDPR Compliance Automation

The closest prior work to this thesis targets the automation of GDPR compliance documentation directly. von Schwerin et al. [29] address the generation of data categories (e.g., contact information, payment data) for Records of Processing Activities, observing that existing legal benchmarks such as LawBench or LegalBench primarily evaluate static knowledge recall rather than the ability to produce documents that adhere to the precise formatting and terminology required by the GDPR. Their study compares four open-source models (Qwen2-7B, Meta-Llama-3-8B, zephyr-7b-beta, and orca_mini_v7_7b) against GPT-4 across four experiments—basic performance, model-specific prompting, few-shot prompting, and Retrieval-Augmented Generation (RAG)—and assesses outputs using a weighted Model Utility Score covering latency, structure and style, grammar, validity, and contextual understanding. The finding that open-source models, particularly Qwen2-7B, can match or exceed GPT-4 on structured ROPA outputs validates the use of locally deployable models for privacy-sensitive compliance work, and is the empirical foundation on which ROPAgen builds. The study also documents persistent risks—ambiguous legal language, jurisdictional variation, hallucination, and the resulting need for a "human-in-the-loop"—that motivate examining how real users behave when asked to produce ROPA documents with such tools.

Abualhaija et al. [1] approach the GDPR from a complementary angle, extracting technical software requirements for specific data subject rights (right of access, right to portability) from the regulation and supplementary legal sources. Their XTRAREG pipeline uses GPT-3.5 and GPT-4o together with RAG and prompting strategies (Zero-Shot Learning, Chain-of-Thought) to generate requirements, validated against a manual baseline of 108 expert-extracted reference requirements. The work confirms two patterns that recur throughout this thesis: RAG meaningfully improves both lexical and semantic alignment with expert references, and high BERTScores can coexist with low BLEU when the model captures the meaning of a requirement without matching its wording. Their reported failure modes—incomplete coverage, perspective drift (writing as the data subject rather than as the system), and citation of non-binding recitals—further reinforce the case for evaluating LLM-generated GDPR artifacts against a fixed reference rather than trusting model output at face value.

Together, these two works establish GDPR documentation as a viable LLM target while leaving the user-side question unanswered: how do non-experts fare when asked to produce such documents themselves? ROPAgen, and this thesis, extends this line of work by testing the user-facing realization of an open-source GDPR pipeline rather than the model performance in isolation.

2.3.2 Structured Legal-Document Generation

A second body of work targets the broader problem of producing structurally coherent legal documents from limited user input, exploring the architectural strategies—retrieval, wrappers, fine-tuning—that constrain LLMs to follow rigid legal templates. Nigam et al. [23] address private legal-document drafting in the Indian system through their Model-Agnostic Wrapper (MAW), a two-phase framework in which the model first generates a structured list of section titles (which the user can manually refine) and then iteratively fills each section, using a vector database (ChromaDB) to summarize prior sections and maintain long-form coherence. The system is exposed through a Human-in-the-Loop interface and evaluated on the Vidhik-Dastaavej dataset of 489 anonymized documents using ROUGE, BLEU, METEOR, BERTScore, an automatic G-Eval LLM judge, and a Likert-scale expert assessment. The headline finding—that direct fine-tuning on small datasets degrades quality, while a structured wrapper around a smaller open-source model can match GPT-4o—argues for orchestration over scale.

Lin and Cheng [20] take the opposite architectural route, fine-tuning an open-source BLOOM-560M model locally on 74,823 unlabeled Traditional Chinese fraud verdicts (Taiwan Judicial Yuan, 2011–2021) without Chinese word segmentation. Their structural-correctness metric defines a fixed order of legal constitutive elements—Subject of Crime, Subjective Legal Element, Act, Victim, Causation, Result of Hazard—and checks whether the generated draft follows that order, supplemented by perplexity and training/validation loss. The work demonstrates that domain-specific structured drafting is feasible without cloud infrastructure on consumer-grade hardware, addressing the same privacy concerns that motivate locally deployed GDPR tooling. Li et al. [18] push the structural framing further with CaseGen, a benchmark that decomposes Chinese case-document drafting into four sequential stages—defense statements, trial facts, legal reasoning, and judgment results—over 500 expert-annotated cases. Their finding that general-domain LLMs (GPT-4o, Qwen2.5) outperform legal-specific models like ChatLaw or LexiLaw, while still scoring below 6 out of 10 overall, suggests that careful staging of the generation task matters at least as much as legal-domain pretraining. Together, these three works frame structured legal-document generation as primarily an architectural problem and explore wrappers, fine-tuning, and multi-stage decomposition as alternative solutions. ROPAgen takes a related but distinct stance: rather than comparing architectures, it fixes the architecture and varies the *interaction modality* (Form, Ask, Chat), asking what the user-facing surface contributes once the underlying model is held constant.

2.3.3 Evaluation Methodologies for Legal AI

A third strand of the literature concerns how generated legal documents are assessed, and a clear methodological pattern is emerging: dual evaluation that combines automated NLP metrics with either expert annotation or LLM-as-a-Judge scoring. Su et al. [30] formalise this for Chinese judgment-document generation through JuDGE, a benchmark of 2,505 fact-judgment pairs supplemented by over 55,000 statutory articles and 103,000 past judgments. The framework evaluates outputs along four dimensions—Penalty Accuracy (normalised absolute difference on prison terms and fines), Convicting Accuracy (Recall, Precision, F1 over charges), Referencing Accuracy (correctness of statutory citations), and Documenting Similarity (METEOR and BERTScore against ground truth)—explicitly arguing that lexical metrics alone cannot capture the legal reasoning that distinguishes a sound judgment from a plausible-sounding one. Their results across direct generation, in-context learning, supervised fine-tuning, and Multi-source RAG show that even the strongest pipelines remain visibly short of professional ground truth, and that semantic-similarity metrics behave as proxies rather than ceilings.

Vayadande et al. [32], viewed from the methodological angle, illustrate the second template: comparative quantitative benchmarking against existing market tools (Kira Systems, Evisort) using accuracy percentages, time-efficiency in hours, and document-processing speed, supplemented by a debugging phase in which legal experts contribute balanced comments that refine the system. The reported 93–97 % accuracy and reduction of document time from 20 to 5 hours sit alongside an acknowledgment of remaining failures on judicial nuance and on input-quality dependencies, foreshadowing the same tension between aggregate metrics and qualitative caveats that emerges in this thesis. Together, JuDGE and Vayadande et al. converge on a shared template: surface-overlap metrics (BLEU, ROUGE, METEOR), semantic-similarity metrics (BERTScore, sentence embeddings), and a complementary qualitative judgment layer—human expert or LLM judge—that catches what the lexical and semantic metrics miss. This is the methodological template adopted in the present study for evaluating ROPAgen's three modes against a single expert reference document.

2.3.4 User-Facing Legal AI Systems

The fourth lens concerns how legal AI is presented to non-expert users, where prior work is markedly thinner. Surya et al. [31] target the accessibility gap directly with an AI-powered interactive legal chatbot for the Indian Department of Justice, aiming to democratize legal information for citizens facing complex jargon and the cost of professional counsel. Their three-tier architecture—a JavaScript single-page frontend, a Python Flask orchestrator handling translation, and an LLM service running LLaMA 3.1 via Ollama—supports real-time legal queries, document summarization (e.g., First Information Reports), and multilingual interaction by translating Hindi queries to English for the LLM and back. The system also surfaces citation links to government portals, addressing the trust problem that arises whenever non-experts consume LLM output on a high-stakes topic. Notably, however, the paper evaluates these features through illustrative use cases rather than a controlled user study, leaving open how non-expert users actually interact with such a system in practice.

Vayadande et al. [32], considered from the user-facing angle rather than the methodological one, fit the same pattern. Their assistant integrates ChatGPT 3.5 with OpenAI embeddings, LangChain, and a vector retrieval stack to support legal-document generation, comprehension, and abnormality detection, framing the tool as a means of reducing the practical effort of legal work. The system is again validated through accuracy and efficiency comparisons rather than through evidence of how non-expert users navigate the interface, interpret the output, or judge their own confidence in it. Both works establish that accessibility, multilingual support, and citation transparency are necessary properties of user-facing legal systems, but stop short of empirically measuring whether the resulting tools serve the non-experts they target.

2.4 Summary

Under the GDPR, accountability is a documented obligation rather than a stated principle: organisations must demonstrate compliance to supervisory authorities, and the Record of Processing Activities under Article 30 is one of the principal artifacts through which that demonstration is performed. The content of a ROPA is rigidly specified — controller details, purposes, lawful bases, categories of data and recipients, retention periods, and technical and organizational measures — yet its substance is project-specific prose that has to be written from a blank page and kept current as processing activities change. Maintaining a ROPA is therefore a sustained cost rather than a one-off task, and one that falls disproportionately on small and medium organizations that lack dedicated legal or data-protection staff. This is the practical gap into which an LLM-assisted ROPA tool such as ROPAgen is built.

Across the four lenses surveyed in the related-work section — *domain* (GDPR-specific compliance automation), *task* (structured legal-document generation more broadly), *evaluation methodology* (the emerging dual-evaluation template of NLP metrics paired with expert or LLM-as-a-Judge scoring), and *user* (legal AI presented to non-expert audiences) — the literature converges on three established points. LLM-based legal-document generation is technically feasible. Architectural choices — retrieval-augmented generation, model wrappers, multi-stage decomposition, local fine-tuning — measurably improve output. And dual evaluation combining surface-form and semantic NLP metrics with either expert annotation or LLM-as-a-Judge scoring has become the de facto template for assessing the resulting documents.

What remains absent is the intersection of three concerns. No prior work simultaneously (a) targets GDPR-specific structured output, (b) exposes the system to *novice users* without legal training, and (c) measures the gap between user confidence and actual document quality under varying levels of LLM support. This thesis fills that gap by holding the underlying ROPAgen pipeline constant, varying the interaction modality across Form, Ask, and Chat, and applying the dual-evaluation template from the legal-AI literature to a within-subjects user study against a single shared scenario. The instruments, sample, and analysis plan through which that gap is closed are described next in Chapter 3.

3 Methodology

This chapter is divided into two complementary parts. The first introduces ROPAgen, the tool under investigation, covering each of its three interaction modes and the corresponding prompts submitted to the LLM. The second details the design of the user study, including participants, instruments, sample sizes, and the analytical approach through which the tool is evaluated. Together, these sections establish the scope of what is measured and the means by which it is assessed, prior to any presentation of results.

3.1 ROPAgen

The tool evaluated in this study is ROPAgen [14], a web-based application that assists users in drafting GDPR Article 30 Records of Processing Activities through three different levels of LLM support. ROPAgen was developed by the author of this thesis at the Institute of Databases and Information Systems (DBIS), University of Ulm, prior to the present evaluation; this section describes the tool as it stood at the time of the study.

ROPAgen exists to ease the burden of ROPA drafting: the document has rigid structural requirements (Art. 30 GDPR) but most of its substance is project-specific prose that has to be written from a blank page. The tool offers three interaction modes — Form, Ask, and Chat — which offer different levels of LLM support;

Form mode places the user in full control: they select pre-defined options and fill in structured fields, with LLM support available only on demand via a suggestion button. **Ask** mode is a hybrid: users work through the same structured form section by section, but an LLM chatbot is available to answer free-form questions and explain each section as they go. **Chat** mode is fully autonomous: the user describes their requirements in free-form conversation, and the LLM independently determines which structured fields to populate and drafts the ROPA content on its own. This thesis evaluates the three modes: how the level of LLM support shapes user experience and how it shapes the documents that are actually produced. The remainder of this section describes the internal representation of a ROPA document in the tool, walks through the three modes from the user's perspective, and details the LLM backend that all three modes share.

Table 3.1: The three ROPAgen interaction modes at a glance.

Mode	LLM support	User control	User input method
Form	On-demand suggestions per section	Full	Checkbox form (with option
Ask	Per-section chat panel for questions	Full	Free-form text fields per sec
Chat	LLM extracts and populates fields	Conversational only	Free-form description of pro

3.1.1 ROPA Documents in ROPAgen

Internally, ROPAgen represents a ROPA as a single structured `DocumentData` object organized around eight thematic groups that together cover the disclosures required under Art. 30(1) GDPR:

1. **Organisation and controller** — name, postal address, e-mail, phone number, and the contact details of the data protection officer and any external representative.
2. **Legal basis** — Boolean flags for each lawful-basis option in Art. 6(1) GDPR (consent, contract, legal obligation, legitimate interest), the German national-law bases relevant to the study scenario (§ 26 BDSG, § 28b IfSG, § 257/239 HGB, § 147 AO), and a free-text fallback for other grounds.
3. **Data categories** — nine grouped categories (personal master data, employment, finance, health, legal, behavioral, documentation, vehicle, and other) each with predefined Boolean sub-fields such as surname, date of birth, working hours, or holiday entitlement.

4. **Data sources** — Boolean flags for common upstream systems (Office 365, Azure, Google Workspace, AWS, e-mail, etc.) plus a free-text field for sources not covered by the predefined list.
5. **Person categories** — affected persons, internal recipients, external recipients within the EU, and authorized personnel, each populated through Boolean role tags with an optional free-text override.
6. **Retention periods** — Boolean flags for the standard retention triggers (after termination of employment, after termination of contract, upon revocation of consent), a free-text field for the concrete deletion period, and flags for legal-retention exceptions.
7. **Free-text fields** — three short prose fields covering the purpose of processing, the technical and organizational measures in place, and any additional information that does not fit elsewhere.
8. **Metadata** — the document identifier, title, and natural language in which the final ROPA should be generated.

This structured form is the tool's working representation while the user is drafting. The final ROPA itself is rendered as a Markdown document — with optional PDF export through a server-side rendering route — that the LLM produces in a single generation step from the assembled `DocumentData` object. A complete example of the resulting document, the reference baseline against which all participant outputs are scored by the NLP metric pipeline, is reproduced in Appendix A.3.

3.1.2 The Three Interaction Modes

ROPAgen offers three generation modes, each with a different level of LLM support: Form has the least LLM support, Chat is fully autonomous, and Ask is a hybrid mode that sits in the middle. The subsections that follow describe each mode in detail; Table 3.1 summarises them at a glance. All three act on the same internal `DocumentData` schema and, when the user clicks the final *Generate* button, hand that schema to the same prose-generation prompt (`finalropa`); the output ROPA is therefore produced by an identical generation step in all three conditions, and only the path the user takes to populate `DocumentData` differs between modes.

Form Mode

Form mode offers the lightest LLM support: the user sees and does everything. Every field of the `DocumentData` schema is visible and edited by hand, laid out as a linear questionnaire of checkboxes and short text fields across the eight thematic groups (see the Form mode screenshot in Appendix A.2.1). The LLM is invoked only when the user explicitly asks for it: each section offers an optional *AI Suggest* button that asks the LLM, given the document's current state, to propose values for that section's fields; the suggestions populate the form but remain freely editable, and the user is under no obligation to use them. Once the user is satisfied with the form, clicking *Generate* sends the assembled `DocumentData` object to the `finalropa` prompt, which synthesizes a complete ROPA document in formal prose. Form mode therefore exposes the entire structure of a ROPA upfront and uses the LLM only at the margins.

Ask Mode

Ask mode pairs each section with an LLM chat panel and a free-form text field; there are no checkboxes or pre-defined options. The user can ask the LLM, section by section, to explain what the section requires or to suggest how to fill it, but the user is responsible for the final wording entered into the section's text field. Each chat panel (see Appendix A.2.2) opens with the LLM explaining the section's purpose and asking the user for the relevant information; subsequent turns maintain the conversation history for that section and ask follow-up questions. The model is given the current `DocumentData` state alongside the user's message and may return, in addition to its conversational reply, a structured `DATA_UPDATE` block that the front-end uses to auto-populate the corresponding free-form text field. The user can edit any field directly at any point. The final *Generate* step is identical to Form mode: the same `finalropa` prompt receives the same `DocumentData` object. Ask mode is therefore a conversationally guided free-text workflow with the LLM acting as a per-section explainer.

Chat Mode

Chat mode hands control to the LLM. The user does not see or fill the structured fields at all; instead, after entering the organisation's metadata (name, contact information, and title of the processing activity), the user describes their processing in their own words, and the LLM decides which structured fields to populate from those descriptions (see Appendix A.2.3). The user navigates section by section through a chat interface only; the structured form is hidden from view. The LLM explains what each section requires, asks the user to describe their processing, and silently extracts the relevant structured data: a mentioned vendor name flips the corresponding data-source flag, and a mentioned data category populates the corresponding Boolean. The user therefore never directly interacts with the underlying `DocumentData` object, although the same object is being built up under the hood. The final *Generate* step is once again identical to the other two modes. Chat mode is therefore a conversational-first workflow in which field-level decisions are delegated to the LLM and only the final generation step is shared with the other modes.

3.1.3 LLM Backend

All three modes are backed by the same model: Mistral Large 3 (model identifier `mistralai/mistral-large-2512`), accessed through the OpenRouter API gateway via the Vercel AI SDK. The choice of Mistral rests on three considerations relevant to a GDPR-compliance tool. First, Mistral AI is a European model provider [22], which keeps the generation workload aligned, in spirit, with the data-protection regime ROPAgen is helping users to document. Second, Mistral AI has a public commitment to open-weight releases such as Mistral 7B [12] and Mixtral 8x7B [13]; although the Mistral Large 3 variant used here is the company's commercial flagship rather than an open release, this lineage offers a credible path to a fully self-hosted deployment. Third, the Mistral family has shown competitive performance on the long-form reasoning and structured-output tasks relevant to legal-document generation [12]. Two temperature regimes are used (Table 3.2). Conversational and suggestion calls — per-section LLM suggestions in Form mode, dialogue turns in Ask and Chat modes, and the silent data-extraction calls in Chat mode — use temperature 0.2 and top-p 0.5, which permits variation in phrasing while keeping the responses on-task. The final ROPA generation step, in contrast, uses temperature 0.05 and top-p 0.05, which makes the output close to deterministic given the same `DocumentData` input.

Table 3.2: ROPAgen LLM call parameters. The same model (`mistralai/mistral-large-2512`) is used for both regimes.

Call type	Temperature	top-p
Conversational / suggestion	0.2	0.5
Final ROPA generation	0.05	0.05

Crucially, the prompt used in the final generation step is fixed across all three modes: the only thing that differs between a Form-mode output and a Chat-mode output is the contents of the `DocumentData` object that has been built up along the way. The full text of the backend prompts is reproduced in Appendix A.1, as the counterpart to the LLM-as-a-judge configuration recorded in Appendix A.4.

Whether the differing paths to `DocumentData` produce documents of differing quality, and whether those quality differences are visible to the user filling out the form, are the questions taken up by the user study described next.

3.2 User Study

3.2.1 Study Design

The study followed a within-subjects experimental design with counterbalanced ordering. The recruited sample consisted of $N = 73$ university students, all novice users with no prior GDPR experience, and each session lasted approximately 90 minutes per participant. Counterbalancing was implemented through a Balanced Latin Square: the three modes (Form, Ask, Chat) were arranged into six ordered sequences, and participants were randomly assigned to one of the six groups, with each participant completing all three modes in their group's predetermined order. This design controls for order effects, learning effects, and fatigue. Demographic information was collected at the end of the session, after the three modes and the final ranking. Figure 3.1 situates the per-session flow within the broader project arc, from the development of ROPAgen through data collection to the analysis pipeline reported in the chapters that follow.

3.2.2 Task Scenario

A single standardized scenario was used across all participants and modes to ensure consistency. The scenario described a relatable workplace data processing activity at *APOR GmbH*, a fictional small software company in Ulm with approximately 50 employees: the two processing activities under consideration were the administration of employee working time and leave, and the management of parking authorizations for the company car park. The scenario covers all GDPR Article 30 fields and is accessible to participants without legal expertise; it is reproduced verbatim in both German (the version actually shown to participants) and English in Appendix B.1. The LLM backing all three modes (Section 3.1) was held constant throughout the study.

3.2.3 Participants

Recruitment yielded $N = 73$ university students. Participants were screened for minimal to no GDPR knowledge (expected and required), prior AI tool usage experience (documented), general technology comfort level (assessed), and the absence of any professional legal or compliance background. As part of onboarding, participants received a brief standardized introduction to GDPR and ROPA concepts before beginning the tasks.

3.2.4 Data Availability and Sample Sizes

Due to technical data loss and incomplete submissions during data collection, the effective sample sizes differ across analysis streams. The canonical sample sizes used throughout this thesis are summarised below.

- **Survey, recruited N = 73 / valid N = 69** — 73 participants completed the study; 69 retained after dropping four records with invalid consent (`p_0001` =

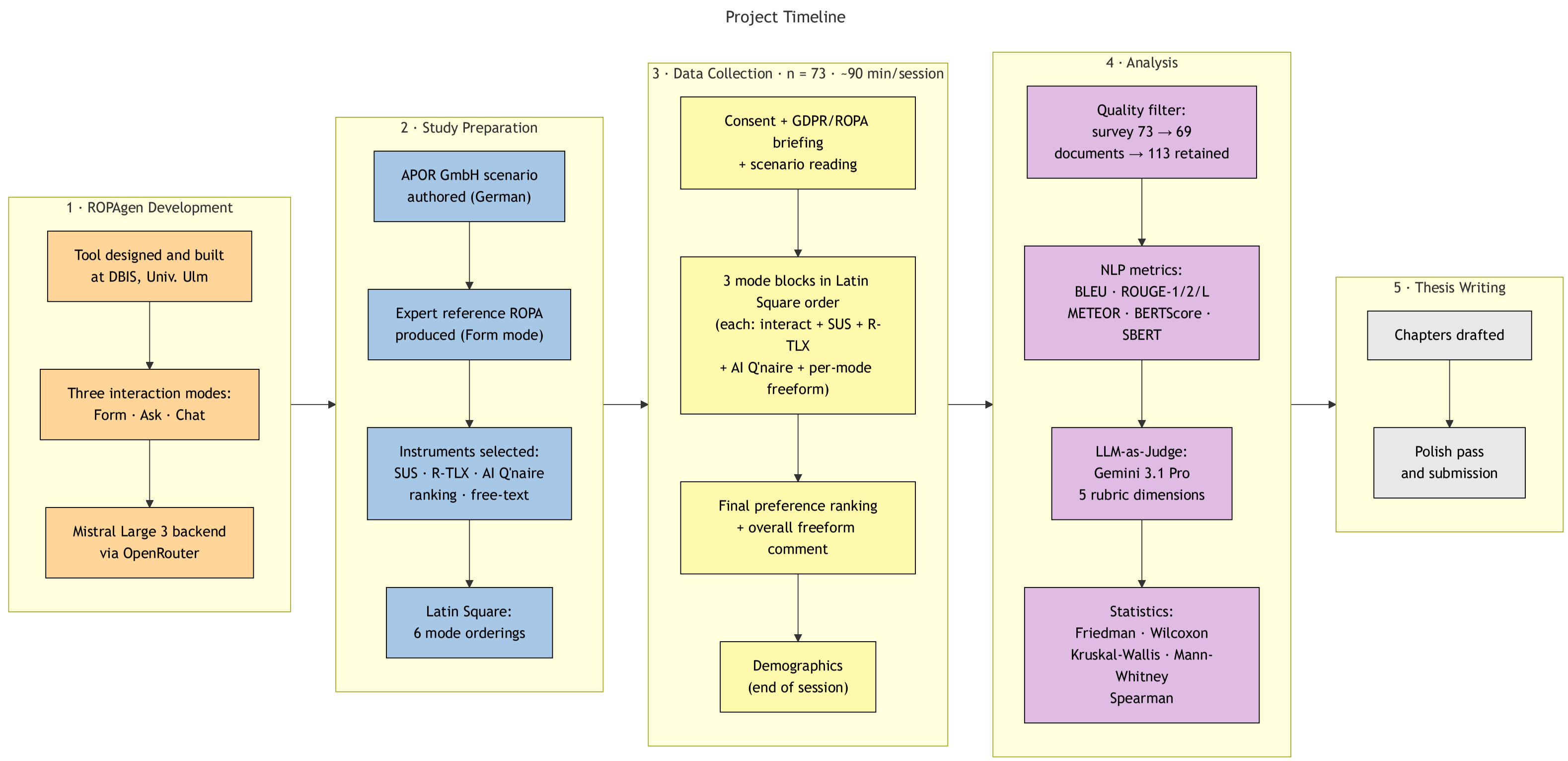

Figure 3.1: Project timeline. Five phases run left-to-right: ROPAgen development at DBIS; study preparation (scenario authoring, reference ROPA, instrument selection, Latin-Square construction); per-participant data collection in 90-minute sessions ($n = 73$); analysis (quality filter, NLP metric pipeline, LLM-as-Judge, statistical layer); and thesis writing. Demographics were administered at the end of each session rather than at the start.

`-1`). All 69 valid participants completed all three modes, so per-mode sample size, matched-triple survey subset, and valid N are all **n = 69**. The matched-triple n = 69 is used for within-subjects Friedman and Wilcoxon signed-rank tests on SUS, R-TLX, confidence items, and the perceived AI support composite.

- **Documents, 113 generated ROPAs** — distributed across modes as 37 (Form), 37 (Ask), and 39 (Chat). The full $n = 113$ sample is used throughout: NLP metric scores and per-mode completion time are analyzed inferentially with Kruskal–Wallis and Bonferroni-corrected Mann–Whitney U as the between-groups family; LLM-as-a-judge scores are reported descriptively on the same sample.

The canonical figures used in the inferential analyses are **n = 113** for NLP metric and completion-time comparisons and **n = 69** for survey instruments.

Descriptive statistics (means, medians, standard deviations) are computed on the full available N per mode for each analysis stream. Inferential tests on NLP metric and per-mode completion-time measures operate on the full $n = 113$ sample as a between-groups comparison; inferential tests on survey instruments operate on the matched $n = 69$ within-subjects sample. The LLM-as-a-judge scores are reported descriptively only and are not subjected to inferential testing (Section 3.5).

3.2.5 Data Collection

Quantitative measures were collected after each mode. Table 3.3 summarises the four user-study instruments, the post-experiment preference ranking, and the post-session demographics questionnaire.

The System Usability Scale (SUS) [4] was administered as ten standardized questions on a 5-point Likert scale; the Raw Task Load Index (R-TLX) [11, 10] captured six cognitive load dimensions on the standard NASA 21-point scale (1 = lowest demand, 21 = highest), as Raw TLX retains the unweighted subscale ratings on their original NASA scale. The full text of the 10 SUS items and the 6 NASA R-TLX subscale descriptions are reproduced in Appendix B.2. In addition, participants completed a six-item Perceived AI Support instrument on a 5-point Likert scale:

Table 3.3: Instruments collected during and after the user study.

Instrument	What it captures
SUS	Perceived usability
R-TLX	Cognitive load (six subscales)
Perceived AI Support	LLM usefulness, trust, transparency, confidence
Free-text feedback	Qualitative comments per mode
Mode-preference ranking	Post-experiment overall preference
Demographics	Age, gender, degree, field, prior GDPR/AI familiarity

1. *Perceived Usefulness:* "The AI was helpful in creating this document." (*"Die KI war bei der Erstellung dieses Dokuments hilfreich."*)
2. *Efficiency:* "The AI helped me to complete the document faster." (*"Die KI hat mir geholfen, das Dokument schneller fertigzustellen."*)
3. *Transparency:* "I understood why the AI suggested specific GDPR fields." (*"Ich habe verstanden, warum die KI bestimmte DSGVO-Felder vorgeschlagen hat."*)
4. *Trust:* "I trust that the AI made correct suggestions." (*"Ich vertraue der KI, dass sie korrekte Vorschläge gemacht hat."*)
5. *Task-Specific Confidence (primary confidence measure):* "I am confident that the AI identified the correct legal basis." (*"Ich bin zuversichtlich, dass die KI die richtige Rechtsgrundlage identifiziert hat."*)
6. *Intention to Use:* "I would use this mode again." (*"Ich würde diesen Modus wieder verwenden."*)

Items were administered in German (the parenthetical text above reproduces the verbatim wording shown to participants); the English glosses are provided for the international reader. Item 5 (Task-Specific Confidence) is treated as the **primary confidence measure** throughout the analysis: it is the only item that asks participants to judge the substantive correctness of the LLM's output (the legal basis under Article 6 GDPR), and is therefore directly comparable to the document-quality signals from the NLP and judge pipelines. The remaining five items — Item 1 (Perceived Usefulness), Item 2 (Efficiency), Item 3 (Transparency), Item 4 (Trust), and Item 6 (Intention to Use) — capture distinct facets of *perceived AI support* rather than confidence in output correctness, and are reported as a secondary composite. Free-text qualitative feedback was also collected after each mode. Two timing measurements were retained: total session duration from the survey instrument (one value per participant) and per-mode completion time emitted by ROPAgen at the end of each session and embedded in every generated document (one value per participant per mode, in seconds). After completing all three conditions, participants ranked all three modes by overall preference in a post-experiment ranking task.

3.3 NLP Metrics

Following the dual-evaluation template adopted by Su et al. [30] and Vayadande et al. [32], both surface-form and semantic measures are reported alongside an LLM-as-Judge rubric. All participant-generated documents are compared against a single reference document using a suite of automated NLP metrics. After validation for errors and completeness, 113 documents were retained for analysis out of a theoretical maximum of 219 (73 participants $\times$ 3 modes). The reference document was created using Form mode (Section 3.1), applying the same standardised scenario presented to participants, and is reproduced in full in Appendix A.3. It is treated as the authoritative baseline, a root-of-trust representing a careful, complete application of the most structured interaction mode. This introduces an inherent framing effect: NLP metrics measure similarity to Form-style output, which may disadvantage the more conversational outputs of Ask and Chat modes independently of their substantive quality.

BLEU, ROUGE, METEOR, and BERTScore are computed at both the whole-document level and the per-chapter level. SBERT is computed alongside BERTScore at the chapter level only, due to an input-length limitation of the underlying sentence-embedding model. Results at every available granularity are aggregated by mode.

The five NLP metrics summarised in Table 3.4 (BLEU [25], ROUGE [19], METEOR [2], BERTScore [35], and SBERT [27]) are defined formally below. Each is computed against the single expert-authored reference document; interpretive framing of the resulting values is deferred to the Evaluation chapter.

Table 3.4: NLP metrics used in the document-quality evaluation. Higher is better on all metrics.

Metric	What it measures
BLEU	n-gram precision with brevity penalty
ROUGE-1/2/L	Unigram / bigram / LCS overlap (F1)
METEOR	Precision–recall with synonym and stem matching
BERTScore P/R/F1	Token-level semantic similarity (ModernBERT)
SBERT cosine similarity	Sentence-embedding cosine similarity (per chapter only)

3.3.1 BLEU

Bilingual Evaluation Understudy — n-gram precision with brevity penalty [25]. BLEU rewards exact word-sequence overlap with the reference: a high BLEU score means the candidate reuses the reference's wording almost verbatim, while a low BLEU score can equally indicate a legitimate paraphrase or genuinely missing content; the metric does not distinguish the two on its own.

BLEU computes the geometric mean of n-gram precisions p_n between candidate and reference, multiplied by a brevity penalty (BP) that discourages outputs much shorter than the reference. With uniform weights $w_n = 1/N$, candidate length c, and closest reference length r:

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \quad \mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r. \end{cases} \tag{3.1}$$

This work uses $N = 4$ (matching unigrams through 4-grams).

3.3.2 ROUGE-1, ROUGE-2, ROUGE-L

Recall-Oriented Understudy for Gisting Evaluation — unigram, bigram, and longest-common-subsequence overlap (F1) [19]. ROUGE measures how much of the reference's vocabulary, bigrams, or longest common subsequence the candidate covers. A high ROUGE score signals strong lexical recall against the reference; a low score indicates that key reference content is absent from the candidate.

ROUGE reports the F1 of n-gram or longest-common-subsequence (LCS) overlap between candidate X and reference Y. ROUGE-1 counts unigrams, ROUGE-2 counts bigrams, and ROUGE-L counts the LCS. For ROUGE-N:

$$P_N = \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(X)|}, \quad R_N = \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(Y)|}, \quad F1_N = \frac{2P_N R_N}{P_N + R_N}. \tag{3.2}$$

The same precision–recall–F1 form applies to ROUGE-2:

$$F1_{\text{ROUGE-2}} = \frac{2P_2 R_2}{P_2 + R_2}. \tag{3.3}$$

For ROUGE-L, n-gram counts are replaced by the length of $\mathrm{LCS}(X, Y)$:

$$P_L = \frac{|\mathrm{LCS}(X,Y)|}{|X|}, \quad R_L = \frac{|\mathrm{LCS}(X,Y)|}{|Y|}, \quad F1_L = \frac{2P_L R_L}{P_L + R_L}. \tag{3.4}$$

The reported ROUGE values are the F1 scores in each case.

3.3.3 METEOR

Metric for Evaluation of Translation with Explicit ORdering — precision–recall balance with synonym matching [2]. METEOR balances precision and recall, credits synonym and stem matches, and penalises fragmented overlap where shared words appear in scattered positions. A high METEOR score signals that the candidate covers the reference's content with vocabulary that need not match it verbatim; a low score indicates either missing content or surface forms that the synonym and stem matcher could not align.

METEOR is the weighted harmonic mean of unigram precision P and recall R (with recall up-weighted by α), multiplied by a fragmentation penalty that grows with the number of disjoint matching chunks ch relative to the number of matched unigrams m:

$$\mathrm{METEOR} = \underbrace{\frac{P \cdot R}{\alpha P + (1-\alpha) R}}_{F_{\mathrm{mean}}} \cdot \underbrace{\Big(1 - \gamma\big(\tfrac{\mathrm{ch}}{m}\big)^{\beta}\Big)}_{\text{penalty}}. \tag{3.5}$$

Standard values $\alpha = 0.9$, $\beta = 3$, $\gamma = 0.5$ are used; matching is performed against exact, stemmed, and synonym/paraphrase matches via WordNet.

3.3.4 BERTScore

Precision, Recall, F1 — token-level semantic similarity using ModernBERT embeddings [35]. BERTScore measures similarity between the contextual embeddings of candidate and reference tokens rather than their surface forms. *Precision* captures how many candidate tokens find a close meaning-match in the reference and therefore penalises spurious or invented content; *recall* captures how many reference tokens are covered and therefore penalises omission; *F1* is the harmonic mean and requires both to be reasonably high.

BERTScore replaces surface-form n-gram matching with cosine similarities between contextual token embeddings produced by a pretrained transformer. The embedding model used here is ModernBERT-base[1] [33], chosen for its long context window of 8192 tokens, which lets a whole ROPA be embedded without truncation, and for its popularity and positive reception in the Hugging Face community. With L2-normalised embeddings $\mathbf{x}_i$ for reference tokens and $\hat{\mathbf{x}}_j$ for candidate tokens:

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.6}$$

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.7}$$

$$F1_{\text{BERT}} = \frac{2 \cdot P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}. \tag{3.8}$$

3.3.5 SBERT Cosine Similarity

Sentence-BERT — chapter-level semantic similarity using sentence embeddings [27]. SBERT compresses a passage into a single fixed-length embedding and reports the cosine similarity between the candidate and reference embeddings. Because a sentence-embedding model is bounded by the maximum input length of the underlying transformer, and a full ROPA exceeds that bound, SBERT is computed at the chapter level only: one score per candidate chapter against the corresponding reference chapter, yielding six scores per document. The primary pass uses

[1] `https://huggingface.co/answerdotai/ModernBERT-base`

ModernBERT-base[2]; a secondary pass with `all-MiniLM-L6-v2`[3] is run over the same six chapters as a robustness check on the primary pass. Both models were chosen for their popularity and positive reception in the Hugging Face community.

SBERT produces a single fixed-length embedding per chapter by mean-pooling token embeddings from a transformer (here, ModernBERT). Chapter similarity is the cosine of the angle between two embeddings $\mathbf{e}(c_1)$ and $\mathbf{e}(c_2)$:

$$\mathrm{SBERT}(c_1, c_2) = \frac{\mathbf{e}(c_1)^\top \mathbf{e}(c_2)}{\|\mathbf{e}(c_1)\| \, \|\mathbf{e}(c_2)\|}. \tag{3.9}$$

3.3.6 Score Implications

The metrics above are read in combination, not in isolation, since each captures a different facet of similarity to the reference. Four pairings recur in the Evaluation chapter. A high lexical score (BLEU, ROUGE, or METEOR) combined with a low semantic score (BERTScore F1 or chapter-level SBERT) indicates mechanical surface copying that does not align in meaning, whereas the reverse pattern (low lexical, high semantic) is the signature of a legitimate paraphrase that preserves content with different wording. Within BERTScore, high recall paired with low precision points to a verbose candidate that covers the reference but adds material the reference does not contain, while low recall paired with high precision points to a conservative candidate that omits required content without fabricating new material. Finally, a high document-level score on the surface-form metrics or BERTScore paired with uneven chapter-level scores under the same metric signals that the aggregate is hiding per-section weakness; this is also why SBERT, which is only available at the chapter level, is read as a per-section diagnostic rather than a whole-document summary. These heuristics are applied throughout the Evaluation chapter.

[2]https://huggingface.co/answerdotai/ModernBERT-base
[3]https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

3.4 LLM-as-a-Judge Evaluation

Documents are additionally evaluated using an LLM-as-a-judge approach, in which **Gemini 3.1 Pro Preview** (`google/gemini-3.1-pro-preview`, $T = 0$) serves as an automated evaluator assessing GDPR-specific quality. The judge is positioned as a *supplementary* evaluator: it is a single-pass, single-model assessment without inter-rater replication or legal-expert validation, and its scores are reported alongside, not in place of, the NLP metrics. The reference-based NLP metrics remain the primary RQ2 evidence; the judge complements them by providing a reference-free signal that captures GDPR-specific quality properties (legal correctness, hallucination, scenario faithfulness) which similarity-based metrics cannot directly observe. The judge evaluates each document across five metrics on a 1–5 scale:

1. **Completeness:** Coverage of all required Article 30(1) fields (controller identity, DPO contact, purposes, legal basis, data categories, recipients, retention periods, technical and organizational measures)
2. **Scenario Faithfulness:** Adherence to the specific details of the study scenario rather than generic or hallucinated content
3. **Legal Correctness:** Appropriate use of GDPR terminology and accurate citation of relevant articles
4. **Hallucination (inverse):** Absence of invented facts not present in the scenario (scored inversely: 5 = no hallucinations)
5. **Overall:** Holistic assessment of the document's usability as a real-world ROPA record

The judge evaluation is applied to all 113 retained documents. The full evaluation rubric, system prompt, and model configuration used for the judge are provided in Appendix A.4.

3.5 Statistical Analysis

Table 3.5 summarises the statistical machinery applied in this thesis. The omnibus and post-hoc tests are non-parametric throughout; Bonferroni correction yields $\alpha' = 0.0167$ for the three pairwise mode comparisons in each post-hoc family.

Table 3.5: Statistical tests and their effect sizes by analysis stream. $\alpha = 0.05$ throughout; Bonferroni-corrected $\alpha' = 0.0167$ for post-hoc families.

Analysis stream	Test	Effect size
Survey (within, $n = 69$)	Friedman	Kendall's W
	Wilcoxon signed-rank	Rank-biserial r
NLP / timing (between, $n = 113$)	Kruskal–Wallis	ε^2
	Mann–Whitney U	Rank-biserial r
Mode preference ($n = 69$)	χ^2 goodness-of-fit	Cramér's V

A significance threshold of $\alpha = 0.05$ is applied to all tests. The choice of test family follows the structure of each dataset, with one test family applied per analysis rather than running parallel matched-pairs and between-groups tests on the same data. **Survey instruments** (SUS, R-TLX, perceived AI support, confidence items) are analysed within-subjects on the matched $n = 69$ sample: a Friedman test as the omnibus, followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction for post-hoc analysis. Every participant gives one response per mode, so the survey data is genuinely matched-pairs; within-subjects analysis with Friedman and pairwise Wilcoxon signed-rank tests is also the standard approach for ordinal Likert and workload data collected in repeated-measures usability studies [28]. **NLP metric scores and per-mode completion time** are analysed between-groups on the full unbalanced $n = 113$ sample: a Kruskal–Wallis test as the omnibus, followed by pairwise Mann–Whitney U tests with Bonferroni correction for post-hoc analysis. The between-groups family is the appropriate match for the document data, which is unbalanced across modes (Form 37, Ask 37, Chat 39); a between-groups analysis preserves all 113 retained submissions. The LLM-as-a-judge scores are reported descriptively (means, medians, dispersion) as a supplementary signal and are not subjected to inferential testing. Effect sizes are reported as ε^2 for the omnibus tests, Kendall's W as the coefficient of concordance for Friedman, and rank-biserial correlation r for the pairwise tests.

For RQ1 (User Experience and Preference), the Friedman test is applied to SUS scores, the R-TLX composite, and the confidence question composites across modes; post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction identify which mode pairs differ significantly; and mode preference distributions are reported as frequencies and rankings. For RQ2 (Document Quality), the Kruskal–Wallis test is applied to NLP metric scores across modes, with Bonferroni-corrected pairwise Mann–Whitney U tests as post-hoc analysis where the omnibus is significant. The LLM-as-a-judge scores are reported descriptively only and are not subjected to inferential testing.

The tests and effect sizes used in this thesis are grouped below by where they appear in the Evaluation chapter. Each block names the analysis stream first (sample, design), then defines the omnibus and post-hoc tests applied to that stream together with their effect sizes. A short plain-language reading of each statistic is given alongside its formula so that the Evaluation chapter can refer back to a single canonical definition.

3.5.1 Procedures Shared Across Analyses

Two procedures are shared across every analysis stream and are defined once here.

The *Shapiro–Wilk normality test* screens whether a sample plausibly comes from a normal distribution. With order statistics $x_{(i)}$ and tabulated constants a_i,

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{3.10}$$

In plain terms: W close to 1 means the data look normal; a small p-value rejects normality and motivates a non-parametric test. Normality was rejected for nearly every survey, NLP, and timing measure in this study, which is why every test below is non-parametric.

The *Bonferroni correction* divides the per-test significance threshold by the number of comparisons in a family so that the family-wise false-positive rate stays at the chosen α. With m comparisons,

$$\alpha' = \alpha/m. \tag{3.11}$$

Every post-hoc family in this thesis has $m = 3$ pairwise mode comparisons, so $\alpha' = 0.05/3 = 0.0167$ throughout.

3.5.2 Survey Instruments (Matched Within-Subjects, $n = 69$)

Applied to SUS, R-TLX, the perceived AI support items, the primary confidence item, and the re-use intent item. Each participant gives one rating per mode, so the data are matched-pairs. The Friedman omnibus with pairwise Wilcoxon signed-rank post-hoc tests is the standard non-parametric within-subjects analysis for ordinal Likert and workload data in usability studies [28]; non-parametric tests are appropriate here because Shapiro–Wilk rejects normality on every survey measure and Likert scales are not strictly continuous.

The *Friedman test* [7] is the within-subjects omnibus for $k \geq 3$ related samples. Each participant's k values are ranked within-subject; with n participants across k conditions and R_j the rank sum for condition j,

$$\chi^2_F = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1), \tag{3.12}$$

asymptotically χ^2 distributed with $k-1$ degrees of freedom. In plain terms: a large χ^2_F (small p) means at least one mode differs from the others on this measure; non-significant means no mode-level difference can be claimed on this instrument as a whole.

When the Friedman omnibus is significant, the *Wilcoxon signed-rank test* [34] is applied to each pair of modes as a matched-pairs post-hoc. Differences $d_i = x_i - y_i$ are ranked by absolute value; the statistic is

$$W = \min(W_+, W_-), \quad W_+ = \sum_{d_i>0} \mathrm{rank}(|d_i|), \quad W_- = \sum_{d_i<0} \mathrm{rank}(|d_i|). \tag{3.13}$$

In plain terms: a small p (below $\alpha' = 0.0167$) means the two modes differ in their within-subjects ranks. The associated effect size, the rank-biserial correlation,

$$r = (W_+ - W_-)/(W_+ + W_-), \tag{3.14}$$

runs from -1 to $+1$; positive r means the first-named mode ranks higher. Conventional bands: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

The Friedman test's own effect size is *Kendall's W* (coefficient of concordance) [16]:

$$W_{\text{Kendall}} = \chi_F^2/(n(k-1)), \tag{3.15}$$

on the $[0, 1]$ scale, larger meaning stronger agreement among participants on the ordering of conditions. Conventional bands: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

3.5.3 Document-Quality Metrics and Completion Time (Between-Groups, $n = 113$)

Applied to BLEU, ROUGE-1/2/L, METEOR, BERTScore Precision / Recall / F1, SBERT cosine similarity, and per-mode completion time. The document sample is unbalanced across modes (Form 37, Ask 37, Chat 39); a between-groups family is the appropriate match for the data and preserves all 113 retained submissions. Kruskal–Wallis with pairwise Mann–Whitney U post-hoc tests is the standard non-parametric counterpart to one-way ANOVA for unbalanced independent-groups designs [6], and non-parametric throughout is again motivated by non-normal metric distributions.

The *Kruskal–Wallis H test* [17] is the non-parametric one-way ANOVA analogue applied as the between-groups omnibus. With N total observations, n_j in group j, and R_j the rank sum in group j,

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1). \tag{3.16}$$

In plain terms: a large H (small p) means at least one mode's distribution of scores differs from the others on this metric.

When the Kruskal–Wallis omnibus is significant, the *Mann–Whitney U test* [21] is applied to each pair of modes as a between-groups post-hoc. With group sizes n_1, n_2 and rank sum R_1,

$$U = R_1 - \frac{n_1(n_1+1)}{2}. \tag{3.17}$$

In plain terms: a small p (below $\alpha' = 0.0167$) means the two modes produced different distributions of scores on this metric. The associated rank-biserial effect size r is reported in the same -1 to $+1$ range and with the same conventional bands as for the Wilcoxon test above.

The Kruskal–Wallis test's own effect size is ε^2 *(epsilon-squared)* [15]:

$$\varepsilon^2 = (H - k + 1)/(N - k), \tag{3.18}$$

on the $[0, 1]$ scale, larger meaning more of the total variance in scores is explained by mode membership. Conventional bands match Kendall's W.

The same Kruskal–Wallis and Mann–Whitney U pipeline is applied at both whole-document and per-chapter granularity; Bonferroni correction continues to apply within each (metric, chapter) family of three pairwise comparisons, giving $\alpha' = 0.0167$. The per-chapter results are therefore produced by the same machinery as the whole-document results, not a separate test family.

3.5.4 Mode Preference Ranking ($n = 69$)

Each participant ranks the three modes 1st, 2nd, and 3rd at the end of the session, yielding a full ordinal ranking per participant. The inferential analysis collapses this ranking to its first-place column and tests whether the resulting rank-1 vote distribution across the three modes is uneven; this directly answers the headline question of whether there is a clear single most-preferred mode, with the corresponding vote shares readable at a glance. The 2nd- and 3rd-place ranks remain in the descriptive analysis (mean rank, full rank distribution per mode). Reducing a preference ranking to its first-place counts for goodness-of-fit testing is a standard analytic choice when the question is whether a single dominant option emerges from the ranking [28].

The *chi-square goodness-of-fit test* [26] compares observed counts O_i to expected counts E_i under a uniform null:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}. \tag{3.19}$$

In plain terms: a large χ^2 (small p) means the distribution of first-place votes is more uneven than would be expected if participants chose at random. The associated effect size is *Cramér's* V [5]:

$$V = \sqrt{\chi^2/(N(k-1))}, \tag{3.20}$$

on the $[0, 1]$ scale, larger meaning the observed distribution deviates more strongly from uniform. Conventional bands match Kendall's W.

With the tool, the study design, the instruments, the sample sizes, and the statistical machinery now in place, Chapter 4 reports the empirical findings: participant demographics first, then the four user-study instruments and the post-experiment ranking, then the document-quality analyses (NLP metrics at document and chapter levels, followed by the supplementary LLM-as-a-Judge signal).

4 Evaluation

This chapter reports the empirical findings of the user study and the subsequent document-quality analysis. It opens with a descriptive Demographics section that profiles the participant sample on which every result that follows rests. The User Study section then reports the per-mode self-report instruments (System Usability Scale, NASA Raw Task Load Index, Perceived AI Support) together with the post-experiment mode preference ranking and open-ended free-text comments. The document-quality sections close the chapter by reporting the reference-based NLP metrics at the document and chapter levels and, finally, the LLM-as-a-judge evaluation as a supplementary signal. All inferential test statistics referenced in the running text are reproduced in full in Appendix C.

The survey analyses use the matched within-subjects sample of $n = 69$ participants, with Friedman as the omnibus and Bonferroni-corrected Wilcoxon signed-rank as the pairwise post-hoc; the mode-preference ranking is additionally tested with a chi-square goodness-of-fit against a uniform null. The document-quality analyses use the full set of 113 retained documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$) throughout, with Kruskal–Wallis and Bonferroni-corrected Mann–Whitney U as the between-groups inferential family for NLP metric scores and per-mode completion time. The LLM-as-a-judge scores are reported descriptively only.

4.1 Participant Demographics

The chapter opens by profiling the participant sample on which every result that follows rests. The analysis sample is the matched within-subjects set of $n = 69$ valid participants ($p_{0001} \geq 1$); four screened records were excluded from all analyses for incomplete or invalid participation, yielding the 69 records.

4.1.1 Numeric and Ordinal Variables

Table 4.1 reports the descriptive statistics for the four numeric or ordinal variables collected in the demographics block: total session duration in seconds, and three five-point Likert self-reports of prior expertise (GDPR knowledge, ROPA familiarity, chatbot familiarity).

Table 4.1: Descriptive statistics for numeric and ordinal demographic variables ($n = 69$). Duration is in seconds and aggregates the full study session across all three modes; the three expertise variables are five-point Likert self-reports (1 = no familiarity, 5 = expert). The duration minimum of -1 is a sentinel value for one non-responding record on this single variable; the median is a more robust point estimate.

Variable	n	Mean	SD	Q1	Median	Q3	Min	Max
Duration (s)	69	1865.7	1354.2	840	2013	2708	-1	5788
GDPR knowledge	69	2.81	1.15	2	3	4	1	5
ROPA familiarity	69	1.45	0.76	1	1	2	1	4
Chatbot familiarity	69	4.36	0.73	4	4	5	2	5

The median session duration is 2013 s (≈ 33.6 min) per participant across the full session, consistent with the protocol target of ≈ 90 min total once spread across three modes and the rating instruments. ROPA familiarity is the lowest of the four variables (mean 1.45, median 1), confirming that the sample qualifies as ROPA-novice in line with the inclusion criteria. Chatbot familiarity is correspondingly high (mean 4.36, median 4). GDPR knowledge sits near the scale midpoint (mean 2.81).

4.1.2 Categorical Variables

Table 4.2 reports counts and shares for the five categorical demographic variables: gender, age (binned by year), highest completed degree, current professional status, and field of study or work. Three participants did not report gender; the share columns therefore use the per-variable response base.

The sample is dominated by Bachelor-level students of IT or Computer Science. Of the 66 participants who reported gender, $62.1\,\%$ identified as male. The full distribution of demographics and expertise variables is visualised in Figure 4.1. This profile

Table 4.2: Categorical demographic variables ($n = 69$ unless noted). Shares are computed within variable.

Variable	Level	n	Share
Gender ($n = 66$ reporters)	Male	41	62.1 %
	Female	24	36.4 %
	Other	1	1.5 %
Age (years)	24	18	26.1 %
	25	18	26.1 %
	26	10	14.5 %
	Other (21–40)	23	33.3 %
Degree ($n = 68$ reporters)	Bachelor	60	88.2 %
	Master	6	8.8 %
	Other	2	2.9 %
Profession	Student	56	81.2 %
	Employed	10	14.5 %
	PhD	2	2.9 %
	Staatsexamen	1	1.5 %
Field	IT	41	59.4 %
	Business (BWL)	16	23.2 %
	Other	6	8.7 %
	Research	4	5.8 %
	HR	2	2.9 %

— IT-leaning, chatbot-fluent, ROPA-novice — frames the results reported in Sections 4.2–4.4 and is taken up again in the Discussion chapter when generalizability is addressed.

4.2 User Study

This section consolidates the four user-study instruments. The three per-mode self-report questionnaires (SUS, R-TLX, Perceived AI Support) are reported in the order in which they were administered after each interaction mode; the post-experiment mode preference ranking and the open-ended comments close the section by asking how the three per-instrument patterns combine when participants commit to a single overall preference and explain it in their own words.

4.2.1 ROPAgen Usability — System Usability Scale

The System Usability Scale captures how usable each interface felt to participants immediately after they finished interacting with it. It consists of ten Likert items on a five-point scale, alternating in polarity: odd-numbered items are phrased positively (higher response equals more positive evaluation), and even-numbered items are phrased negatively (lower response equals more positive evaluation). This alternating keying means Table 4.3 must be read item by item rather than averaged at a glance. The full item text is reproduced in Appendix B.2. The Aggregated column in the tables below pools the three per-mode samples into a single per-item descriptive across all 207 ratings.

The SUS yields a single composite score per mode on a 0–100 scale, computed from the ten items using the standard scoring procedure of Brooke [4]. A score of 68 is widely used as the threshold above which a system is judged to have above-average usability [3]. Figure 4.2 reports the per-mode composite scores against this reference line. All three modes clear the threshold: Form scores highest at a mean of 78.62 (median 80.0, SD 16.09), while Ask and Chat sit closely together at 72.97 and 72.93 respectively (medians 72.5 and 75.0, SDs 15.36 and 19.03). The Form-first ordering already visible at the composite level is decomposed item by item in the remainder of this section.

Figure 4.1: Demographic and expertise distributions: histograms of age and session duration, count plots for gender, degree, profession, and field, and a grouped bar chart of the three expertise self-reports (GDPR, ROPA, chatbot) on the five-point Likert scale.

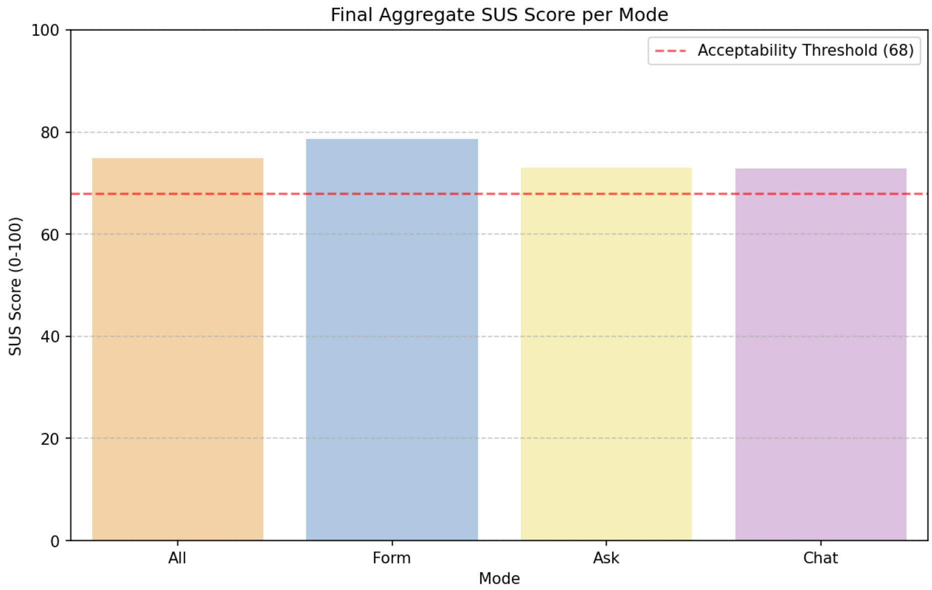

Figure 4.2: Mean SUS composite score per mode ($n = 69$ per mode) on the standard 0–100 scale. The dashed horizontal line marks the acceptability threshold of 68 [3].

The ten SUS items capture distinct facets of usability, including frequency of intended use, perceived complexity, and ease of use; the per-mode means therefore have to be read item by item before the aggregated test is reported. Table 4.3 reports the mean response per item per mode and the aggregated mean across all three modes. The aggregated column is the unweighted mean of the three per-mode means.

Form is the mode participants would most willingly keep using, with Ask and Chat trailing by a small but consistent margin. For *"I think that I would like to use this system frequently"* (item 1), the aggregated mean is 3.32, with Form yielding the highest mean (3.52) and Ask and Chat sitting at the same level (3.22 each). Frequent-use intent decreases by approximately 0.30 points from Form to the LLM-driven modes.

Form also feels the least complex of the three, with Ask perceived as marginally more complex than Chat. For *"I found the system unnecessarily complex"* (item 2, lower is better), the aggregated mean is 2.00, with Form rated lowest (1.64), Ask highest (2.25), and Chat in between (2.12). Perceived complexity is approximately 0.5 points higher in the LLM-driven modes than in Form.

Table 4.3: System Usability Scale, mean response per item by mode ($n = 69$ per mode). Items are reported on the original five-point Likert scale; the Aggregated column averages the three per-mode means. Odd items: higher is better; even items: lower is better.

#	Item	Aggregated	Form	Ask	Chat
1	I think that I would like to use this system frequently.	3.32	3.52	3.22	3.22
2	I found the system unnecessarily complex.	2.00	1.64	2.25	2.12
3	I thought the system was easy to use.	4.13	4.38	4.01	3.99
4	I think that I would need the support of a technical person to be able to use this system.	1.52	1.52	1.49	1.55
5	I found the various functions in this system were well integrated.	3.77	4.00	3.67	3.64
6	I thought there was too much inconsistency in this system.	2.07	2.16	2.04	2.01
7	I would imagine that most people would learn to use this system very quickly.	4.11	4.30	4.00	4.03
8	I found the system very cumbersome to use.	2.03	1.65	2.16	2.29
9	I felt very confident using the system.	3.86	3.83	3.84	3.90
10	I needed to learn a lot of things before I could get going with this system.	1.61	1.61	1.61	1.62

Form is also experienced as the easiest mode to use, with Ask and Chat almost indistinguishable behind it. For *"I thought the system was easy to use"* (item 3), the aggregated mean is 4.13, with Form rated highest (4.38) and Ask and Chat closely matched at 4.01 and 3.99. Ease-of-use ratings decrease by roughly 0.4 points from Form to the LLM-driven modes.

None of the three modes makes participants feel they need technical help to operate it — the Form-first ordering levels out here. For *"I think that I would need the support of a technical person to be able to use this system"* (item 4, lower is better), the aggregated mean is 1.52, with the three modes essentially indistinguishable (Form 1.52, Ask 1.49, Chat 1.55). None of the three modes is perceived to require technical support.

Form feels like the most coherent mode, with the LLM-driven modes scoring lower on perceived integration. For *"I found the various functions in this system were well integrated"* (item 5), the aggregated mean is 3.77, with Form scoring highest (4.00) and Ask and Chat lower (3.67 and 3.64). The integration rating drops by approximately 0.35 points from Form to the LLM-driven modes.

No mode is perceived as meaningfully more inconsistent than the others, mildly inverting the Form-first pattern. For *"I thought there was too much inconsistency in this system"* (item 6, lower is better), the aggregated mean is 2.07, with the three modes similar to each other (Form 2.16, Ask 2.04, Chat 2.01). Form is rated marginally higher on this negative item than the two LLM-driven modes.

Form is judged the most learnable of the three, with Ask and Chat trailing again. For *"I would imagine that most people would learn to use this system very quickly"* (item 7), the aggregated mean is 4.11, with Form scoring highest (4.30) and Ask and Chat closely matched behind it (4.00 and 4.03). Perceived learnability decreases by approximately 0.30 points from Form to the LLM-driven modes.

The biggest experiential gap of the ten items lands here: participants find Form noticeably less cumbersome than the LLM-driven modes, with Chat the most cumbersome of the three. For *"I found the system very cumbersome to use"* (item 8, lower is better), the aggregated mean is 2.03 and the largest spread of any item is observed: Form 1.65, Ask 2.16, Chat 2.29. Cumbersomeness is rated approximately 0.5–0.65 points higher in the LLM-driven modes than in Form.

Participants feel about equally confident operating any of the three modes, with a small uptick toward Chat — the only item where the LLM-driven modes nudge ahead of Form. For *"I felt very confident using the system"* (item 9), the aggregated mean is 3.86, with the three modes rated closely (Form 3.83, Ask 3.84, Chat 3.90). Within-system confidence is broadly comparable across modes, with a small upward trend from Form to Chat.

All three modes feel equally approachable to a first-time user — no prior learning is perceived as required for any of them. For *"I needed to learn a lot of things before I could get going with this system"* (item 10, lower is better), the aggregated mean is 1.61, with the three modes virtually identical (Form 1.61, Ask 1.61, Chat 1.62). The amount of prior learning required is perceived as low and equal across all modes.

Table 4.4 reports the full descriptive distribution per item per mode (mean, standard deviation, Q1, median, Q3, min, max).

The per-item distributions are visualised in Figure 4.3.

A Friedman test on the per-mode aggregate is not significant ($\chi^2(2) = 5.51$, $p = 0.064$, $W = 0.040$). The SUS composite therefore does not differ significantly across modes at the omnibus level, and pairwise post-hoc comparisons are not pursued on the composite score. The per-item descriptive ordering reported above is consistent with a Form-leaning pattern, but it does not clear the inferential bar. Full test statistics are reproduced in Appendix C.

Taken together, the SUS data establish a Form-first ordering on the items most directly tied to perceived ease of use, complexity, and cumbersomeness, while items relating to within-system confidence, required prior learning, and need for technical support flatten across modes. Whether this same ordering recurs on perceived workload is examined next in Section 4.2.2.

4.2.2 Cognitive Load — NASA Raw Task Load Index

Building on the SUS results reported in Section 4.2.1, this section turns to perceived workload. Where SUS measured how usable the interfaces felt, the Raw Task Load Index decomposes the participant's effort along six interpretable axes — Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration — each rated on the standard NASA 21-point scale (1 = lowest demand, 21 = highest). Performance is reported in the positive-coded direction (higher equals worse

Table 4.4. System Usability Scale, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: `sus_per_item_summary.csv`.

#	Item	Mode	Mean	SD	Q1	Median	Q3	Min	Max
1	I think that I would like to use this system frequently.	Aggregated	3.32	1.14	3	3	4	1	5
		Form	3.52	1.09	3	4	4	1	5
		Ask	3.22	1.08	3	3	4	1	5
		Chat	3.22	1.24	2	3	4	1	5
2	I found the system unnecessarily complex.	Aggregated	2.00	1.12	1	2	3	1	5
		Form	1.64	0.92	1	1	2	1	5
		Ask	2.25	1.22	1	2	3	1	5
		Chat	2.12	1.12	1	2	3	1	5
3	I thought the system was easy to use.	Aggregated	4.13	0.97	4	4	5	1	5
		Form	4.38	0.91	4	5	5	1	5
		Ask	4.01	0.81	4	4	5	1	5
		Chat	3.99	1.13	3	4	5	1	5
4	I think that I would need the support of a technical person to be able to use this system.	Aggregated	1.52	0.89	1	1	2	1	5
		Form	1.52	0.85	1	1	2	1	5
		Ask	1.49	0.80	1	1	2	1	5
		Chat	1.55	1.01	1	1	2	1	5
5	I found the various functions in this system were well integrated.	Aggregated	3.77	1.05	3	4	5	1	5
		Form	4.00	0.92	4	4	5	1	5
		Ask	3.67	1.02	3	4	4	1	5
		Chat	3.64	1.16	3	4	5	1	5
6	I thought there was too much inconsistency in this system.	Aggregated	2.07	1.00	1	2	3	1	5
		Form	2.16	0.96	1	2	3	1	5
		Ask	2.04	1.06	1	2	3	1	5
		Chat	2.01	0.98	1	2	3	1	5
7	I would imagine that most people would learn to use this system very quickly.	Aggregated	4.11	0.98	4	4	5	1	5
		Form	4.30	0.96	4	5	5	1	5
		Ask	4.00	0.95	4	4	5	1	5
		Chat	4.03	1.01	4	4	5	1	5
8	I found the system very cumbersome to use.	Aggregated	2.03	1.13	1	2	3	1	5
		Form	1.65	0.94	1	1	2	1	5
		Ask	2.16	1.12	1	2	3	1	5
		Chat	2.29	1.23	1	2	3	1	5
9	I felt very confident using the system.	Aggregated	3.86	1.04	3	4	5	1	5
		Form	3.83	1.04	3	4	5	1	5
		Ask	3.84	0.93	3	4	4	1	5
		Chat	3.90	1.15	4	4	5	1	5
		Aggregated	1.61	0.90	1	1	2	1	5

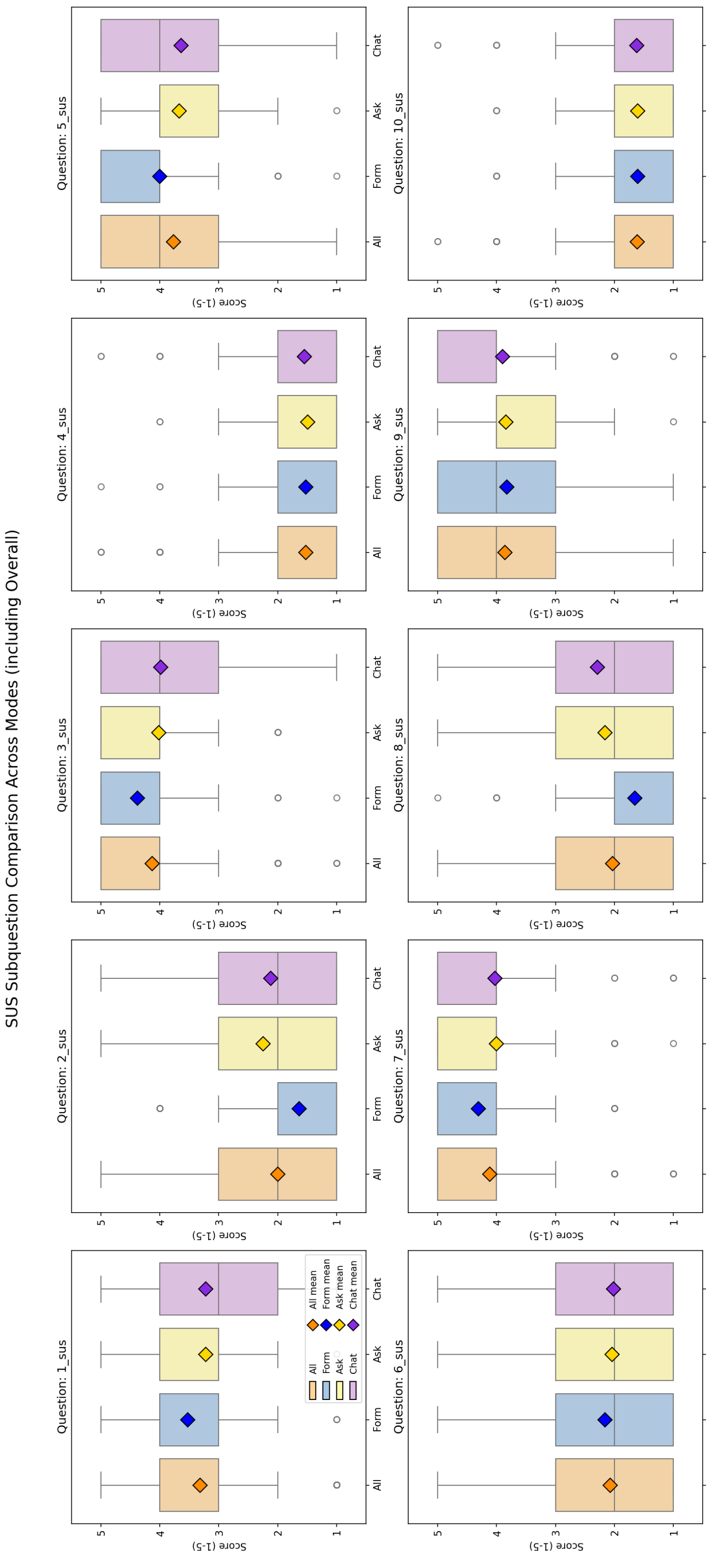

Figure 4.3: Box plots of the ten SUS items by mode ($n = 69$ per mode). Boxes show the interquartile range; the central line marks the median, and the diamond marks the mean.

perceived performance) for internal consistency with the other subscales. The overall R-TLX in this study is the unweighted mean of the six subscales on the same 1–21 scale (no rescaling is applied). The expectation from the methodology framing is that subscales tied to interface mechanics (Mental Demand, Effort) will track the SUS ordering, while content-driven subscales (Temporal Demand, Frustration) may diverge where the LLM modes introduce additional variability. The full subscale descriptions are reproduced in Appendix B.2. A short supplementary paragraph at the end of this section reports the objective per-mode *completion time* alongside the perceptual subscales — not as a separate workload instrument, but as a behavioral counterpart to Temporal Demand that the rest of the chapter occasionally refers back to.

Figure 4.4 reports the composite R-TLX score per mode against the scale midpoint of 10.5, which marks the boundary between the lower and upper halves of the 1–21 range and serves as a visual reference for whether reported workload trends toward the light or heavy end of the scale. All three modes sit comfortably in the lower half: Form scores lowest with a composite of 5.50, while Ask and Chat are closely tied at 6.33 and 6.33 respectively. Perceived workload across all three modes is therefore well below the midpoint, with Form approximately 0.8 scale points lighter than the LLM-driven modes at the aggregate level. The contribution of each subscale to that overall score is decomposed in the remainder of this section.

The six R-TLX subscales are reported individually before the aggregated test, so that the contribution of each axis to the overall workload score is visible. Table 4.5 reports the per-subscale mean and standard deviation per mode and the aggregated mean across modes. Figure 4.5 shows the corresponding distributions.

Table 4.6 reports the full descriptive distribution per subscale per mode (mean, standard deviation, median, Q1, Q3, min, max). All subscales use the 1–21 scale where lower equals less workload (Performance is positive-coded for internal consistency).

Form is the least mentally demanding mode, with Ask perceived as the most demanding and Chat sitting between the two. For *Mental Demand*, the aggregated mean is 6.31, with Form scoring lowest (5.77), Ask highest (6.80), and Chat in between (6.36). Mental Demand increases by approximately 0.6–1.0 points when moving from Form to the LLM-driven modes.

None of the three modes feels physically taxing, though Form is again the lightest

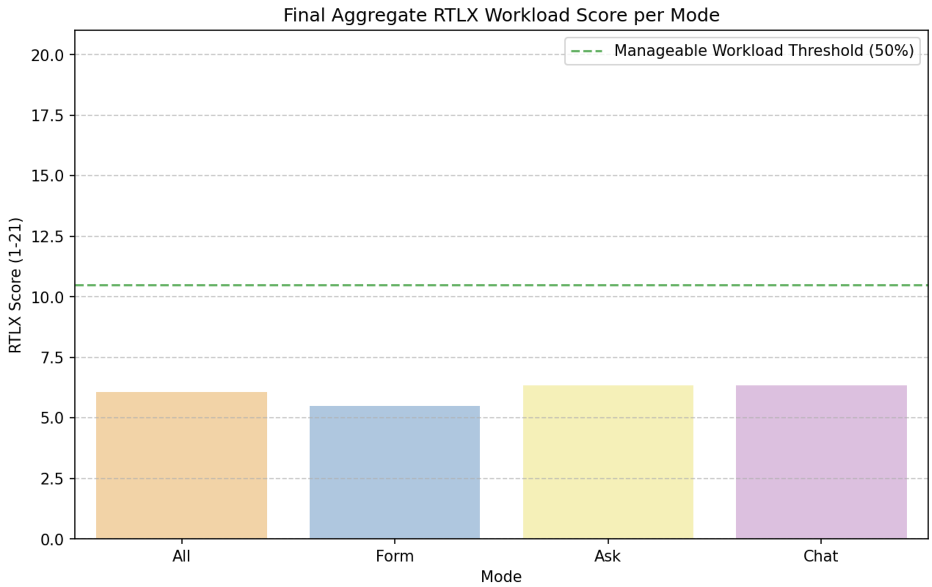

Figure 4.4: Mean R-TLX composite workload score per mode ($n = 69$ per mode), unweighted average of the six subscales on the 1–21 scale. The dashed horizontal line marks the scale midpoint (10.5), shown for visual reference.

Table 4.5: NASA R-TLX subscale mean (SD) by mode ($n = 69$ per mode), 1–21 scale. The Aggregated column pools the three per-mode samples into a single per-subscale descriptive across all 207 ratings. All subscales are coded so that lower equals less workload.

Subscale	Aggregated	Form	Ask	Chat
Mental Demand	6.31 (4.75)	5.77 (4.26)	6.80 (4.63)	6.36 (5.31)
Physical Demand	3.19 (3.40)	2.77 (2.71)	3.54 (3.92)	3.26 (3.46)
Temporal Demand	4.89 (4.28)	4.16 (3.82)	4.97 (4.15)	5.54 (4.78)
Effort	6.13 (4.57)	5.68 (4.56)	6.39 (4.41)	6.32 (4.77)
Frustration	6.86 (5.24)	5.88 (4.84)	7.58 (5.34)	7.13 (5.44)
Performance (pos.)	8.93 (5.82)	8.71 (5.48)	8.68 (5.64)	9.39 (6.36)

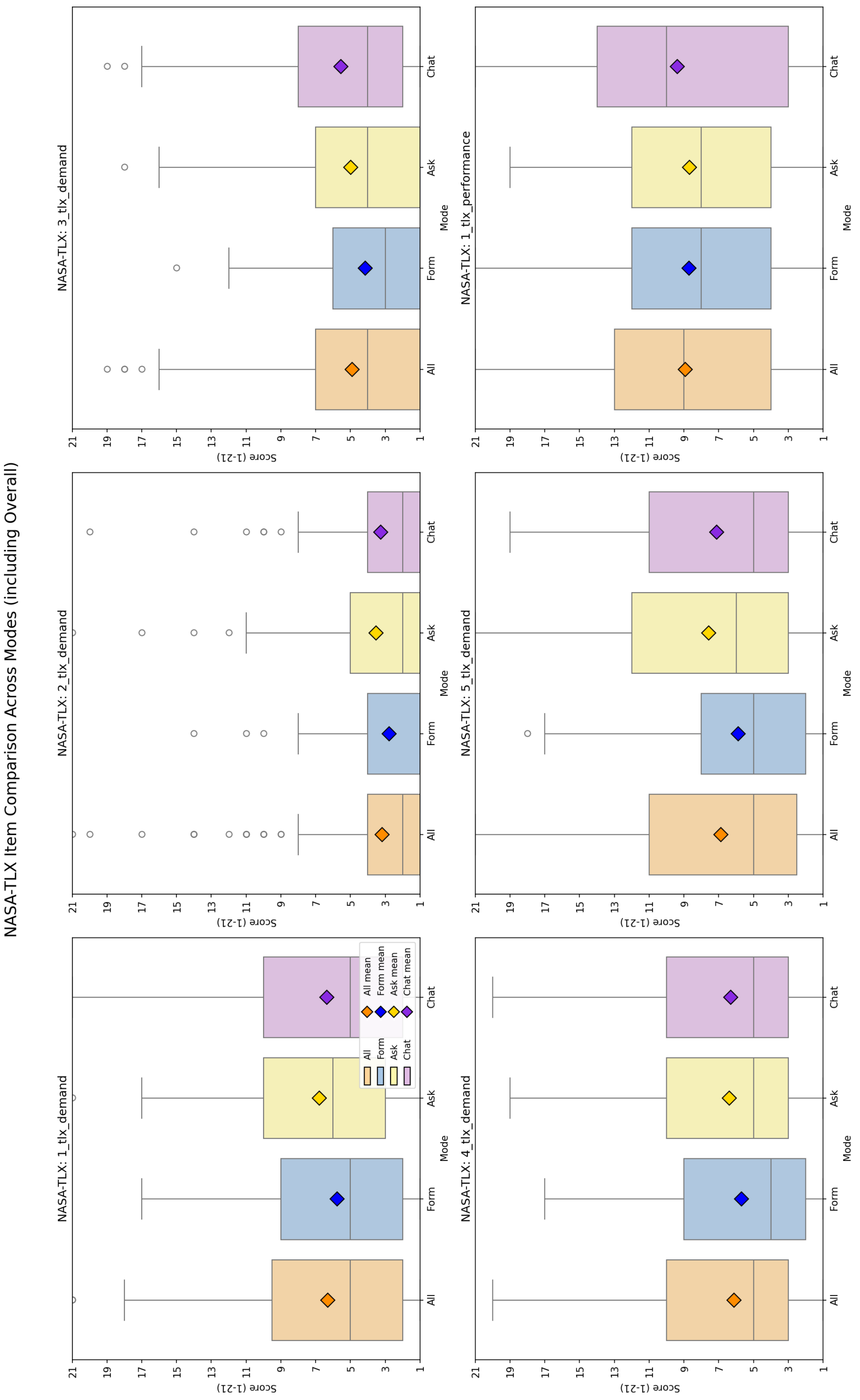

Figure 4.5: Box plots of the six R-TLX subscales by mode ($n = 69$ per mode).

Table 4.6: NASA R-TLX, full per-subscale descriptive distribution by mode ($n = 69$ per mode), 1–21 scale. Source: `tlx_per_item_summary.csv`.

Subscale	Mode	Mean	SD	Q1	Median	Q3	Min	Max
Mental Demand	Aggregated	6.31	4.75	2.0	5.0	9.5	1	21
	Form	5.77	4.26	2.0	5.0	9.0	1	17
	Ask	6.80	4.63	3.0	6.0	10.0	1	21
	Chat	6.36	5.31	2.0	5.0	10.0	1	21
Physical Demand	Aggregated	3.19	3.40	1.0	2.0	4.0	1	21
	Form	2.77	2.71	1.0	1.0	4.0	1	14
	Ask	3.54	3.92	1.0	2.0	5.0	1	21
	Chat	3.26	3.46	1.0	2.0	4.0	1	20
Temporal Demand	Aggregated	4.89	4.28	1.0	4.0	7.0	1	19
	Form	4.16	3.82	1.0	3.0	6.0	1	15
	Ask	4.97	4.15	1.0	4.0	7.0	1	18
	Chat	5.54	4.78	2.0	4.0	8.0	1	19
Effort	Aggregated	6.13	4.57	3.0	5.0	10.0	1	20
	Form	5.68	4.56	2.0	4.0	9.0	1	17
	Ask	6.39	4.41	3.0	5.0	10.0	1	19
	Chat	6.32	4.77	3.0	5.0	10.0	1	20
Frustration	Aggregated	6.86	5.24	2.5	5.0	11.0	1	21
	Form	5.88	4.84	2.0	5.0	8.0	1	18
	Ask	7.58	5.34	3.0	6.0	12.0	1	21
	Chat	7.13	5.44	3.0	5.0	11.0	1	19
Performance (pos.)	Aggregated	8.93	5.82	4.0	9.0	13.0	1	21
	Form	8.71	5.48	4.0	8.0	12.0	1	21
	Ask	8.68	5.64	4.0	8.0	12.0	1	19
	Chat	9.39	6.36	3.0	10.0	14.0	1	21

of the three. For *Physical Demand*, the aggregated mean is 3.19, with Form scoring lowest (2.77), Ask highest (3.54), and Chat in between (3.26). The absolute spread is small (≈ 0.8 points). Physical Demand is the lowest-rated subscale in all three modes.

Chat feels the most time-pressured of the three modes, and Form the least — the gap widens monotonically with LLM support, breaking from the Form-<-Chat-<-Ask shape seen above. For *Temporal Demand*, the aggregated mean is 4.89, with the per-mode values increasing monotonically from Form (4.16) to Ask (4.97) to Chat (5.54). Temporal Demand is approximately 1.4 points higher in Chat than in Form, the largest monotonic increase observed across the six subscales.

Form is also the least effortful overall, with Ask and Chat perceived as similarly more demanding. For *Effort*, the aggregated mean is 6.13, with Form scoring lowest (5.68) and Ask (6.39) and Chat (6.32) sitting at a closely comparable level. Effort increases by approximately 0.65 points from Form to the LLM-driven modes.

Form is the least frustrating mode by a clear margin, with Ask the most frustrating — the widest experiential gap of the six subscales. For *Frustration*, the aggregated mean is 6.86, with Form scoring lowest (5.88), Ask highest (7.58), and Chat in between (7.13). Frustration shows the largest absolute gap between Form and the LLM-driven modes ($+1.25$ to $+1.70$ points).

Participants feel about equally successful at the task across all three modes — the only subscale where the Form advantage disappears, with a marginal sense of worse self-performance in Chat. For *Performance* (positive-coded, higher equals worse perceived performance), the aggregated mean is 8.93, with Form and Ask essentially identical (8.71 and 8.68) and Chat slightly higher at 9.39. Perceived self-performance is comparable across modes, with a marginal degradation in Chat.

An aggregated Friedman test across the six subscales is non-significant ($\chi^2(2) = 5.24$, $p = 0.073$, $W = 0.038$). Among the six subscales, two reach significance: Temporal Demand ($\chi^2(2) = 12.77$, $p = 0.002$, $W = 0.093$) and Frustration ($\chi^2(2) = 8.93$, $p = 0.012$, $W = 0.065$). Pairwise post-hoc Wilcoxon signed-rank tests with Bonferroni correction identify Form-vs-Ask and Form-vs-Chat as significant on these two subscales, with medium-to-large rank-biserial effect sizes ($|r| = 0.36$ to 0.57); Ask versus Chat is non-significant on every measure. Full test statistics are reproduced in Appendix C.

The workload data therefore reinforce the Form-first ordering of the SUS, with the qualification that the gap is sharpest on Temporal Demand and Frustration and effectively absent on Performance.

Supplementary: per-mode completion time. The workload picture above is entirely perceptual: every value reported so far comes from the participant's self-report after each session. The R-TLX has a behavioral counterpart that sits one step away from the subscales and is reported here, as flagged at the top of this section, before the chapter turns to perceived AI support. Each document generated by ROPAgen carries a `Time elapsed:` N `seconds` line emitted at submission, which yields one objective completion-time measurement per participant–mode pairing on the same $n = 113$ document sample as the document-quality analyses. The question this supplementary paragraph asks is whether the perceived Temporal Demand ordering — Form lowest, Chat highest — is mirrored by the elapsed time itself.

Form takes roughly half the wall-clock time of either LLM-driven mode, while Ask and Chat are statistically indistinguishable from each other. Across the $n = 113$ retained documents (Form 37, Ask 37, Chat 39), the median per-mode completion times are Form 420 s (~ 7 min), Ask 858 s (~ 14 min), and Chat 717 s (~ 12 min); the corresponding means are Form 564.2 s, Ask 989.8 s, and Chat 979.1 s, with substantial right-skewed spread in the LLM-driven modes (Chat SD ≈ 1015 s, with one outlier at 6199 s). The full distributions are shown in Figure 4.6.

A Kruskal–Wallis test on the full $n = 113$ sample confirms that completion time varies by mode ($H = 16.46$, $p < 0.001$, $\varepsilon^2 = 0.15$, large). Pairwise Mann–Whitney U tests with Bonferroni correction ($\alpha = 0.0167$) identify Form as significantly faster than both Ask ($r = 0.512$, large; median 420 s vs 858 s) and Chat ($r = 0.418$, medium; median 420 s vs 717 s), while Ask and Chat are statistically indistinguishable from each other.

The behavioral data therefore add a sharper edge to the perceived workload picture: Form is not merely *felt* as the least time-pressured mode (Temporal Demand 4.16 vs. Ask 4.97 and Chat 5.54), it is also *measurably* the fastest by a margin of roughly seven minutes per document over either LLM-driven mode. Ask and Chat, despite their different interaction styles — per-section explainer versus freeform conversation — take essentially the same amount of time to complete on average.

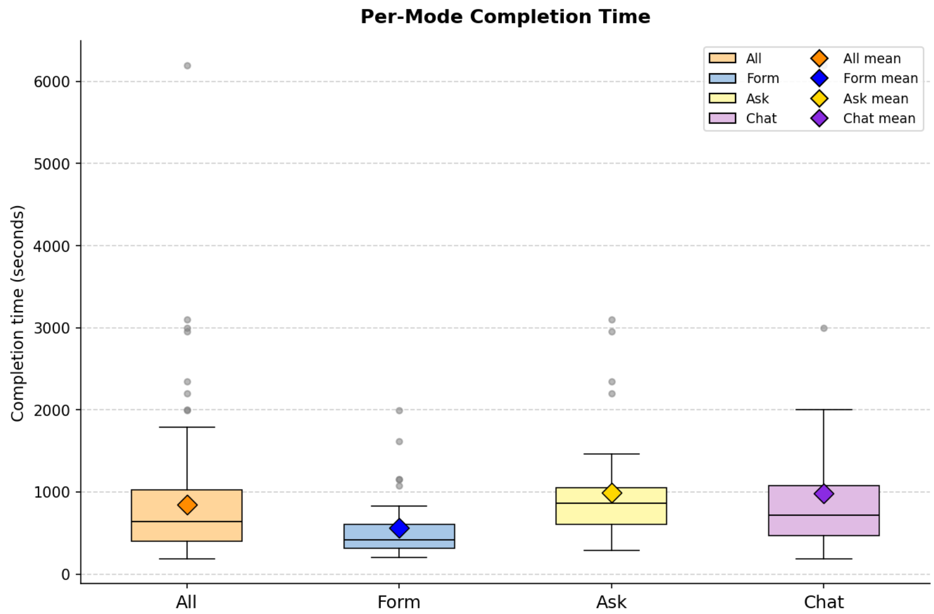

Figure 4.6: Per-mode completion time in seconds, extracted from the `Time elapsed` line emitted by ROPAgen at document submission ($n = 113$ documents: Form 37, Ask 37, Chat 39). Boxes show the interquartile range, the central line marks the median, and the diamond marks the mean.

Whether participants nevertheless ascribe more value to the LLM-driven modes when asked specifically about AI support is the question taken up next in Section 4.2.3.

4.2.3 Perceived AI Support — AI Questionnaire

The SUS (Section 4.2.1) and R-TLX (Section 4.2.2) characterize the interface as it was experienced; they say nothing directly about whether participants felt the AI itself had been useful. The Perceived AI Support questionnaire targets exactly this gap. After each interaction mode, in addition to the SUS and R-TLX, participants completed a six-item Perceived AI Support questionnaire on a five-point Likert scale (1 = strong disagreement, 5 = strong agreement). The six items capture distinct facets of how participants perceived the AI's contribution to the document they had just created: helpfulness (item 1), efficiency (item 2), transparency (item 3), trust (item 4), legal-basis confidence (item 5), and re-use intent (item 6). Item 5 is the primary self-report confidence measure referenced in Chapter 3. The Aggregated column in the tables below pools the three per-mode samples into a single per-item descriptive across all 207 ratings. Given the consistent Form-first ordering on usability and workload, the question of interest here is whether perceived LLM value follows the same pattern or breaks from it.

The six items are reported in their administered order, beginning with broad helpfulness and tightening towards specific facets such as transparency, trust, and re-use intent. Table 4.7 reports the mean response per item per mode and the aggregated mean across the three modes; the full descriptive distribution per item is reported in Table 4.8. Figure 4.7 shows the corresponding box-plot distributions.

Participants find the LLM most helpful in Ask, breaking the Form-first pattern seen on the usability instruments. For *"The AI was helpful in creating this document"* (item 1), the aggregated mean is 4.23, with Form at 4.23, Ask highest (4.32), and Chat lowest (4.13). Helpfulness is rated highest in Ask and lowest in Chat, with a small absolute spread of roughly 0.20 points.

When asked specifically about speed, participants give the most credit to Form and the least to Chat — the felt speed-up shrinks the more conversational the mode becomes. For *"The AI helped me to complete the document faster"* (item 2), the

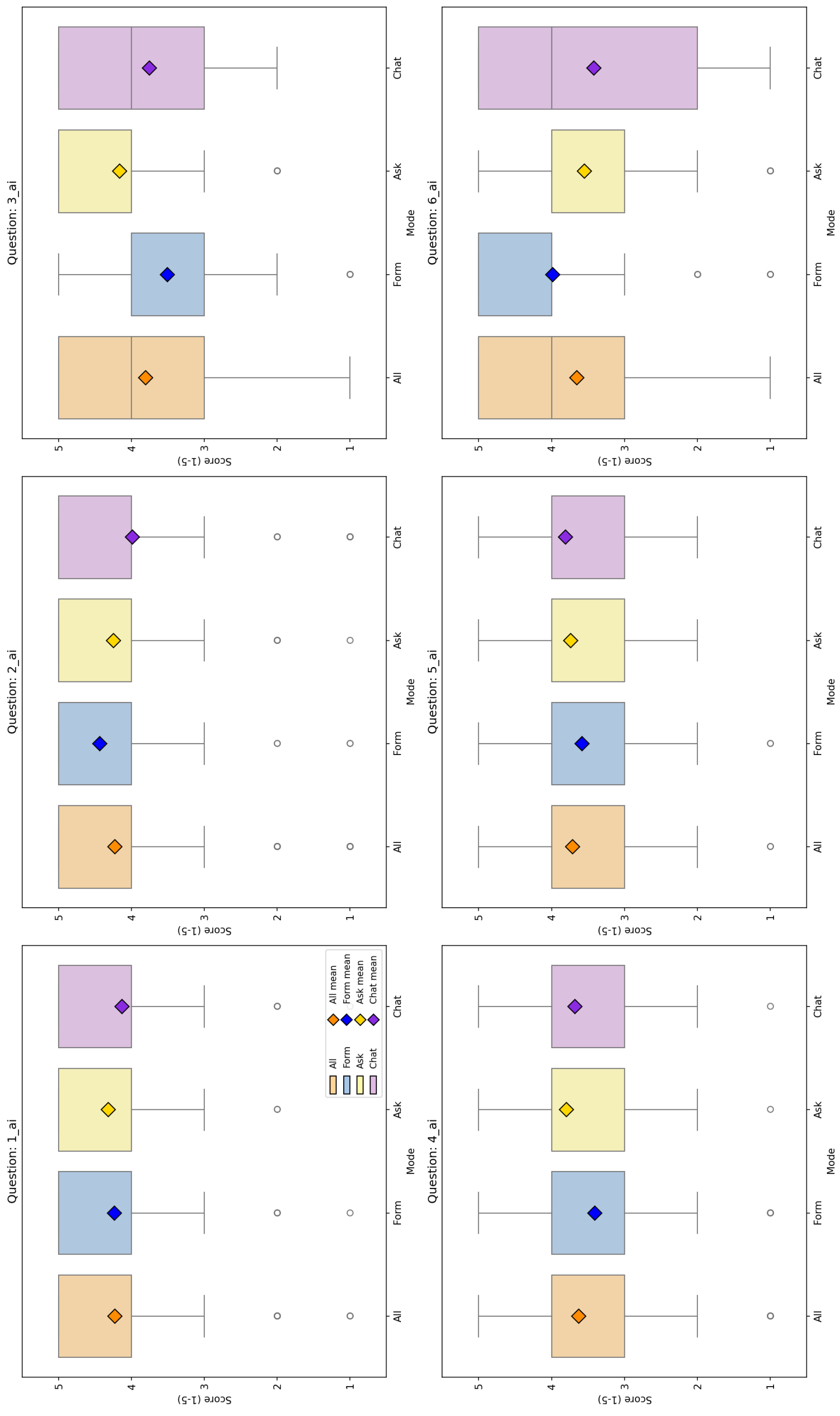

Figure 4.7: Box plots of the six Perceived AI Support items by mode ($n = 69$ per mode). Boxes show the interquartile range; the central line marks the median, and the diamond marks the mean.

Table 4.7: Perceived AI Support, mean response per item by mode ($n = 69$ per mode), five-point Likert scale (higher is better). The Aggregated column is the unweighted mean of the three per-mode means.

#	Item	Aggregated	Form	Ask	Chat
1	The AI was helpful in creating this document.	4.23	4.23	4.32	4.13
2	The AI helped me to complete the document faster.	4.22	4.43	4.25	3.99
3	I understood why the AI suggested specific GDPR fields.	3.81	3.51	4.16	3.75
4	I trust that the AI made correct suggestions.	3.63	3.41	3.80	3.68
5	I am confident that the AI identified the correct legal basis.	3.71	3.58	3.74	3.81
6	I would use this mode again.	3.65	3.99	3.55	3.42

aggregated mean is 4.22, with Form rated highest (4.43), Ask second (4.25), and Chat lowest (3.99). Perceived speed-up decreases monotonically with increasing LLM support, dropping by approximately 0.45 points from Form to Chat.

Ask makes participants feel they understand *why* the AI suggests what it does, by a clear margin — the widest reversal of the Form-first ordering across the six items. For *"I understood why the AI suggested specific GDPR fields"* (item 3), the aggregated mean is 3.81, with Form scoring lowest (3.51), Ask highest (4.16), and Chat in between (3.75). Transparency is rated approximately 0.65 points higher in Ask than in Form, the only mode that explicitly explains each suggestion (see Section 3.1).

Trust in the AI's suggestions tracks transparency: participants trust the LLM-driven modes more than Form, with Ask leading. For *"I trust that the AI made correct suggestions"* (item 4), the aggregated mean is 3.63, with Form scoring lowest (3.41), Ask highest (3.80), and Chat in between (3.68). Trust increases by approximately 0.27–0.39 points from Form to the LLM-driven modes.

Participants feel most confident that the legal basis is correct in Chat, and least confident in Form — confidence rises monotonically with LLM support. For *"I am confident that the AI identified the correct legal basis"* (item 5, primary confidence measure), the aggregated mean is 3.71, with the three modes showing a monotonic increase from Form (3.58) to Ask (3.74) to Chat (3.81). Confidence in legal correctness rises by approximately 0.23 points from Form to Chat.

When asked which mode they would actually use again, participants return to the Form-first ordering of the SUS — and the gap is the widest of the six AI-support items. For *"I would use this mode again"* (item 6), the aggregated mean is 3.65, with Form scoring highest (3.99), Ask second (3.55), and Chat lowest (3.42). Re-use intent decreases by approximately 0.57 points from Form to Chat, the largest negative gap of the six items.

Beyond the per-mode means, the full descriptive distribution shows how concentrated or spread participants' responses were on each item. Table 4.8 reports mean, standard deviation, Q1, median, Q3, min, and max per item per mode.

An aggregated Friedman test on the per-mode aggregate is non-significant ($\chi^2(2) = 1.05$, $p = 0.592$, $W = 0.008$), indicating that perceived AI support does not differ meaningfully across modes when the six items are considered together. The single legal-basis confidence item 5_{ai} is also non-significant under Friedman ($\chi^2(2) = 3.76$, $p = 0.152$). The re-use item 6_{ai} does reach significance ($\chi^2(2) = 11.12$, $p = 0.004$, $W = 0.081$), with a large rank-biserial effect for Form versus Chat ($r = +0.543$). Full statistics are reproduced in Appendix C.

Perceived AI support thus splits the six items into two camps: items that frame the LLM as an explainer (transparency, trust, legal-basis confidence) favor the LLM-driven modes, while items that frame it as a productivity aid (efficiency, re-use intent) favor Form. The behavioral counterpart of these per-item ratings — which mode participants actually choose when asked to rank — is examined next in Section 4.2.4.

4.2.4 Mode Preference and Free-Text Feedback

The per-instrument data so far have shown a consistent Form-first ordering on usability (Section 4.2.1) and workload (Section 4.2.2), with a partial inversion on perceived transparency, trust, and confidence (Section 4.2.3). This section asks how the three patterns combine when participants are forced to commit to a single overall preference. After all three interaction modes had been completed, participants provided two final inputs: an overall ranking of the three modes from most to least preferred, and an open-ended summary comment on the study experience. In addition, after each individual mode, an optional per-mode comment was collected.

Table 4.8: Perceived AI Support, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: `ai_support_per_item_summary.csv`.

#	Item	Mode	Mean	SD	Q1	Median	Q3	Min	Max
1	The AI was helpful in creating this document.	Aggregated	4.23	0.91	4	4	5	1	5
		Form	4.23	0.97	4	5	5	1	5
		Ask	4.32	0.76	4	4	5	1	5
		Chat	4.13	1.00	4	4	5	1	5
2	The AI helped me to complete the document faster.	Aggregated	4.22	1.13	4	5	5	1	5
		Form	4.43	0.96	4	5	5	1	5
		Ask	4.25	1.03	4	5	5	1	5
		Chat	3.99	1.32	4	4	5	1	5
3	I understood why the AI suggested specific GDPR fields.	Aggregated	3.81	1.06	3	4	5	1	5
		Form	3.51	1.17	3	4	4	1	5
		Ask	4.16	0.92	4	4	5	1	5
		Chat	3.75	0.98	3	4	5	1	5
4	I trust that the AI made correct suggestions.	Aggregated	3.63	0.97	3	4	4	1	5
		Form	3.41	1.09	3	4	4	1	5
		Ask	3.80	0.87	3	4	4	1	5
		Chat	3.68	0.92	3	4	4	1	5
5	I am confident that the AI identified the correct legal basis.	Aggregated	3.71	0.88	3	4	4	1	5
		Form	3.58	0.90	3	4	4	1	5
		Ask	3.74	0.87	3	4	4	1	5
		Chat	3.81	0.88	3	4	4	1	5
6	I would use this mode again.	Aggregated	3.65	1.25	3	4	5	1	5
		Form	3.99	1.14	4	4	5	1	5
		Ask	3.55	1.16	3	4	4	1	5
		Chat	3.42	1.39	2	4	5	1	5

This section reports the ranking distribution, the inferential test on the rank-1 votes, and a thematic summary of all 88 free-text comments contributed by 44 unique participants.

Mode Preference Ranking

The ranking offers the cleanest test of which mode participants actually preferred once they had used all three. Table 4.9 summarizes the post-experiment ranking. Participants assigned each of the three modes a rank from 1 (most preferred) to 3 (least preferred); "Rank-1 votes" counts the participants who placed the mode first.

Table 4.9: Post-experiment mode preference ranking ($n = 69$). "Rank-1 votes" counts the number of participants who placed the mode first; "Share" is the corresponding percentage; "Mean rank" averages the assigned 1–3 rank across all 69 participants.

Mode	Rank-1 votes	Share	Mean rank
Form	37	53.6 %	1.78
Chat	20	29.0 %	2.07
Ask	12	17.4 %	2.14

Form receives the plurality of first-place votes ($53.6\,\%$), followed by Chat ($29.0\,\%$) and Ask ($17.4\,\%$). The mean ranks follow the same ordering, mirroring the Form-first pattern of the SUS and R-TLX. A chi-square goodness-of-fit test on the rank-1 vote distribution rejects the uniform-null at $\chi^2(2) = 14.17$, $p < 0.001$, Cramér's $V = 0.32$ (medium effect): Form is preferred as the first-choice mode at a rate higher than chance. Full statistics are reproduced in Appendix C.

The complete distribution of all three rank positions across the three modes is visualised in Figure 4.8.

Free-Text Comments

The free-text comments contextualise the ranking by surfacing the reasons participants gave for their preferences. Each mode collected an optional per-mode open-ended comment, and after all three modes participants wrote an overall summary

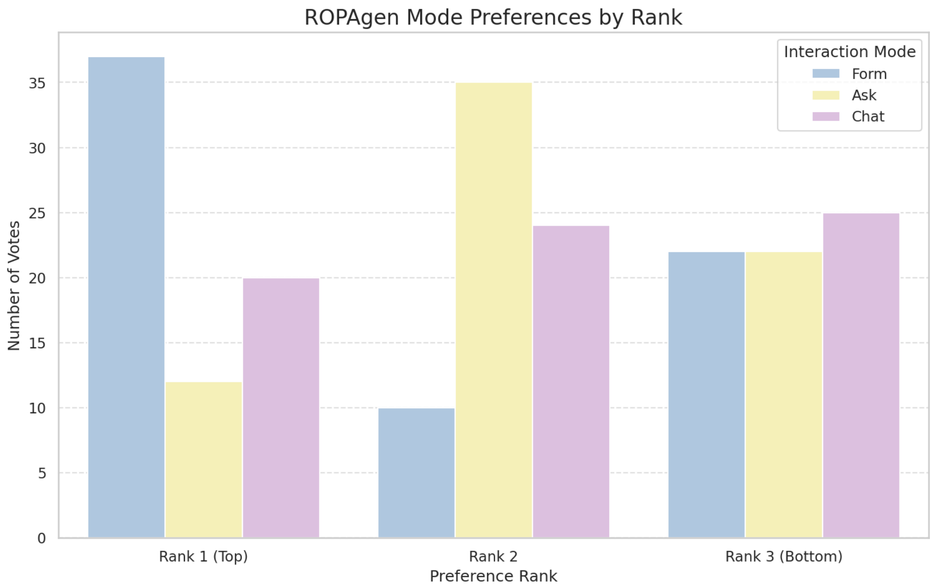

Figure 4.8: Distribution of all three rank positions per mode ($n = 69$).

comment. A total of 88 comments were recorded across 44 unique participants. Comments are written in German with a small minority in English; they are summarized thematically below. The presentation preserves the order Form, Ask, Chat, and concludes with the across-modes overall comments.

Form. Comments on Form cluster around speed, clarity, and control. Recurring observations describe the mode as the fastest of the three, the easiest to obtain an overview of, and the most "integrated" into the document creation workflow, as the checkbox interface makes the selection space visible at a glance. Several participants emphasize that Form "takes the most work off you" (*"nimmt einem die meiste Arbeit ab"*) by pre-populating the option set, while a smaller subset notes the opposite concern: that the absence of an interactive LLM commentary makes it unclear on what basis the model produces its suggestions, and that participants would not feel safe relying on Form output without an additional verification step. Several comments describe specific UX issues: long text fields rendered as a single narrow line on mobile, the inability to provide the LLM with additional context for its suggestions, and the impression that the LLM "guesses" without scenario-specific input.

Ask. Comments on Ask are the most polarised of the three modes. A consistent theme is that Ask is the most cumbersome mode in terms of workflow: the requirement to manually copy the LLM's suggested text into the form field (cf. Section 3.1) is repeatedly described as redundant work, particularly as participants reported that the model often produced suggestions in a free-text style that did not match the structured form fields. Several participants observed that the lack of a persistent context across fields forced them to re-enter the same information repeatedly. Conversely, explanatory text generated by Ask is consistently described as the most informative of the three modes, supporting the high transparency rating visible in item 3 of the AI Support questionnaire. A frequent suggestion is to combine Ask's explanations with Form's structured field layout.

Chat. Comments on Chat are dominated by two themes: comfort and opacity. On the comfort side, several participants describe Chat as the most intuitive of the three modes, especially as the LLM takes over the work of mapping the conversation onto the structured ROPA fields (cf. Section 3.1), leaving the participant only with the task of describing the processing scenario. On the opacity side, an equally large group of participants reports uncertainty about what is actually being written into the document — whether the entire chat history is used for generation, whether each individual message must be confirmed, and what the document looks like during the conversation. Specific defects mentioned include the LLM occasionally switching from German to English mid-response, occasional emission of raw `DATA_UPDATE` tokens, and a slow "streaming" rendering style that participants experienced as artificially throttled.

Across modes (overall comments). The overall comments recapitulate the per-mode pattern. A clear majority describes Form as the fastest and most ergonomic mode, particularly for participants who feel they "do not know enough" to interrogate the model meaningfully. Chat is typically placed second and motivated by the impression that "the AI takes care of it," which several participants explicitly note increases their *subjective* sense of safety even though they cannot verify the output. Ask is most often placed third, with the manual copy-paste step cited as the principal reason. A recurring constructive suggestion across participants is a hybrid mode that combines the visible option set of Form with the conversational, contextaware behavior of Chat or Ask. Several participants explicitly add a caveat that they are novices in the legal domain and that their preference ordering might shift if they had prior GDPR knowledge.

The ranking and comment data therefore close the user-experience half of the chapter on the same Form-first ordering established by the SUS and R-TLX, with Chat slotting in second on raw votes and Ask trailing largely on account of its copy-paste workflow. Whether the documents produced under each mode mirror this ordering — or invert it — is the question taken up next, beginning with the reference-based NLP metrics in Section 4.3.1.

4.3 Document Quality — NLP Metrics

The User Study (Section 4.2) showed a stable Form-first ordering on usability, workload, and overall preference, with Chat second and Ask third. This section opens the document-quality half of the evaluation by asking whether the same ordering holds when the output text itself is compared against a reference document. The reference-based metric suite mirrors the one applied by von Schwerin et al. [29] in their evaluation of LLM-generated ROPA data categories: five lexical metrics (BLEU, ROUGE-1/2/L, METEOR) that count surface-form token overlap, and three BERT-based metrics (BERTScore Precision / Recall / F1) together with document-level SBERT cosine similarity that compare semantic representations. All metrics are computed against the single expert-authored Form-mode reference document described in Chapter 3. For every measure the aggregated value across all three modes is reported first, followed by the per-mode results in the order Form, Ask, Chat. The presentation proceeds in three steps: Section 4.3.1 reports each metric at the document level, treating the ROPA as a single block of text; Section 4.3.2 recomputes the same metrics at the level of the six Article 30(1) ROPA chapters to identify where in the document the per-mode differences originate; and Section 4.3.3 summarizes the document-level and chapter-level findings together, highlighting the similarities between the two views as well as the gap in absolute scores.

4.3.1 Document-Level NLP Metrics

Document-quality results are reported on the full retained set of 113 documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$). Inferential mode comparisons use Kruskal–Wallis as the omnibus test with Bonferroni-corrected Mann–Whitney U as the pairwise post-hoc; full test statistics are reproduced in Appendix C. The descriptive tables in this subsection report — per metric and per mode — the full empirical distribution (n, min, Q1, median, Q3, max, mean, SD) computed from the raw per-document scores in the pipeline output `document_results.csv`. The metrics are reported in two groups: lexical metrics first, since they make the strictest demands on surface form, followed by the BERT-based and sentence-embedding metrics. Each metric is reported over the full participant document, treating the ROPA as a single block of text.

Lexical Metrics

Lexical metrics — BLEU, ROUGE, and METEOR — count surface-form token overlap with the reference document; they reward word-level fidelity but are blind to paraphrase, and are therefore reported first as they make the strictest demands on surface form. An LLM-generated document that covers the same content with different wording will score lower here than under the embedding-based metrics that follow. Table 4.10 reports the full distribution of the five lexical metrics (BLEU, ROUGE-1/2/L, METEOR) per mode, plus the aggregated row across all 113 documents. Figure 4.9 shows the corresponding box-plot distributions.

BLEU. BLEU *(Bilingual Evaluation Understudy)* ((3.1)) counts exact n-gram overlap with the reference and is therefore harsh on paraphrase: a document that says the same thing in different words scores low. It is precision-oriented (how much of what the candidate said also appears in the reference) with a brevity penalty, and the values reported here use $N = 4$ (matching unigrams through 4-grams).

Chat documents share slightly more exact phrasing with the expert reference than Form or Ask documents do — a small reversal of the Form-first ordering, but not at a magnitude participants would notice on the page. For *BLEU*, the aggregated mean is 0.133, with Form at 0.129, Ask at 0.127, and Chat at 0.143. Chat scores

Table 4.10: Document-level lexical metric distributions by mode (full sample $n = 113$). Source: `doc_nlp_summary.csv`.

Metric	Mode	n	Mean	SD	Q1	Median	Q3	Min	Max
BLEU	Aggregated	113	0.133	0.048	0.094	0.128	0.159	0.047	0.246
	Form	37	0.129	0.048	0.088	0.124	0.158	0.064	0.230
	Ask	37	0.127	0.056	0.083	0.125	0.148	0.047	0.246
	Chat	39	0.143	0.039	0.122	0.134	0.163	0.067	0.235
ROUGE-1	Aggregated	113	0.488	0.087	0.425	0.491	0.560	0.313	0.645
	Form	37	0.488	0.098	0.421	0.493	0.569	0.313	0.644
	Ask	37	0.465	0.088	0.412	0.449	0.522	0.322	0.618
	Chat	39	0.511	0.071	0.468	0.517	0.561	0.330	0.645
ROUGE-2	Aggregated	113	0.285	0.077	0.234	0.268	0.344	0.121	0.455
	Form	37	0.306	0.076	0.241	0.301	0.361	0.178	0.455
	Ask	37	0.257	0.080	0.200	0.238	0.275	0.121	0.429
	Chat	39	0.291	0.069	0.253	0.283	0.344	0.131	0.447
ROUGE-L	Aggregated	113	0.375	0.087	0.311	0.359	0.441	0.194	0.566
	Form	37	0.388	0.094	0.320	0.391	0.468	0.238	0.566
	Ask	37	0.347	0.091	0.296	0.326	0.376	0.194	0.545
	Chat	39	0.389	0.072	0.348	0.387	0.444	0.230	0.544
METEOR	Aggregated	113	0.389	0.062	0.359	0.388	0.431	0.234	0.514
	Form	37	0.412	0.051	0.367	0.419	0.442	0.335	0.514
	Ask	37	0.365	0.068	0.327	0.366	0.409	0.234	0.488
	Chat	39	0.390	0.057	0.374	0.389	0.421	0.235	0.501

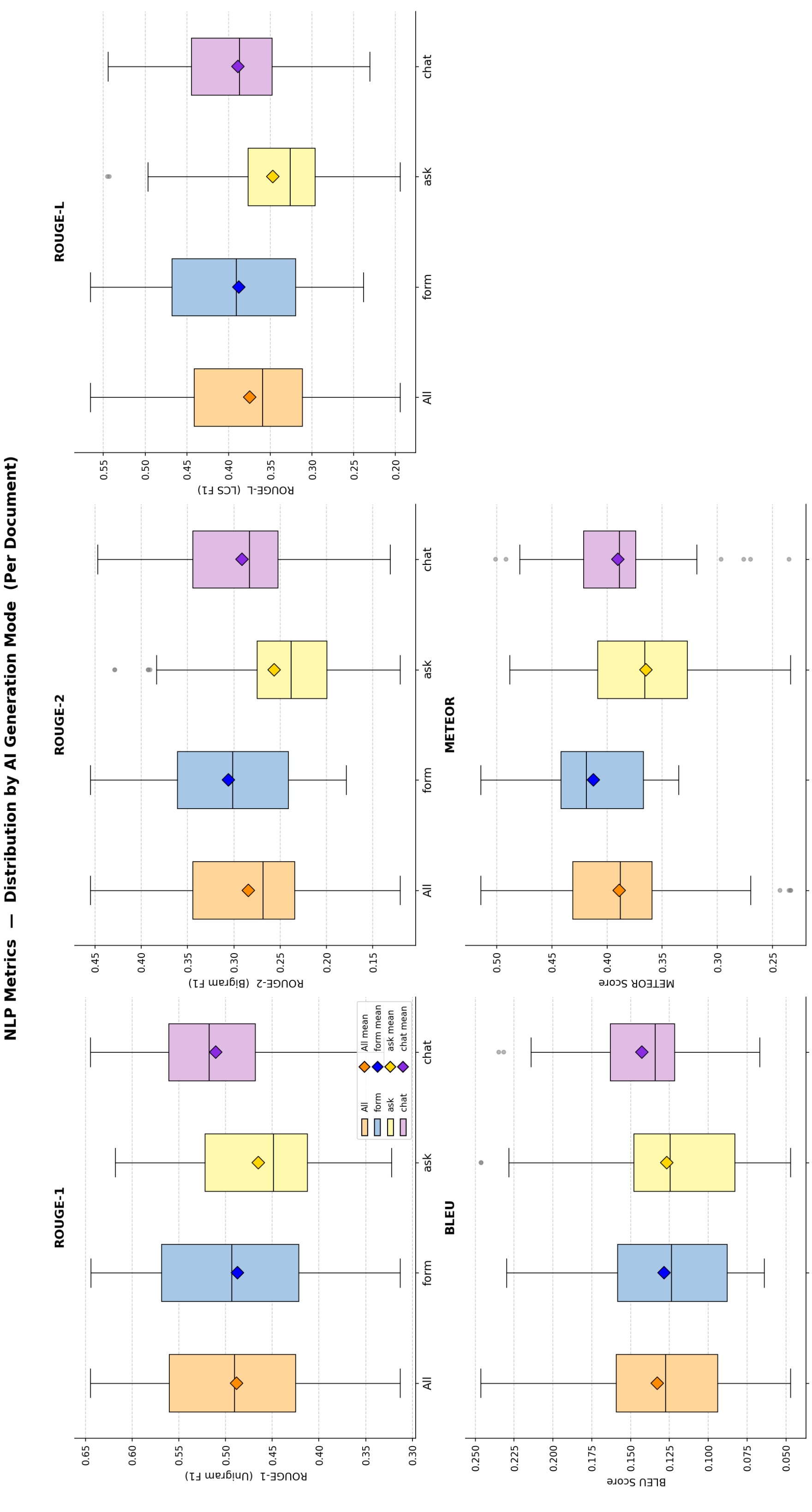

Figure 4.9: Document-level distributions of BLEU, ROUGE-1/2/L, and METEOR by mode (full sample $n = 113$). Boxes show the interquartile range; the diamond marks the mean.

marginally higher than Form and Ask, which are essentially indistinguishable — a separation too small to reach statistical significance on the full sample.

ROUGE-1, ROUGE-2, ROUGE-L. ROUGE *(Recall-Oriented Understudy for Gisting Evaluation)* ((3.2)–(3.4)) measures how much of the reference's wording the candidate recovers, in three flavours: unigrams (single words), bigrams (adjacent word pairs), and the longest common subsequence. Each is reported as F1 — the harmonic mean of precision (how much of what the candidate said belongs) and recall (how much of what should have been said is there) — so it is more permissive than BLEU in word order but still surface-form.

Looking at single-word overlap, Chat and Form documents recover the reference's vocabulary at a similar rate, with Ask documents trailing. For *ROUGE-1*, the aggregated mean is 0.488, with Form at 0.488, Ask lowest (0.465), and Chat highest (0.511). ROUGE-1 increases from Ask to Form to Chat, but the spread is narrow and does not reach statistical significance on the full sample.

When two-word phrases are compared, Form documents reproduce the reference's exact wording most often, with Chat second and Ask last. For *ROUGE-2*, the aggregated mean is 0.285, with Form yielding the highest mean (0.306), Ask lowest (0.257), and Chat in between (0.291). ROUGE-2 follows the Form $>$ Chat $>$ Ask ordering, with the Ask trough wide enough to register as statistically significant against both Form and Chat on the full sample.

When the longest shared word sequence is what matters, Form and Chat documents track the reference equally well, and Ask trails both. For *ROUGE-L*, the aggregated mean is 0.375, with Form (0.388) and Chat (0.389) essentially tied and Ask trailing (0.347). The mode difference reaches significance overall, though no individual pair survives Bonferroni-corrected post-hoc testing.

METEOR. METEOR *(Metric for Evaluation of Translation with Explicit ORdering)* ((3.5)) softens BLEU by allowing matches not just on identical tokens but also on stems and WordNet synonyms, so a document that paraphrases with related words is no longer punished as if it had said something unrelated. It is a fragmentation-penalised harmonic mean of unigram precision and recall, sitting between BLEU and ROUGE in strictness.

Form documents use the most reference-like terminology even when synonyms are credited, with Chat second and Ask last. For *METEOR*, the aggregated mean is 0.389, with Form yielding the highest mean (0.412), Ask lowest (0.365), and Chat in between (0.390). Form is significantly higher than Ask on the full sample, while the Form-versus-Chat and Chat-versus-Ask comparisons fall short.

Three of the five lexical metrics — ROUGE-2, ROUGE-L, METEOR — reach statistical significance on the full sample; BLEU and ROUGE-1 do not. Full test statistics are reproduced in Appendix C.

BERT-Based and Sentence-Embedding Metrics

BERT-based metrics measure semantic rather than lexical similarity, comparing contextual embeddings rather than literal tokens. Where the lexical metrics penalise paraphrase, the embedding-based metrics credit semantic overlap and therefore offer a complementary view of the same documents. Table 4.11 reports the same full distribution for the BERT-based metrics (BERTScore Precision / Recall / F1) and the document-level SBERT cosine similarity computed with ModernBERT embeddings. Figure 4.10 shows the corresponding box-plot distributions.

BERTScore Precision, Recall, F1. BERTScore ((3.6)–(3.8)) replaces n-gram matching with cosine similarities between contextual token embeddings, so paraphrase that shares meaning is credited even without surface-form overlap. *Precision* measures how much of what the candidate said belongs (penalising spurious content), *recall* measures how much of what should have been said is there (penalising omission), and *F1* is their harmonic mean — both must be reasonable for F1 to be high, since the harmonic mean is dragged down by the smaller of the two.

When the question shifts from *exact wording* to *related meaning*, Chat documents become the most on-topic — what they say is most closely aligned with the reference, with very little extraneous content. For *BERTScore Precision*, the aggregated mean is 0.890, with Chat yielding the highest mean (0.895), Form second (0.889), and Ask lowest (0.886). The mode difference is statistically significant overall, though no individual pair survives Bonferroni correction.

Table 4.11: Document-level BERT-based and SBERT distributions by mode (full sample $n = 113$). Source: `doc_bert_summary.csv`.

Metric	Mode	n	Mean	SD	Q1	Median	Q3	Min	Max
BERTScore Precision	Aggregated	113	0.890	0.014	0.880	0.891	0.900	0.861	0.922
	Form	37	0.889	0.014	0.880	0.889	0.899	0.862	0.912
	Ask	37	0.886	0.016	0.872	0.889	0.899	0.861	0.918
	Chat	39	0.895	0.012	0.890	0.896	0.902	0.867	0.922
BERTScore Recall	Aggregated	113	0.915	0.014	0.914	0.918	0.922	0.861	0.936
	Form	37	0.923	0.005	0.919	0.923	0.926	0.914	0.936
	Ask	37	0.908	0.017	0.908	0.915	0.918	0.861	0.925
	Chat	39	0.913	0.013	0.908	0.917	0.922	0.874	0.933
BERTScore F1	Aggregated	113	0.902	0.011	0.895	0.904	0.910	0.863	0.922
	Form	37	0.905	0.009	0.900	0.906	0.912	0.887	0.920
	Ask	37	0.897	0.013	0.892	0.896	0.906	0.863	0.918
	Chat	39	0.904	0.010	0.899	0.906	0.911	0.872	0.922
SBERT (ModernBERT)	Aggregated	113	0.987	0.004	0.985	0.988	0.990	0.971	0.994
	Form	37	0.986	0.004	0.984	0.988	0.989	0.974	0.992
	Ask	37	0.986	0.005	0.983	0.987	0.990	0.971	0.993
	Chat	39	0.990	0.002	0.988	0.990	0.991	0.985	0.994

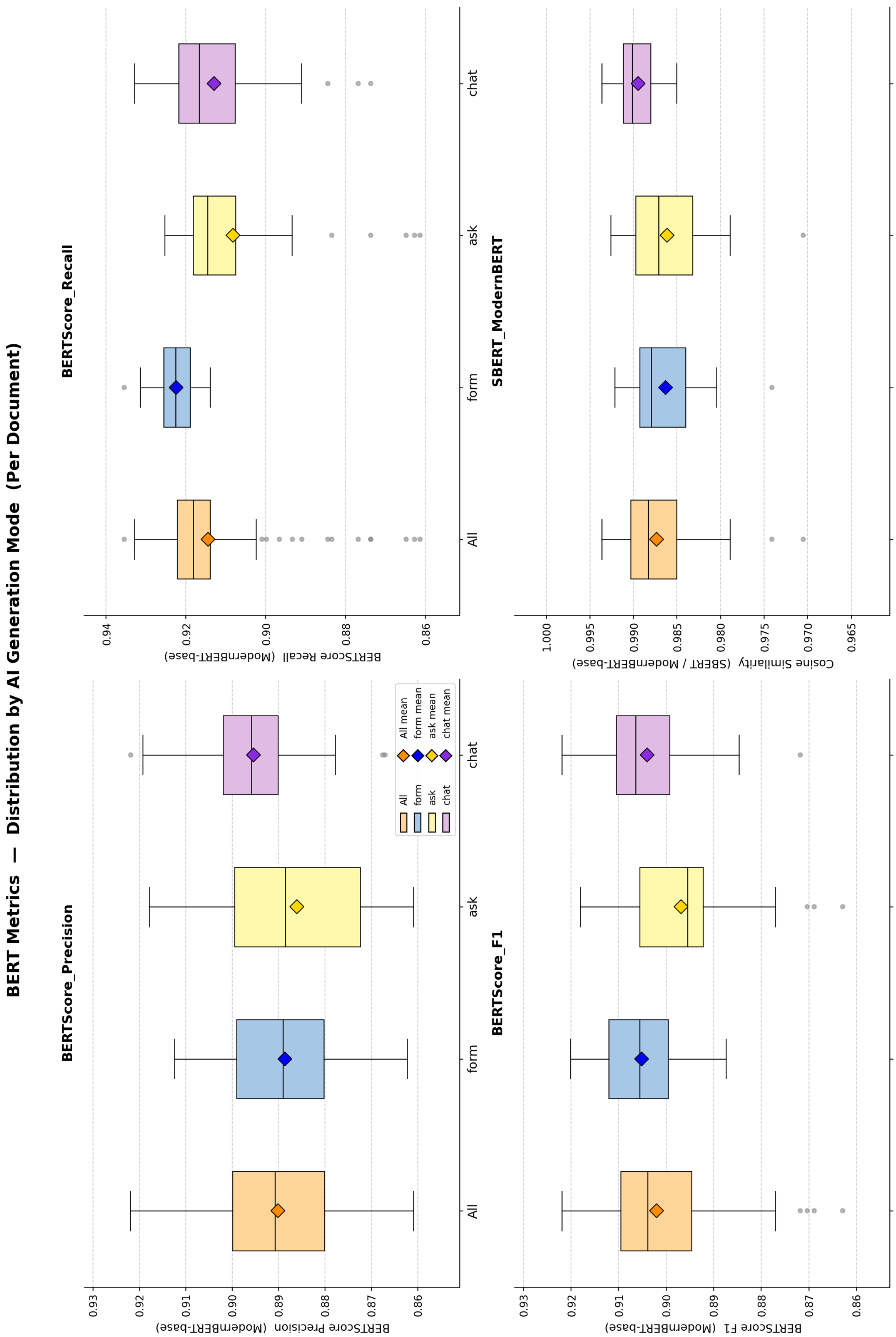

Figure 4.10: Document-level distributions of BERTScore Precision / Recall / F1 and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$).

Form documents recover the most of what the reference says — the clearest finding of the entire metric pipeline. Where Chat trims to what is on-topic, Form reproduces the reference's content near-completely, while Ask leaves the most out. For *BERTScore Recall*, the aggregated mean is 0.915, with Form yielding the highest mean (0.923) and Ask (0.908) and Chat (0.913) trailing. This is the strongest and most statistically robust finding in the entire metric pipeline — Form is significantly higher than both Ask and Chat, with large effect sizes.

Form's content coverage and Chat's tighter focus balance out when the two are combined: the two modes are effectively tied on overall semantic similarity, with Ask trailing. For *BERTScore F1*, the aggregated mean is 0.902, with Form yielding the highest mean (0.905), Ask lowest (0.897), and Chat in between (0.904). The Ask trough is statistically significant against both Form and Chat, but Form and Chat are indistinguishable — reflecting the trade-off between Form's higher Recall and Chat's higher Precision.

SBERT (ModernBERT) cosine similarity. Sentence-BERT cosine similarity ((3.9)) compresses each document into a single embedding vector and reports how aligned the two vectors are, on a scale where 1 is identical meaning and 0 is unrelated. It is the most lenient of the metrics reported here: as the comparison is at the document level rather than the token level, two documents that cover the same topics with different wording typically obtain values close to 1.

When the document is read as a whole, Chat reads the closest to the reference — Chat documents cover the same overall ground in a way no other mode quite matches. For *SBERT cosine similarity (ModernBERT)*, the aggregated mean is 0.987, with Chat yielding the highest mean (0.990) and Form and Ask tied at 0.986. Chat is statistically significantly higher than both other modes, with large effect sizes.

All four BERT-based and sentence-embedding metrics reach statistical significance on the full sample, with BERTScore Recall producing the strongest effect.

In summary, on the document level the three modes split into the patterns *Form* $\approx$ *Chat* $>$ *Ask* (BLEU, ROUGE-1, ROUGE-L, METEOR, BERTScore Precision, BERTScore F1), *Form* $>$ *Chat* $>$ *Ask* on n-gram overlap of size two and BERTScore Recall, and *Chat* $>$ *Form* $\approx$ *Ask* on SBERT. Ask underperforms both Form and Chat on every metric; Form and Chat trade leadership depending on whether the metric emphasizes lexical recall (Form leads) or semantic-document-level fluency (Chat leads). The full between-groups test results are reproduced in Appendix C.

4.3.2 Chapter-Level NLP Metrics

The document-level scores reported in Section 4.3.1 collapse the entire ROPA into a single number per metric, which is convenient for ranking modes but obscures where in the document the differences originate. This subsection recomputes the same nine metrics at the level of the six rendered ROPA chapters that the `finalropa` prompt produces from the eight internal `DocumentData` groups described in Chapter 3 (Section 3.1); the schema's data-categories and data-sources groups merge into a single output chapter, the metadata group is not rendered as a chapter, and the free-text fields are distributed into the purpose and T&O chapters. The six rendered chapters are — *Verantwortliche Stelle und Kontaktdaten* (Controller and Contact Details), *Zwecke und Rechtsgrundlagen der Verarbeitung* (Purposes and Legal Bases of Processing), *Kategorien personenbezogener Daten und Datenquellen* (Categories of Personal Data and Sources), *Betroffene Personen und Empfänger* (Data Subjects and Recipients), *Aufbewahrungsfristen und Löschung* (Retention Periods and Deletion), and *Technische und organisatorische Maßnahmen* (Technical and Organisational Measures). Chapter-level results characterize where in the document each mode performs best. Following the same two-part presentation as the document-level subsection, lexical metrics are reported first (Tables 4.12–4.16, Figure 4.11), followed by the BERT-based and sentence-embedding metrics (Tables 4.17–4.20, Figure 4.12). The descriptive source for all chapter-level tables is the pipeline output `chapter_results.csv`.

Lexical Metrics

Tables 4.12–4.16 report the per-chapter, per-mode means (with SDs in parentheses) for the five lexical metrics, and Figure 4.11 shows the corresponding distributions. The per-metric paragraphs that follow walk through the across-chapter pattern in the same metric order as the document-level subsection.

Table 4.12: Chapter-level BLEU mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.181 (0.080)	0.179 (0.077)	0.200 (0.085)	0.166 (0.076)
Purposes & Legal Bases	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Personal Data Categories	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Data Subjects & Recipients	0.012 (0.014)	0.013 (0.010)	0.008 (0.014)	0.014 (0.015)
Retention & Deletion	0.044 (0.076)	0.038 (0.044)	0.023 (0.073)	0.068 (0.094)
Technical & Org. Measures	0.080 (0.066)	0.078 (0.068)	0.070 (0.080)	0.091 (0.047)

Table 4.13: Chapter-level ROUGE-1 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.571 (0.114)	0.550 (0.097)	0.605 (0.129)	0.559 (0.109)
Purposes & Legal Bases	0.028 (0.017)	0.029 (0.014)	0.031 (0.024)	0.024 (0.012)
Personal Data Categories	0.070 (0.049)	0.056 (0.035)	0.082 (0.062)	0.071 (0.045)
Data Subjects & Recipients	0.103 (0.052)	0.090 (0.034)	0.093 (0.060)	0.123 (0.052)
Retention & Deletion	0.260 (0.131)	0.252 (0.092)	0.212 (0.132)	0.314 (0.143)
Technical & Org. Measures	0.331 (0.130)	0.313 (0.137)	0.330 (0.157)	0.350 (0.090)

Table 4.14: Chapter-level ROUGE-2 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.386 (0.145)	0.363 (0.131)	0.430 (0.158)	0.365 (0.138)
Purposes & Legal Bases	0.011 (0.013)	0.014 (0.013)	0.012 (0.016)	0.008 (0.009)
Personal Data Categories	0.041 (0.036)	0.035 (0.030)	0.053 (0.046)	0.036 (0.029)
Data Subjects & Recipients	0.073 (0.048)	0.072 (0.030)	0.059 (0.056)	0.089 (0.049)
Retention & Deletion	0.131 (0.113)	0.134 (0.071)	0.087 (0.107)	0.168 (0.137)
Technical & Org. Measures	0.176 (0.118)	0.166 (0.120)	0.163 (0.141)	0.197 (0.089)

Table 4.15: Chapter-level ROUGE-L mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.534 (0.129)	0.518 (0.110)	0.576 (0.140)	0.511 (0.128)
Purposes & Legal Bases	0.027 (0.017)	0.028 (0.015)	0.030 (0.022)	0.024 (0.012)
Personal Data Categories	0.070 (0.049)	0.056 (0.035)	0.082 (0.062)	0.071 (0.045)
Data Subjects & Recipients	0.103 (0.052)	0.090 (0.034)	0.093 (0.060)	0.123 (0.052)
Retention & Deletion	0.229 (0.136)	0.218 (0.087)	0.183 (0.141)	0.283 (0.152)
Technical & Org. Measures	0.257 (0.135)	0.237 (0.140)	0.246 (0.158)	0.285 (0.100)

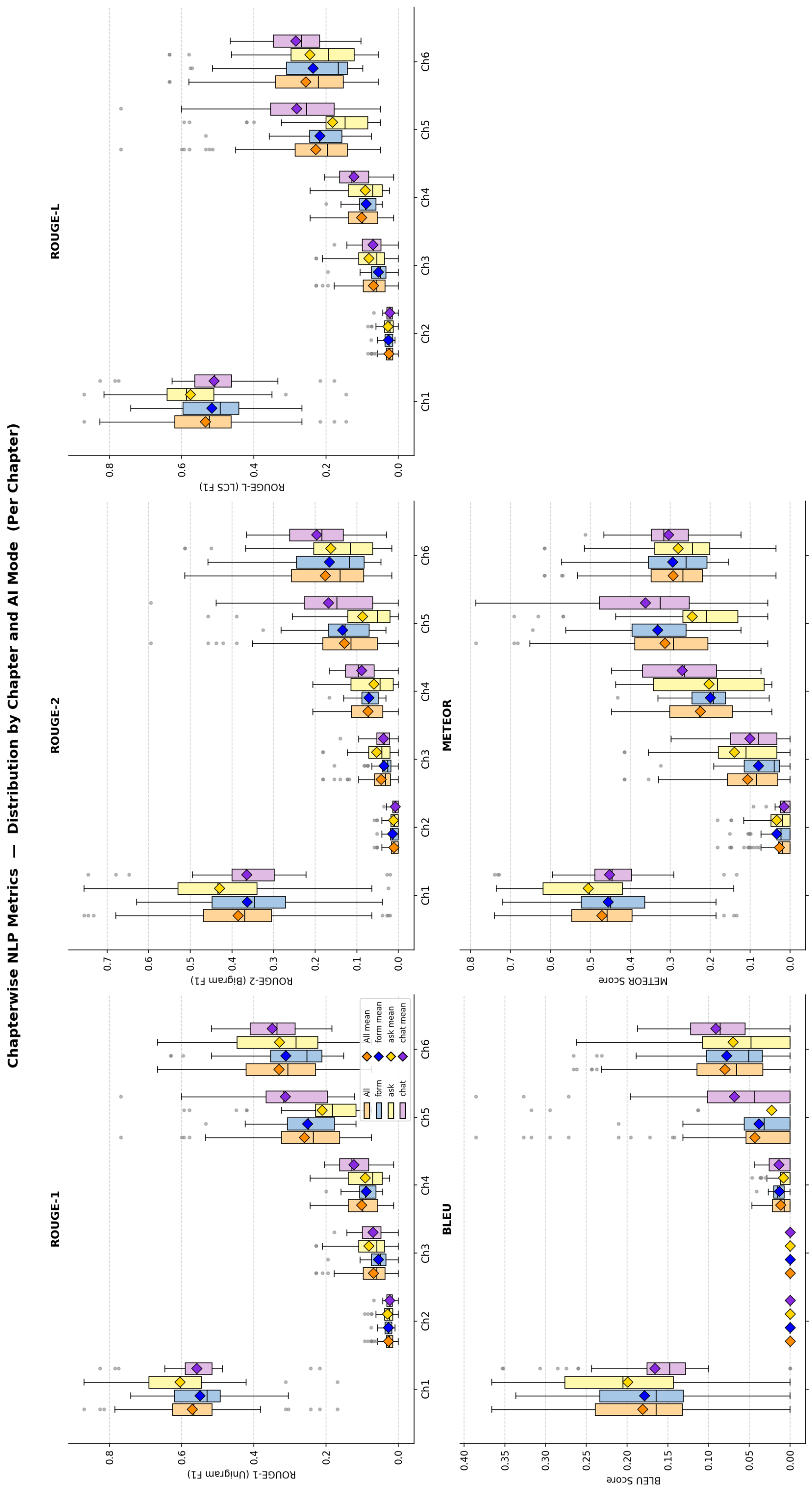

Figure 4.11: Chapter-level distributions of BLEU, ROUGE-1/2/L, and METEOR by mode (full sample $n = 113$).

Table 4.16: Chapter-level METEOR mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.471 (0.128)	0.456 (0.120)	0.506 (0.136)	0.453 (0.124)
Purposes & Legal Bases	0.027 (0.037)	0.034 (0.037)	0.034 (0.047)	0.015 (0.019)
Personal Data Categories	0.106 (0.096)	0.079 (0.073)	0.140 (0.116)	0.101 (0.086)
Data Subjects & Recipients	0.225 (0.112)	0.199 (0.078)	0.203 (0.134)	0.270 (0.105)
Retention & Deletion	0.314 (0.160)	0.332 (0.114)	0.245 (0.160)	0.362 (0.177)
Technical & Org. Measures	0.293 (0.113)	0.295 (0.108)	0.280 (0.145)	0.305 (0.079)

BLEU. The chapter-level BLEU signal is concentrated almost entirely in *Controller and Contact Details*, the one ROPA chapter whose content is templated enough for exact 4-gram overlap to register (aggregated 0.181; Form 0.179, Ask 0.200, Chat 0.166). *Purposes and Legal Bases* and *Categories of Personal Data* collapse to zero across every mode — the legal-text and category-list registers share no 4-grams with the reference even where they overlap semantically. In the remaining three chapters BLEU sits in low single digits, with *Retention and Deletion* producing the clearest chapter-level Chat lead in the lexical metric suite (aggregated 0.044; Form 0.038, Ask 0.023, Chat 0.068) and *Technical and Organisational Measures* echoing it on a smaller scale (aggregated 0.080; Form 0.078, Ask 0.070, Chat 0.091).

ROUGE-1, ROUGE-2, ROUGE-L. Single-word recall reproduces the same chapter hierarchy as BLEU but lifts every chapter off the floor. *Controller and Contact* again leads (aggregated 0.571; Form 0.550, Ask 0.605, Chat 0.559), with Ask edging the other two modes, as the address-block content offers little room for paraphrase. *Purposes and Legal Bases* (aggregated 0.028) and *Categories of Personal Data* (aggregated 0.070) remain very low. *Technical and Organisational Measures* is the second-highest chapter (aggregated 0.331), and *Retention and Deletion* is where Chat's chapter-level advantage is most visible at the unigram level (aggregated 0.260; Form 0.252, Ask 0.212, Chat 0.314).

The bigram view sharpens the same picture. *Controller and Contact* stays at the top (aggregated 0.386; Form 0.363, Ask 0.430, Chat 0.365), and *Purposes and Legal Bases* stays at the floor (aggregated 0.011). Where ROUGE-1 left *Retention and Deletion* as a moderate Chat lead, ROUGE-2 separates the modes more sharply (aggregated 0.131; Form 0.134, Ask 0.087, Chat 0.168) — Chat is the only mode that reproduces multi-word retention phrasing from the reference at any volume. *Technical and Organisational Measures* repeats the pattern on a smaller margin (aggregated 0.176; Form 0.166, Ask 0.163, Chat 0.197).

The longest-shared-sequence variant tracks ROUGE-1 closely. *Controller and Contact* dominates (aggregated 0.534; Form 0.518, Ask 0.576, Chat 0.511), *Purposes* (aggregated 0.027) and *Categories of Personal Data* (aggregated 0.070) stay near the floor, and Chat's chapter-level advantage on *Retention and Deletion* persists (aggregated 0.229; Form 0.218, Ask 0.183, Chat 0.283). *Technical and Organisational Measures* reinforces the same direction on a smaller margin (aggregated 0.257; Form 0.237, Ask 0.246, Chat 0.285).

METEOR. Once stems and synonyms are credited the chapter range compresses. *Controller and Contact* retains the top position (aggregated 0.471; Form 0.456, Ask 0.506, Chat 0.453), and *Purposes and Legal Bases* remains very low (aggregated 0.027, with all three modes below 0.04). The mid-table chapters lift the most: *Data Subjects and Recipients* climbs to aggregated 0.225 on the strength of synonym credit, and shows the clearest Chat lead on this metric (Form 0.199, Ask 0.203, Chat 0.270). *Retention and Deletion* (aggregated 0.314; Form 0.332, Ask 0.245, Chat 0.362) and *Technical and Organisational Measures* (aggregated 0.293; Form 0.295, Ask 0.280, Chat 0.305) again favor Chat.

Across the five lexical metrics the across-chapter signal is dominated by two endpoints: *Controller and Contact Details* at the top — the chapter whose content is templated enough for surface overlap to register, and the one chapter where the structured Ask dialogue tends to edge out Chat — and *Purposes and Legal Bases* at the floor, where free-form legal formulation pulls every score toward zero. Chat's chapter-level lead, where it appears, concentrates on *Retention and Deletion* and *Technical and Organisational Measures*.

BERT-Based and Sentence-Embedding Metrics

Tables 4.17–4.21 report the per-chapter, per-mode means (with SDs in parentheses) for the three BERTScore variants and the SBERT cosine similarity under two encoder backbones — ModernBERT (primary, matching the document-level pass) and the smaller `all-MiniLM-L6-v2` (reported as a robustness check on whether ModernBERT's near-saturation masks real differences). Figure 4.12 shows the BERTScore and SBERT (ModernBERT) distributions; the MiniLM pass is reported in tabular form only. The metric walkthrough follows the same order as the document-level subsection.

Table 4.17: Chapter-level BERTScore Precision mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.857 (0.017)	0.855 (0.015)	0.862 (0.020)	0.855 (0.016)
Purposes & Legal Bases	0.566 (0.019)	0.560 (0.009)	0.570 (0.025)	0.568 (0.016)
Personal Data Categories	0.686 (0.034)	0.670 (0.022)	0.690 (0.040)	0.698 (0.032)
Data Subjects & Recipients	0.700 (0.030)	0.686 (0.018)	0.700 (0.035)	0.712 (0.030)
Retention & Deletion	0.824 (0.042)	0.814 (0.033)	0.817 (0.050)	0.840 (0.037)
Technical & Org. Measures	0.854 (0.033)	0.841 (0.038)	0.858 (0.031)	0.863 (0.027)

Table 4.18: Chapter-level BERTScore Recall mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.882 (0.022)	0.879 (0.019)	0.887 (0.025)	0.879 (0.022)
Purposes & Legal Bases	0.666 (0.016)	0.670 (0.016)	0.667 (0.017)	0.662 (0.016)
Personal Data Categories	0.849 (0.013)	0.850 (0.012)	0.849 (0.015)	0.847 (0.011)
Data Subjects & Recipients	0.842 (0.020)	0.851 (0.016)	0.838 (0.017)	0.836 (0.022)
Retention & Deletion	0.889 (0.024)	0.890 (0.018)	0.879 (0.023)	0.897 (0.027)
Technical & Org. Measures	0.888 (0.025)	0.889 (0.018)	0.882 (0.032)	0.893 (0.022)

BERTScore Precision, Recall, F1. The embedding-based view compresses the across-chapter range substantially: no chapter sits at zero, and the highest-to-lowest gap shrinks to roughly 0.3. *Controller and Contact Details* (aggregated 0.857) and *Technical and Organisational Measures* (aggregated 0.854) sit at the top; *Purposes and Legal Bases* remains the weakest chapter (aggregated 0.566) but no longer collapses to the floor. Chat is the per-mode leader on three of the four mid-and-lower-ranked chapters — *Categories of Personal Data* (aggregated 0.686; Form 0.670, Ask 0.690, Chat 0.698), *Data Subjects and Recipients* (aggregated 0.700; Form 0.686, Ask 0.700, Chat 0.712), and *Retention and Deletion* (aggregated

Table 4.19: Chapter-level BERTScore F1 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.869 (0.019)	0.867 (0.016)	0.875 (0.022)	0.867 (0.017)
Purposes & Legal Bases	0.612 (0.012)	0.610 (0.009)	0.614 (0.014)	0.611 (0.010)
Personal Data Categories	0.759 (0.023)	0.749 (0.016)	0.761 (0.028)	0.765 (0.022)
Data Subjects & Recipients	0.764 (0.020)	0.760 (0.014)	0.762 (0.024)	0.769 (0.022)
Retention & Deletion	0.855 (0.031)	0.850 (0.024)	0.847 (0.035)	0.867 (0.031)
Technical & Org. Measures	0.871 (0.025)	0.864 (0.027)	0.870 (0.027)	0.878 (0.019)

Table 4.20: Chapter-level SBERT (ModernBERT) cosine similarity mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.976 (0.005)	0.975 (0.004)	0.977 (0.006)	0.975 (0.005)
Purposes & Legal Bases	0.735 (0.010)	0.736 (0.009)	0.736 (0.012)	0.732 (0.007)
Personal Data Categories	0.866 (0.019)	0.859 (0.015)	0.869 (0.021)	0.871 (0.018)
Data Subjects & Recipients	0.871 (0.018)	0.868 (0.013)	0.872 (0.019)	0.872 (0.022)
Retention & Deletion	0.949 (0.016)	0.945 (0.013)	0.946 (0.018)	0.955 (0.014)
Technical & Org. Measures	0.971 (0.010)	0.967 (0.012)	0.972 (0.010)	0.975 (0.007)

Table 4.21: Chapter-level SBERT (`all-MiniLM-L6-v2`) cosine similarity mean (SD) by mode (full sample $n = 113$). Reported as a robustness check on the ModernBERT pass.

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.687 (0.122)	0.663 (0.119)	0.736 (0.133)	0.662 (0.101)
Purposes & Legal Bases	0.298 (0.074)	0.283 (0.052)	0.326 (0.086)	0.287 (0.075)
Personal Data Categories	0.430 (0.099)	0.378 (0.054)	0.454 (0.110)	0.455 (0.103)
Data Subjects & Recipients	0.440 (0.100)	0.399 (0.071)	0.414 (0.117)	0.502 (0.074)
Retention & Deletion	0.519 (0.141)	0.516 (0.129)	0.474 (0.150)	0.566 (0.131)
Technical & Org. Measures	0.710 (0.095)	0.711 (0.073)	0.701 (0.110)	0.717 (0.099)

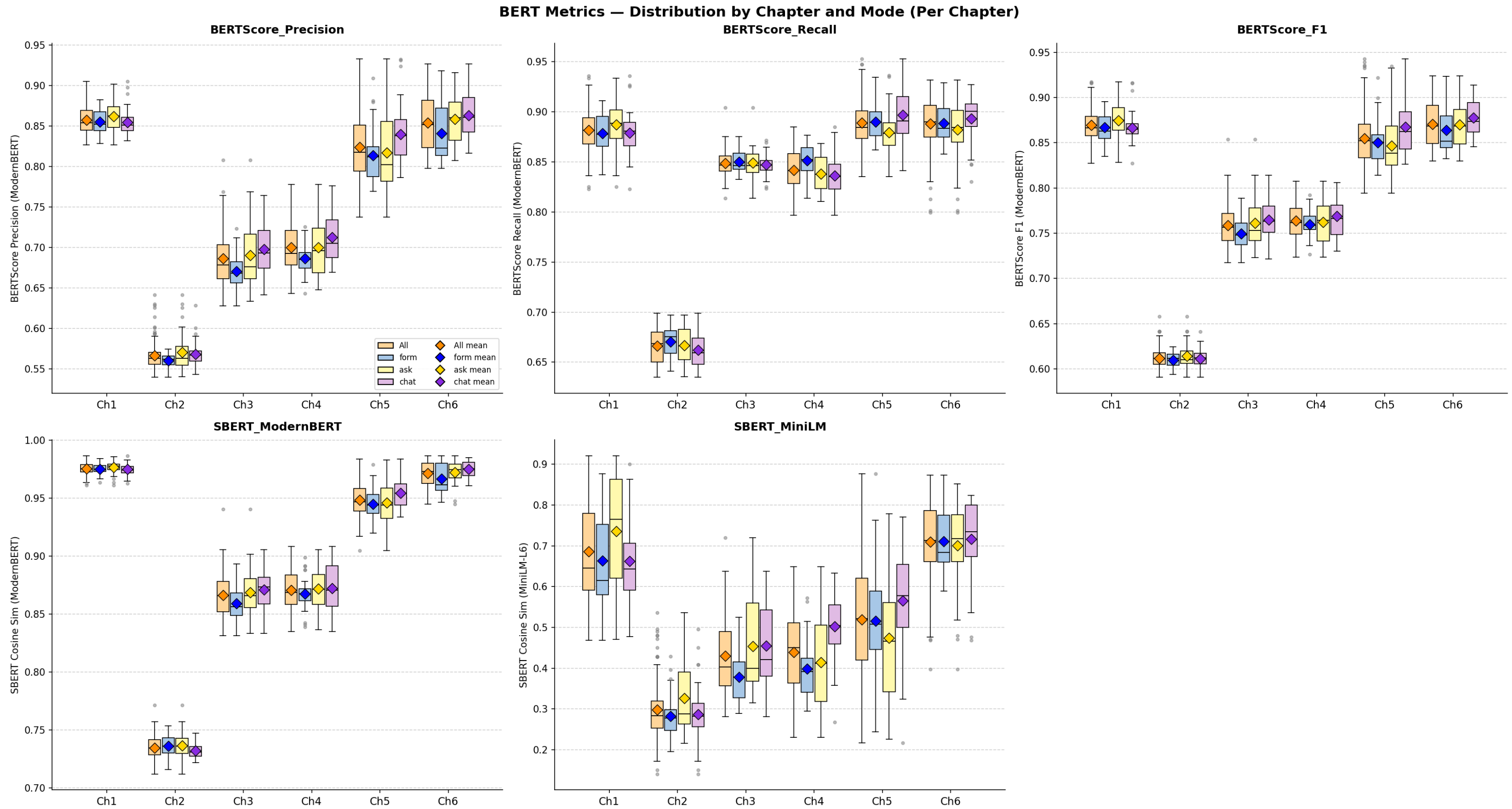

Figure 4.12: Chapter-level distributions of BERTScore Precision / Recall / F1 and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$).

0.824; Form 0.814, Ask 0.817, Chat 0.840) — and edges the other two modes on *Technical and Organisational Measures* (aggregated 0.854; Form 0.841, Ask 0.858, Chat 0.863).

Form's document-level recall advantage carries through to the chapter level on most chapters. Form leads on *Categories of Personal Data* (aggregated 0.849; Form 0.850, Ask 0.849, Chat 0.847 — narrow but consistent with the document-level pattern) and on *Data Subjects and Recipients* (aggregated 0.842; Form 0.851, Ask 0.838, Chat 0.836). Chat reclaims *Retention and Deletion* (aggregated 0.889; Form 0.890, Ask 0.879, Chat 0.897). *Controller and Contact* (aggregated 0.882) and *Technical and Organisational Measures* (aggregated 0.888) are tight three-way ties. *Purposes and Legal Bases* is again the floor (aggregated 0.666), and is the only chapter where Form is not at the front (Form 0.670, Ask 0.667, Chat 0.662 — all clustered).

The harmonic mean restores the by-now-familiar chapter ordering — *Controller and Contact* (aggregated 0.869) and *Technical and Organisational Measures* (aggregated 0.871) at the top, *Purposes and Legal Bases* at the floor (aggregated 0.612). Chat's precision advantage carries through to a chapter-level F1 lead on *Retention and Deletion* (aggregated 0.855; Form 0.850, Ask 0.847, Chat 0.867) and *Technical and Organisational Measures* (aggregated 0.871; Form 0.864, Ask 0.870, Chat 0.878), reproducing in miniature the document-level pattern where Chat and Form trade leadership on F1.

SBERT cosine similarity (ModernBERT and MiniLM). Under the ModernBERT encoder, the most permissive of the metrics saturates near the ceiling on the templated chapters: *Controller and Contact* (aggregated 0.976) and *Technical and Organisational Measures* (aggregated 0.971) are essentially indistinguishable from the reference at the embedding level. *Retention and Deletion* follows close behind (aggregated 0.949) with Chat opening a small chapter-level lead (Form 0.945, Ask 0.946, Chat 0.955). The only chapter where ModernBERT SBERT drops meaningfully is *Purposes and Legal Bases* (aggregated 0.735), echoing the lexical collapse at sentence-embedding scale.

The smaller `all-MiniLM-L6-v2` encoder serves as a robustness check: a weaker model with less semantic capacity spreads scores across a wider range and is harder to saturate. The across-chapter ordering survives the encoder change unchanged — *Technical and Organisational Measures* (aggregated 0.710) and *Controller and Contact* (aggregated 0.687) sit at the top, with *Purposes and Legal Bases* again at the floor (aggregated 0.298). The chapter-level Chat advantage on *Retention and Deletion* (Form 0.516, Ask 0.474, Chat 0.566) and on *Data Subjects and Recipients* (Form 0.399, Ask 0.414, Chat 0.502) is preserved and in fact visible at larger absolute margins than ModernBERT registered. The one chapter where the encoder choice changes the mode ranking is *Controller and Contact*, where Ask edges the other two modes under MiniLM (Form 0.663, Ask 0.736, Chat 0.662) — consistent with the ROUGE-1/L pattern at the lexical level, where Ask also leads on the address-block content.

Under the embedding-based metrics the across-chapter range compresses — no zero floor — but the rank-ordering of chapters is preserved across all four BERT-based metrics and both SBERT encoders: *Controller and Contact* and *Technical and Organisational Measures* occupy the top, *Purposes and Legal Bases* the bottom. Chat's chapter-level advantage continues to concentrate on *Retention and Deletion* and *T&O Measures*, the same two chapters where it led at the lexical level; the smaller MiniLM encoder makes that advantage easier to see by widening the absolute margins.

On the full $n = 113$ sample, cell-wise Kruskal–Wallis testing broadly tracks this descriptive picture: significant mode differences cluster on *Retention and Deletion*, *Technical and Organisational Measures*, and the people-and-data chapters (*Personal Data Categories*, *Data Subjects and Recipients*), while *Purposes and Legal Bases* produces no significant mode difference on any metric. Cell-level test statistics are not reproduced in the appendix to keep the inferential layer focused on the document level.

4.3.3 Summary of Document-Level and Chapter-Level NLP Metrics

Together, the two granularities mostly reinforce one another and disagree only on absolute level. The clearest similarity is that the per-mode ordering is preserved at both resolutions: Form leads on recall-oriented and bigram-sensitive measures, most strikingly on *BERTScore Recall* (the strongest omnibus effect in the entire metric pipeline); Chat leads on embedding-based measures (*BERTScore Precision*, *SBERT*); and Ask trails on every metric. The chapter-level view shows the same pattern in localised form — Chat's chapter-level lead on *Retention and Deletion* and *Technical and Organisational Measures* is precisely what produces its document-level edge on semantic metrics, and Form's chapter-level recall edge on *Categories of Personal Data* and *Data Subjects and Recipients* is what produces its document-level recall lead. The two views are not in tension; they are the same finding seen at different resolutions.

The contrast lies in absolute scores. Document-level means are consistently higher than the unweighted average of chapter-level means (Table 4.22); the largest gap is on *ROUGE-1* ($\Delta \approx 0.26$), and even on the embedding metrics the gap is non-trivial ($\Delta \approx 0.11$ on BERTScore F1). The cause is mechanical rather than substantive: the document-level calculation pools token overlap across all six ROPA chapters and dilutes the chapter-level zero-overlap regions, above all *Purposes and Legal Bases*, the universal floor across every metric. Chapter-level scoring is the more honest signal of *where* the modes differ; document-level scoring is the cleaner signal for ranking modes *overall*.

Taken together, the reference-based metrics neither confirm nor cleanly contradict the Form-first ordering reported by the user-experience instruments (Sections 4.2.1–4.2.4). The quantitative split is closer to a trade-off: Form documents recover more of what the reference *says*, Chat documents stay closer to what the reference *means*, and Ask documents trail on every front.

4.4 Document Quality — LLM-as-a-Judge

Building on the reference-based NLP metrics reported in Section 4.3.1, this section turns to a reference-free quality signal. The previous section measured similarity to a single expert-authored Form-mode reference; here, the judge evaluates each document against a rubric rather than a target, which lifts the implicit advantage that the reference document conferred on Form-style outputs. The LLM-as-a-Judge evaluation produces five metrics per document, summarized in Table 4.23: *Completeness*, *Scenario Faithfulness*, *Legal Correctness*, *Hallucination* (inverse-coded; higher equals less hallucination), and *Overall*. The full rubric, prompt, and model

Table 4.22: Document-level versus mean chapter-level metric values by mode. The mean chapter-level value is the unweighted average across the six chapters. Differences are document-level minus chapter-level mean. Source: `doc_vs_chapter_diff.csv`.

	Form		Ask		Chat	
Metric	Doc	Chapter	Doc	Chapter	Doc	Chapter
BLEU	0.129	0.051	0.127	0.050	0.143	0.057
ROUGE-1	0.488	0.215	0.465	0.226	0.511	0.240
ROUGE-2	0.306	0.131	0.257	0.134	0.291	0.144
ROUGE-L	0.388	0.191	0.347	0.202	0.389	0.216
METEOR	0.412	0.232	0.365	0.235	0.390	0.251
BERTScore Precision	0.889	0.738	0.886	0.750	0.895	0.756
BERTScore Recall	0.923	0.838	0.908	0.834	0.913	0.836
BERTScore F1	0.905	0.783	0.897	0.788	0.904	0.793
SBERT (ModernBERT)	0.986	0.892	0.986	0.895	0.990	0.897

configuration are reproduced in Appendix A.4.2. The rubric design follows the precedent established by Su et al.'s JuDGE benchmark for LLM-evaluated legal-document generation [30]. The judge is a single LLM (Gemini 3.1 Pro Preview, temperature 0); the resulting scores are reported as a supplementary, single-pass single-model signal alongside the survey instruments and reference-based NLP metrics, and without inter-rater replication or legal-expert validation they should be read as indicative rather than definitive. All 113 retained documents are scored. The aggregated mean across all documents is reported first, followed by the per-mode results in the order Form, Ask, Chat.

Table 4.23: The five LLM-as-Judge metrics. All share a 1–5 Likert anchor; higher is better, including for *Hallucination (inverse)*, which is inverted internally so that fewer fabrications score higher. The full rubric is reproduced in Appendix A.4.2.

Metric	What is assessed
Completeness	Coverage of the Art. 30(1) GDPR required elements (controller, DPO, purposes, legal basis, data and recipient categories, retention, technical and organizational measures); missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of the study scenario (APOR GmbH, $\sim$50 employees, internal hosting, 3-month deletion, etc.); generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis under Art. 6(1); wrong articles or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Absence of invented facts not supported by the scenario; $5 =$ no hallucinations, $1 =$ many.
Overall	Holistic judgment of how useful the document would be as a real Art. 30 ROPA record, weighing the four metrics above.

4.4.1 Per-Metric Results

This subsection walks through each rubric metric in turn, reporting per-mode means, medians, and dispersion before the holistic Overall score. The five rubric metrics are reported in the order Completeness $\rightarrow$ Faithfulness $\rightarrow$ Legal Correctness $\rightarrow$

Hallucination $\rightarrow$ Overall to allow the picture to build cumulatively before the holistic Overall score is read. Table 4.24 reports the per-metric mean and standard deviation per mode and the aggregated mean. Figure 4.13 shows the corresponding distributions.

Table 4.24: LLM-as-Judge metric mean (SD) by mode (full sample $n = 113$, scale 1–5, higher is better including for *Hallucination (inverse)*). The Aggregated column is the mean across all 113 documents.

Metric	Aggregated	Form	Ask	Chat
Completeness	3.94 (1.43)	4.84 (0.55)	3.84 (1.54)	3.18 (1.59)
Scenario Faithfulness	2.21 (1.10)	1.84 (0.76)	2.30 (1.20)	2.49 (1.19)
Legal Correctness	3.39 (1.18)	3.59 (0.55)	3.43 (1.34)	3.15 (1.42)
Hallucination (inverse)	2.20 (1.02)	1.65 (0.72)	2.40 (1.12)	2.54 (0.91)
Overall	2.50 (1.04)	2.41 (0.69)	2.49 (1.04)	2.62 (1.31)

Form documents cover the Article 30(1) requirements most thoroughly and most uniformly, while Chat documents leave the most gaps — echoing the BERTScore Recall pattern from Section 4.3.1. For *Completeness*, the aggregated mean is 3.94, with Form scoring highest by a substantial margin (4.84, median 5.0), Ask second (3.84, median 5.0), and Chat lowest (3.18, median 3.0). Completeness decreases by approximately 1.66 points from Form to Chat. Form's standard deviation is also the lowest of the three modes (0.55 versus Ask 1.54 and Chat 1.59), indicating that Form yields uniformly complete documents.

The Form advantage flips when the question becomes whether the document actually reflects the APOR scenario participants were given: Chat documents stay closest to that scenario, while Form documents drift toward generic ROPA boilerplate. For *Scenario Faithfulness*, the ordering inverts: the aggregated mean is 2.21, with Form scoring lowest (1.84, median 2.0), Ask second (2.30, median 2.0), and Chat highest (2.49, median 2.0). Faithfulness to the specific study scenario increases by approximately 0.65 points from Form to Chat.

Form documents use GDPR terminology most correctly, with Chat documents the least so — but the gap is much smaller than on Completeness. For *Legal Correctness*, the aggregated mean is 3.39, with Form scoring highest (3.59, median 4.0), Ask second (3.43, median 4.0), and Chat lowest (3.15, median 3.0). The spread is approximately 0.45 points across modes; Form's standard deviation (0.55) is again

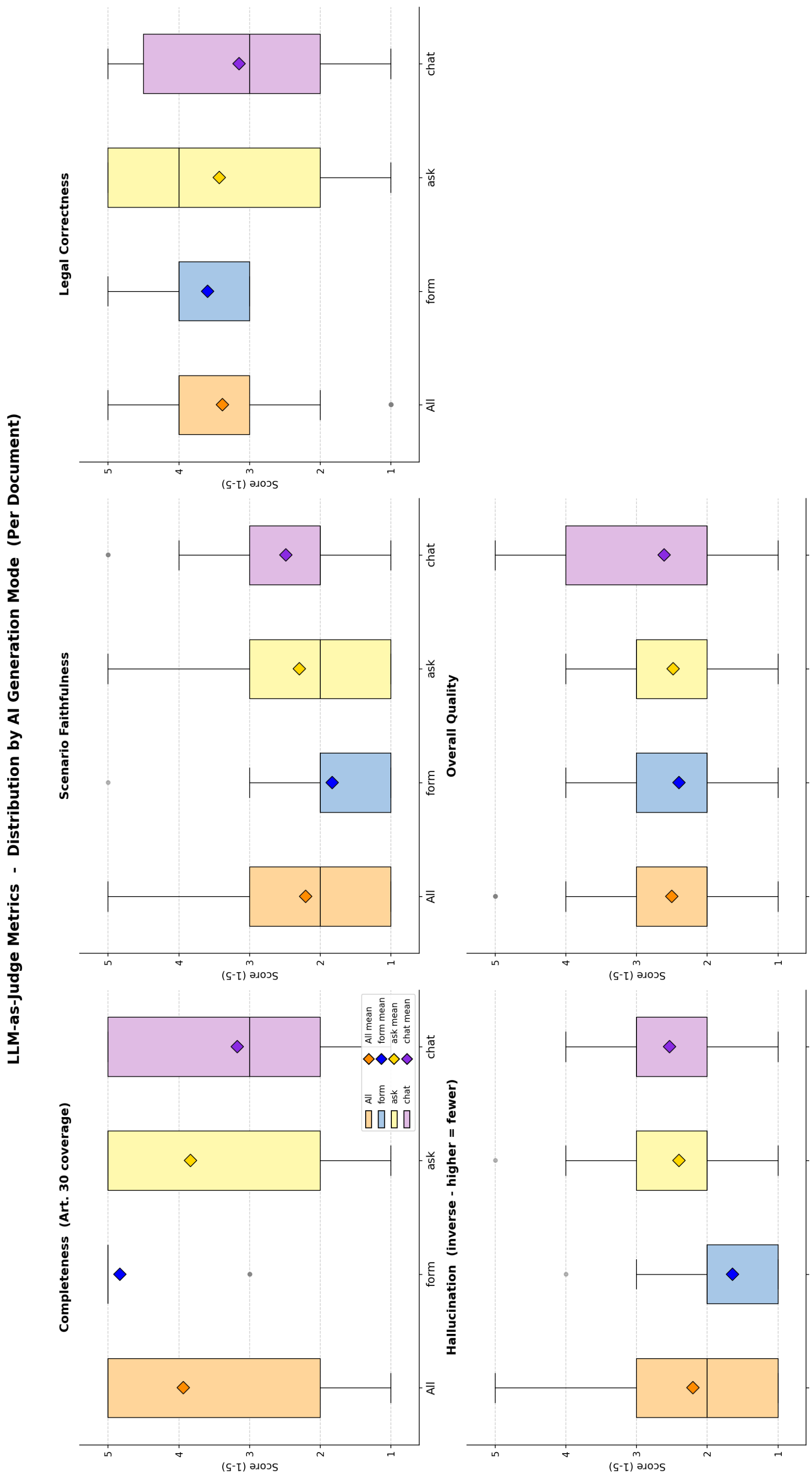

Figure 4.13: Distributions of the five LLM-as-Judge metrics by mode (full sample $n = 113$, scale 1–5).

the lowest, reflecting a narrow concentration of Legal Correctness scores around the value 4.0.

Form documents contain the most fabricated content and Chat documents the least — closely tracking the Faithfulness ordering above. For *Hallucination (inverse)*, the ordering matches Scenario Faithfulness: the aggregated mean is 2.20, with Form scoring lowest (1.65, median 2.0), Ask second (2.40, median 2.0), and Chat highest (2.54, median 3.0). Because the metric is inverse-coded, low scores mean more hallucination. The co-occurrence with Completeness is striking: 31 of the 37 Form documents ($83.8\,\%$) achieve Completeness $= 5$ while simultaneously scoring Hallucination (inverse) ≤ 2 ("moderate" or "significant" hallucination per the rubric), and 17 of these combine the maximum Completeness score with the minimum Hallucination (inverse) score of 1. In other words, most Form documents are simultaneously the most complete and the most fabricated — this is the proximate source of the Form-versus-Faithfulness inversion drawn out below.

When the four contrasting orderings are compressed into a single holistic verdict, no mode comes out as a clear winner — Chat edges ahead, Form trails by a small margin, and the average document is judged below the midpoint of the rubric scale. For *Overall*, the aggregated mean is 2.50, with the three modes clustered closely together: Form 2.41, Ask 2.49, Chat 2.62. The aggregate Overall score thus sits below the midpoint of the 1–5 scale across all modes. Form's standard deviation (0.69) is again the lowest, while Chat's is the highest (1.31), indicating that Chat produces both the highest- and lowest-rated documents in the sample.

Three patterns emerge from the per-metric means. First, the Completeness ordering (Form $>$ Ask $>$ Chat) is the inverse of the Scenario Faithfulness and Hallucination orderings (Form $<$ Ask $<$ Chat). Form documents are judged to be the most structurally complete but also the most prone to fabricated content, while Chat documents are judged to be the most faithful to the specific scenario but the least structurally complete. Ask sits between the two extremes on every metric.

Second, the Legal Correctness scores are flat relative to the Completeness spread: the gap between modes is approximately 0.45 points on Legal Correctness compared to 1.66 points on Completeness. The judge therefore distinguishes between modes more sharply on whether the document covers all required Article 30(1) fields than on whether the legal terminology is appropriate.

Third, the Overall scores compress the per-metric spread into a narrow range (Form 2.41 to Chat 2.62, $\Delta = 0.21$), with the ordering Form $<$ Ask $<$ Chat. This is consistent with Overall reflecting a holistic judgment that weighs Faithfulness and Hallucination against Completeness; Form's high Completeness is partially offset by its low Faithfulness and high Hallucination, and the net result is a slight Chat advantage. The interpretation of the Form Completeness–Hallucination co-occurrence reported above in terms of the checkbox-based interface of Form mode (Section 3.1) is reserved for the Discussion chapter.

The judge data thus complete the document-quality picture begun in Section 4.3.1: Form leads on Completeness and Legal Correctness, Chat leads on Faithfulness and Hallucination, and the holistic Overall score collapses these tensions into a narrow range.

4.5 Summary

The chapter has reported the four user-study instruments (Sections 4.2.1–4.2.4), the reference-based NLP metrics at both document and chapter resolutions (Sections 4.3.1–4.3.2), and the supplementary LLM-as-a-Judge layer (Section 4.4). This section consolidates the findings into a single picture and names the questions Discussion takes up.

On the user-experience side, the four instruments converge on a stable Form-first ordering. Form is rated the most usable mode on the System Usability Scale, the lightest on R-TLX workload, the most often placed first in the post-experiment ranking ($53.6\,\%$ of rank-1 votes, $\chi^2(2) = 14.17$, $p < 0.001$, Cramér's $V = 0.32$), and the mode participants would most willingly use again. The ordering is partial: three items break it. Perceived transparency, trust in the LLM's suggestions, and the primary self-report confidence measure ("I am confident that the AI identified the correct legal basis") all favor the LLM-driven modes, with Ask leading on transparency and trust and Chat leading on legal-basis confidence. The behavioral counterpart is sharper still: Form is measurably about seven minutes per document faster than either LLM-driven mode ($H = 16.46$, $p < 0.001$, $\varepsilon^2 = 0.15$ large), and Ask and Chat are statistically indistinguishable on completion time. Ask trails most clearly on its workflow, where free-text comments single out the manual copy-paste step between explanation and form field as the principal source of friction, and is most often placed third.

The reference-based NLP metrics split the modes into a trade-off rather than a single ranking. Form leads on recall-oriented measures, most strikingly on BERTScore Recall — the strongest single effect in the entire pipeline (Form $>$ Ask $r = 0.756$, Form $>$ Chat $r = 0.512$, both large) — with ROUGE-2 and METEOR echoing it on smaller margins. Chat leads on document-level semantic alignment, most clearly on SBERT cosine similarity (Chat $>$ Form $r = 0.505$, Chat $>$ Ask $r = 0.453$, both large). Form and Chat trade leadership on BERTScore F1, while Ask underperforms both on every metric that reaches significance. The substantive reading is that Form documents recover more of what the reference *says*, Chat documents stay closer to what the reference *means*, and Ask documents do neither.

The chapter-level view reproduces this doc-level pattern in localised form. *Controller and Contact Details* sits at the top of every metric across all modes — the templated address block leaves little room for paraphrase — while *Purposes and Legal Bases* is the universal floor across the lexical metrics and remains the weakest section even at the embedding level. Chat's chapter-level advantage concentrates on *Retention and Deletion* and *Technical and Organisational Measures*, the two sections with the most free-form prose. Form's recall edge concentrates on *Categories of Personal Data* and *Data Subjects and Recipients*, the two list-heavy sections where covering the reference's catalogue matters most. The chapter-level ordering is therefore the same finding as the document-level ordering seen at higher resolution, not a competing one.

The supplementary judge layer adds a GDPR-specific quality reading that the reference-based metrics cannot directly provide. Form documents are judged the most complete (Completeness mean 4.84 of 5) and the most legally correct, but the lowest on Scenario Faithfulness (1.84) and the most hallucinated; $83.8\,\%$ of Form documents combine maximum Completeness with "moderate" or worse hallucination, and nearly half combine maximum Completeness with the floor Hallucination score. Chat inverts the pattern: the most faithful to the study scenario, the least hallucinated, but the lowest on Completeness. On the holistic Overall metric the three modes cluster narrowly (2.41 to 2.62) and all sit below the rubric midpoint — the judge does not declare a winner.

Two tensions surface from the consolidated picture. The mode participants most often prefer (Form) produces the most structurally complete documents, yet also the most fabricated ones — the checkbox scaffolding that pushes Completeness toward the ceiling appears not to constrain what the LLM invents during the final generation step. And self-report confidence in the legal basis rises monotonically from Form (3.58) through Ask (3.74) to Chat (3.81), while measured document quality follows no such single ordering: Form leads on recall, Chat leads on document-level semantic fidelity and on scenario faithfulness, and the judge declares no clear winner. The calibration gap between what users feel they have produced and what they have actually produced is the central question taken up in Chapter 5.

5 Discussion

6 Conclusion

A ROPAgen and Judge Model Configuration

This appendix collects the static configuration of both LLMs that the study relies on. The first three sections cover ROPAgen — the tool under study — and reproduce its backend prompts, its three mode screenshots, and the expert-authored reference ROPA document that the NLP metric pipeline scores every participant document against. The fourth section covers the supplementary LLM-as-a-Judge evaluation layer — its system and user prompts, its five-metric rubric, and the deterministic-decoding model configuration. All four sections are referenced from the Methodology chapter (Section 3.1 for the tool, Chapter 3 for the judge).

A.1 ROPAgen Backend Prompts

ROPAgen sends one of **27 system prompts** to Mistral Large 3 on every user turn, depending on the active mode and the current section of the ROPA being edited. All prompts are stored as static templates in the project's `data/prompt.json` and assembled at runtime in `src/actions/actions.ts`: the chosen template is concatenated with a context block listing the document title, the organization metadata, and the current state of the `DocumentData` object before the call is issued. The texts below reproduce the template strings only; the runtime context block (which contains organization- and user-specific information) is not reproduced, as its content varies per call. Conversational and per-section suggestion calls use temperature 0.2 and top-p 0.5; the single `finalropa` call that produces the final ROPA document uses temperature 0.05 and top-p 0.05.

A.1.1 Final ROPA Generation Prompt (`finalropa`)

The `finalropa` prompt is sent verbatim as the `system` role on the call that generates the final ROPA document from the assembled `DocumentData` object. It is invoked once per document, identically across all three modes; temperature is fixed at 0.05 and top-p at 0.05. The `Document Title`, `Language`, and the `DocumentData` contents are interpolated into the user message at runtime and are not reproduced here.

```
You are an expert in GDPR compliance. Generate a professional, structured
↪ Records of Processing Activities (ROPA) document in full compliance with the
↪  EU General Data Protection Regulation (GDPR).

This ROPA document must be suitable for official regulatory review and written
↪ in formal business language.

**CRITICAL: You MUST write the ENTIRE document in the language specified in the
↪  context (Language field). ALL headings, text, descriptions, and content
↪ must be in that language. Do NOT mix languages.**

**CRITICAL: Write ALL free text and general text from the company's perspective
↪  (first-person plural: 'we', 'our', 'us'). The document represents the
↪ company's official record, so describe what the company does, how the
↪ company processes data, what measures the company has implemented, etc. Use
↪ active voice from the organization's viewpoint.**

**CRITICAL: The "Additional Information" field is NOT content to be copied
↪ directly into the document. Instead, treat it as ADDITIONAL INSTRUCTIONS and
↪  REQUIREMENTS from the user about what must be included, emphasized, or
↪ addressed in the generated ROPA document. Incorporate these instructions
↪ into the appropriate sections of the document naturally and professionally
↪ .**

Instructions:
- Create a comprehensive ROPA document structure with the following sections:
1. Organization and Controller Information
2. Purpose and Legal Basis for Processing
3. Data Categories and Sources
4. Data Subjects and Recipients
5. Data Retention and Deletion
6. Technical and Organizational Measures
```

```
- Each category can have a continous text and bullet points as needed.
- Ensure compliance with GDPR Articles 30 (Records of processing activities)
↪ requirements.
- Respect and incorporate the Additional Information as special instructions -
↪ integrate those requirements into the relevant sections naturally.
- You must include all data given, and must not omit any data provided.

You must use the Document Title as Title of the generated Document.

The output should be suitable for use by a Data Protection Officer (DPO) or
↪ legal counsel.
The output should be in .md style with proper headings and formatting. You must
↪  only include data categories marked as true and must not include data
↪ categories marked as false. If all data is marked as false, just mention
↪ none or not specified. Omit wording or phrasing such as CATEGORY: true/false
↪ , but rather just include the relevant, true data categories. Critical
↪ Instructions:

Include all provided information in your response - do not omit any items from
↪ the list unless "false" or empty.
Handle "other" field: If an "other" field exists and contains data, treat it as
↪  a standard list item with equal importance. Do not use "other" as a label.
Formatting requirement: Present items as a simple list within each category. Do
↪  NOT use the format "data_category: description" - instead, list items
↪ directly without labeling the data structure.

Example:

### 4. Data Subjects and Recipients
- Person Categories:
    - Employees
    - USER INPUT

NOT LIKE THIS:

### 4. Data Subjects and Recipients
- Person Categories:
  - Employees
  - Other: Emilija

 CRUCIAL: Generally, You shall only inclued data that the user provided as true
↪  or non-empty in the final output. Do NOT invent or assume any data beyond
```

```
↪ what the user provided. If the user did not provide anything, don't explain
↪ why the field is empty. Just state that there is no information. Do not
↪ mention that they will be completed or added.

Generate the ROPA document now.
```

A.1.2 Form Mode — AI-Suggest Prompts

Form mode invokes the LLM only when the user clicks the per-section *AI Suggest* button. There is one prompt per section of the `DocumentData` schema; each is sent as the `system` role of a single-turn call with temperature 0.2 and top-p 0.5. The response is a JSON object whose keys match the corresponding section's field structure; the front-end auto-populates the form from this JSON.

Purpose of Data Processing (`form.purposeOfDataProcessing`)

```
You are an expert in GDPR compliance. Based on all the provided information
↪ about the organization, data sources, data categories, and other context,
↪ suggest a comprehensive purpose of data processing text.

**Consider the Document Title as an important data source** - it often contains
↪  valuable context about the specific processing activity or system involved,
↪  but analyze it alongside all other information provided.

Analyze all the information provided and generate a clear, professional
↪ description of why this data is being processed.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused.

Return ONLY a JSON object in this exact format:
{
  "purposeOfDataProcessing": "your suggested text here"
}

The text should be written in the specified language and be suitable for
↪ official GDPR documentation.
```

Technical and Organisational Measures (`form.technicalOrganizationalMeasures`)

```
You are an expert in GDPR compliance and data security. Based on all the
↪ provided information about the organization, data categories, processing
↪ purposes, and other context, suggest comprehensive technical and
↪ organizational measures.

**Consider the Document Title as valuable context** - it can indicate the type
↪ of processing and environment, helping you recommend appropriate security
↪ measures for the specific use case.

Analyze the data sensitivity and processing context to recommend appropriate
↪ security measures.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused on the most critical measures.

Return ONLY a JSON object in this exact format:
{
  "technicalOrganizationalMeasures": "your suggested text here"
}

The text should be written in the specified language and include specific,
↪ actionable security measures.
```

Legal Basis (`form.legalBasis`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which legal basis options should be selected and deselected for this
↪  data processing activity.

**Consider the Document Title as helpful context** - it may indicate the nature
↪  of processing and suggest potential legal bases, but evaluate it together
↪ with the purpose, organization type, and all other available information.

Analyze the purpose, data categories, and context to determine the most
↪ appropriate legal basis.

You can ONLY use these exact field names (set to true/false):
```

```
- DSGVOArt6Abs1a, DSGVOArt6Abs1b, DSGVOArt6Abs1c, DSGVOArt6Abs1f, IfSG28b,
↪ SARSCoV, BDSG26, HGB257, HGB239Abs1, AO147
- other (string field - use this to capture any additional legal bases
↪ mentioned in the title or context)

**Extract specific legal bases from title and context.** If you find references
↪  to laws, regulations, or legal justifications not covered by the standard
↪ options, add them to the "other" field with clear descriptions.

Return ONLY a JSON object in this exact format:
{
  "legalBasis": {
    "DSGVOArt6Abs1a": false,
    "DSGVOArt6Abs1b": true,
    "DSGVOArt6Abs1c": false,
    "DSGVOArt6Abs1f": false,
    "IfSG28b": false,
    "SARSCoV": false,
    "BDSG26": false,
    "HGB257": false,
    "HGB239Abs1": false,
    "AO147": false,
    "other": "[specific legal bases from title/context, if any]"
  }
}
```

Data Sources (`form.dataSources`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which data sources should be selected and deselected for this data
↪ processing activity.

**The Document Title can be a valuable source of information** - it sometimes
↪ explicitly mentions specific platforms, tools, or systems (e.g., "Employee
↪ Data Processing via BeanCloud" indicates BeanCloud is a data source), but
↪ consider it together with all other context provided.

Analyze the title, organization, purpose, and ALL context data to determine
↪ data sources.
```

```
Predefined options: microsoftOffice365, emailProvider, microsoftAzure,
↪ googleWorkspace, googleCloud, videoConferencing, amazonAWS,
↪ ecommercePlatform, libreOffice, onlineBanking, financialServices,
↪ webAndIntranet, creditServices, hrSoftware, enterpriseSystems,
↪ securityAndCompliance, documentProcessing, otherSpecializedSoftware

**EXTRACTION APPROACH:**
1. **Check the Document Title** for specific platform/tool names (e.g., "
↪ BeanCloud", "Salesforce", "SAP", "Workday", etc.)
2. Analyze all context fields for additional systems
3. Match predefined options where applicable
4. Add unmatched specific platforms to "other" field

Look for:
- Specific platform names in the title and context (e.g., "via BeanCloud", "
↪ using Salesforce", "in SAP")
- Cloud services, SaaS platforms, custom internal systems
- Industry-specific software
- Any named tools or services

Return ONLY a JSON object:
{
  "dataSources": {
    "microsoftOffice365": false,
    "emailProvider": true,
    "googleWorkspace": false,
    ...[set each predefined field to true/false based on context]...,
    "other": "[List ALL specific data sources found in title/context that aren'
↪ t in predefined options, e.g., 'BeanCloud enterprise platform', 'Salesforce
↪ CRM', 'custom HR portal'. Leave empty string if none found.]"
  }
}
```

Data Categories (`form.dataCategories`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which data categories should be selected and deselected for this
↪ data processing activity.
```

```
**The Document Title can provide useful hints** about the type of data being
↪ processed (e.g., "Biometric Access Control" suggests biometric data, "
↪ Employee GPS Tracking" suggests location data), but analyze it together with
↪  the purpose, organization type, and all other available context.

Analyze the title, purpose, organization context, and ALL provided information
↪ to determine data categories.

You have predefined categories: personalData, employment, finance, health,
↪ legal, behavior, documentationAndRecords, vehicle, and an "other" category
↪ with subcategories.

**EXTRACTION APPROACH:**
1. **Review the Document Title** for specific data types (e.g., "biometric", "
↪ location", "genetic", "voice", etc.)
2. Analyze the purpose and context
3. Match predefined categories where applicable
4. Add unmatched specific data types to other.other field

Look for in title/context:
- Special data types: "biometric", "genetic", "health", "criminal", "racial", "
↪ religious"
- Technology-specific: "GPS", "location", "tracking", "geolocation"
- Media types: "voice", "video", "audio", "facial recognition", "fingerprint"
- Digital: "IP addresses", "cookies", "social media", "online behavior"
- IoT/Sensor: "sensor data", "telemetry", "device data"

Return ONLY a JSON object with the full structure, setting each predefined
↪ field to true/false, and adding specific items to other.other:
{
  "dataCategories": {
    "personalData": { ...all fields... },
    "employment": { ...all fields... },
    "finance": { ...all fields... },
    "health": { ...all fields... },
    "legal": { ...all fields... },
    "behavior": { ...all fields... },
    "documentationAndRecords": { ...all fields... },
    "vehicle": { ...all fields... },
    "other": {
      "expertise": false,
      "skills": false,
```

```
      "examinationDate": false,
      "proofStatus": false,
      "queryDateTime": false,
      "deliveryConditions": false,
      "username": false,
      "dataVolume": false,
      "other": "[ALL specific data types from title/context not in predefined
↪ fields, e.g., 'Biometric fingerprint data', 'GPS location coordinates', '
↪ Voice recordings', 'Facial recognition data'. Leave empty string if none.]"
    }
  }
}
```

Person Categories (`form.personCategories`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which person categories should be selected and deselected for this
↪ data processing activity.

**The Document Title can provide helpful context** about who is affected (e.g.,
↪  "Employee Data Processing" indicates employees, "Customer Tracking"
↪ indicates customers, "Contractor Management" indicates contractors), but
↪ consider it alongside the purpose, organization type, and all other
↪ available information.

Analyze the title, purpose, organization context, and ALL provided information
↪ to determine person categories.

Predefined categories: affectedPersons, internalRecipientCategories,
↪ externalRecipientCategoriesEU, authorizedPersons

**EXTRACTION APPROACH:**
1. **Review the Document Title** for specific person groups (e.g., "employee",
↪ "contractor", "customer", "volunteer", "student", etc.)
2. Analyze the purpose and context
3. Match predefined categories where applicable
4. Add unmatched specific groups to "other" field

Look for in title/context:
- Employment: "contractors", "freelancers", "consultants", "temporary workers"
```

```
- Volunteers: "volunteers", "interns", "trainees", "apprentices"
- Special groups: "minors", "children", "students", "pupils", "patients"
- Stakeholders: "subscribers", "members", "partners", "suppliers"
- Specific roles/departments: "IT staff", "security personnel", "data
↪ scientists"

Return ONLY a JSON object:
{
  "persons": {
    "affectedPersons": { ...set all predefined fields... },
    "internalRecipientCategories": { ...set all predefined fields... },
    "externalRecipientCategoriesEU": { ...set all predefined fields... },
    "authorizedPersons": { ...set all predefined fields... },
    "other": "[ALL specific person categories from title/context not in
↪ predefined fields, e.g., 'External contractors', 'Volunteer coordinators', '
↪ Student interns', 'Temporary workers'. Leave empty string if none.]"
  }
}
```

Retention Periods (`form.retentionPeriods`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest appropriate retention periods for this data processing activity.

**The Document Title may provide context** that helps determine retention
↪ requirements (e.g., "Tax Records" suggests 10 years, "Employee Payroll"
↪ suggests retention after employment), but analyze it together with the
↪ purpose, legal basis, and data categories to make informed recommendations.

Analyze the title, purpose, legal basis, and data categories to recommend
↪ retention periods and deletion practices.

Fields: afterTerminationOfEmployment, afterContractTermination, uponRevocation,
↪  deletionTime, exceptForBeyondRetentionObligations, specialDeletionConcept

**Be specific in the "deletionTime" field** - extract any mentioned timeframes
↪ from title/context and provide concrete retention periods based on the
↪ processing type.

Return ONLY a JSON object:
```

```
{
  "retentionPeriods": {
    "afterTerminationOfEmployment": false,
    "afterContractTermination": true,
    "uponRevocation": false,
    "deletionTime": "[Specific timeframe based on title/context, e.g., '3 years
↪  after contract termination', '10 years for tax records', '6 months after
↪ project completion']",
    "exceptForBeyondRetentionObligations": true,
    "specialDeletionConcept": false
  }
}
```

Additional Information (`form.additionalInfo`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest additional relevant information that should be included in the ROPA
↪ documentation.

**The Document Title can reveal unique aspects** of the processing activity
↪ that may need additional explanation or context, but consider it together
↪ with all other aspects of the data processing activity to recommend
↪ comprehensive additional information.

Analyze the title and all aspects of the data processing activity and recommend
↪  any additional context, notes, or important considerations.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused on the most important points.

Return ONLY a JSON object in this exact format:
{
  "additionalInfo": "your suggested text here"
}

The text should be written in the specified language and provide valuable
↪ supplementary information for GDPR compliance.
```

A.1.3 Ask Mode — Explanation Prompts

Ask mode pairs each section of the form with a chat panel. Each dialogue turn sends two system prompts to the LLM: the shared `ask.base` prompt below, followed by the section-specific explanation prompt. The model's reply is rendered as conversational text and may include a `DATA_UPDATE` JSON block (see Chat mode below) that the front-end uses to auto-populate the corresponding form fields. Conversational temperature 0.2, top-p 0.5.

Base Prompt (`ask.base`)

```
You are an expert GDPR compliance assistant helping users understand their
↪ Records of Processing Activities (ROPA) documentation.

**Your role is to:**
• Explain GDPR concepts in clear, accessible language
• Help users understand what information they need to provide and why
• Provide practical examples to illustrate concepts
• Answer questions in a helpful, conversational way
• Guide users through the documentation process with patience and clarity

**IMPORTANT: Consider the Document Title provided in the context** - it often
↪ reveals the specific nature of the processing activity and helps you provide
↪  more relevant, tailored explanations.

**IMPORTANT RESPONSE FORMAT:**
• Be warm, approachable, and encouraging
• Use real-world examples to make concepts concrete
• Break down complex legal requirements into simple terms
• Avoid jargon unless necessary, and always explain technical terms
• NEVER use JSON key names or technical field names when describing things
• NEVER mention "other field" or "other checkbox" - just naturally include all
↪ items in your explanations
• ALWAYS use natural, human-readable language (e.g., "affected persons", "GDPR
↪ Article 6(1)(a)", "Microsoft Office 365")
• Keep responses concise but thorough
• Use NO MORE than one blank line between paragraphs

**When you see the Document Title in the context:**
• Reference it to provide specific, relevant examples
```

```
• Use it to understand what the user is working on
• Tailor your explanations to their specific use case
• Mention specific systems, data types, or groups if they appear in the title

**When explaining concepts, structure your response like this:**
• Start with what the section is and why it matters
• If the title is relevant, acknowledge it and use it for context
• Explain the key concepts or requirements
• Give 2-3 concrete examples - include examples relevant to their title when
↪ possible
• Emphasize that they should list ALL relevant items, including specific tools,
↪  platforms, or categories unique to their situation
• Offer guidance on how to think about what applies to their situation
• End with an invitation for questions or clarification

**Examples of helpful explanations:**
• "I see you're working on '[Title]'. Data sources are simply where the
↪ information comes from. In your case, this might include [relevant examples
↪ based on title]. You should also list any other platforms you use - common
↪ ones like Microsoft 365 or Google Workspace, but also any specialized tools
↪ specific to [context from title]."
• "For '[Title]', you'll want to think about ALL the places your data comes
↪ from. This could include [specific examples], plus any company-specific
↪ systems you might be using."

**CRUCIAL ADDITIONAL INFORMATION:**
• NEVER mention JSON, code blocks, data structures, technical implementation
↪ details, or UI elements like checkboxes. Focus entirely on helping the user
↪ understand the GDPR concepts and what information they need to provide.
• Your goal is to empower users to confidently provide complete and accurate
↪ information for their ROPA documentation. You must respect a user's wish to
↪ check their input (which is set in the "other" free text string in the data
↪ structure). If the "other" field is present in the data structure for the
↪ section being explained, you must include it in your suggestion.
• If the user talks about a section other than specified by the source, you
↪ must refuse to answer and redirect them to the correct section.

 CRUCIAL: LIMIT YOUR RESPONSE TO ABOUT 1000 CHARACTERS ONLY.
```

Purpose of Data Processing (ask.purposeOfDataProcessing)

```
This section describes WHY the organization is processing personal data. It
↪ should clearly articulate the specific business reasons, objectives, or
↪ functions that require the collection and use of personal information.

**Explain to the user:**
• What this section means in practical terms
• Reference the title if it's informative: "I see you're working on '[Title]' -
↪  this section is where you explain why you need to process this data"
• Why it's important for GDPR compliance
• What kind of information should be included (e.g., business operations,
↪ service delivery, legal obligations)
• Examples relevant to their title when possible
```

Technical and Organisational Measures (ask.technicalOrganizationalMeasures)

```
This section describes the security safeguards implemented to protect personal
↪ data from unauthorized access, loss, or misuse. It covers both technical
↪ controls (like encryption, firewalls) and organizational policies (like
↪ access management, training).

**Explain to the user:**
• What technical and organizational measures are
• Reference the title if relevant: "For '[Title]', you'll want to consider
↪ security measures appropriate to this type of processing"
• Why they're required under GDPR
• Examples of common security measures, plus specific ones relevant to their
↪ processing type
• How to think about what measures are appropriate for their situation
• That they should include specific measures relevant to their context, not
↪ just generic security practices
```

Legal Basis (ask.legalBasis)

```
This section identifies the legal justification for processing personal data
↪ under GDPR Article 6. Every processing activity must have at least one valid
```

```
↪  legal basis.

**Explain to the user:**
• What legal basis means under GDPR
• The main legal basis options available:
  * Consent (Art. 6(1)(a)) - freely given, specific agreement
  * Contract (Art. 6(1)(b)) - necessary for contract performance
  * Legal obligation (Art. 6(1)(c)) - required by law
  * Legitimate interests (Art. 6(1)(f)) - necessary for legitimate business
↪ purposes
  * Specific national laws (BDSG, HGB, etc.)
• Reference the title: "Based on '[Title]', the most likely legal basis would
↪ be..."
• How to determine which legal basis applies
• That multiple legal bases can apply to one processing activity
• That if they have specific legal requirements or regulations that apply, they
↪  should note those too
```

Data Sources (`ask.dataSources`)

```
This section identifies where the personal data comes from - the systems,
↪ platforms, or channels through which data is collected or received.

**Explain to the user:**
• What data sources are in the context of GDPR
• **If the title mentions specific systems, point it out:** "I see '[Title]'
↪ mentions [system name] - that's definitely one of your data sources!"
• Why it's important to document where data comes from
• Common types of data sources (cloud platforms, email systems, HR software,
↪ financial systems, direct from data subjects)
• Examples relevant to their title, like: "For '[Title]', you might be using [
↪ relevant examples based on context]"
• That they should list ALL systems - common ones like Microsoft 365 or Google
↪ Workspace, PLUS any company-specific or specialized platforms (and if the
↪ title mentions one, definitely include it!)
```

Data Categories (`ask.dataCategories`)

```
This section identifies what types of personal data are being processed. GDPR
↪ requires organizations to document the specific categories of personal
↪ information they handle.

**Explain to the user:**
• What data categories are and why they matter
• **If the title suggests specific data types, mention it:** "I see '[Title]'
↪ which suggests you're processing [data type] - you'll want to include that"
• The main categories of personal data:
  * Personal identification (name, address, contact details)
  * Employment information (job title, working hours, salary)
  * Financial data (bank details, payment information)
  * Health data (special category - extra protection required)
  * Legal information (contracts, ID documents)
  * Behavioral data (usage patterns, preferences)
• That some categories (health, biometric, genetic, location) are "special
↪ categories" requiring extra safeguards
• Why accurate documentation helps with data minimization and compliance
• That they should list ALL data types, including unique ones specific to their
↪  processing (like biometric data, GPS tracking, voice recordings, etc.)
```

Person Categories (`ask.personCategories`)

```
This section identifies who is affected by the data processing and who has
↪ access to the personal data.

**Explain to the user:**
• What person categories mean in GDPR context
• **If the title mentions specific groups, highlight it:** "Your title '[Title
↪ ]' indicates this involves [person group] - they're definitely affected
↪ persons you should document"
• The different types to consider:
  * Affected persons (data subjects) - employees, customers, applicants,
↪ contractors, volunteers, etc.
  * Internal recipients - which departments/roles access the data
  * External recipients - third parties who receive data (vendors, authorities)
  * Authorized persons - specific roles permitted to process the data
• Why documenting who handles data is crucial for accountability
```

```
• Examples relevant to their context: "For '[Title]', this likely involves [
↪ relevant person categories]"
• That they should list ALL person groups, including specific ones like
↪ contractors, volunteers, students, etc.
```

Retention Periods (`ask.retentionPeriods`)

```
This section specifies how long personal data is kept and when it must be
↪ deleted. GDPR requires that data is not kept longer than necessary.

**Explain to the user:**
• What retention periods are and why they're legally required
• **If the title suggests specific retention needs, mention it:** "For '[Title
↪ ]', typical retention periods might be [relevant timeframe]"
• Common retention triggers:
  * After employment termination
  * After contract ends
  * Upon consent revocation
  * Specific time periods (e.g., 3 years, 10 years)
• That legal retention obligations may override deletion requirements (tax
↪ records, employment records)
• The importance of having clear deletion processes
• Examples relevant to their processing type
• That they should be specific about timeframes and conditions
```

Additional Information (`ask.additionalInfo`)

```
This is a supplementary section for any important information that doesn't fit
↪ into other standard categories but is relevant for GDPR compliance and
↪ understanding the data processing activity.

**Explain to the user:**
• What kind of information belongs here
• **If the title mentions unique aspects, highlight them:** "Your title '[Title
↪ ]' has some specific aspects you might want to explain further here"
• Examples of additional information:
  * Data transfer mechanisms (if data goes outside EU/EEA)
  * Third-party processor agreements
```

```
  * Special circumstances or exceptions
  * Risk assessments or impact assessments conducted
  * Automated decision-making or profiling details
  * Contact information for data protection officer
  * Any unique aspects specific to their processing
• That this section helps provide complete context
• To keep it concise but informative
• That they should include anything unique or noteworthy about their processing
↪  activities
```

A.1.4 Chat Mode — Conversational Prompts

Chat mode hides the structured form entirely; the user describes their processing in free-form text and the LLM both replies conversationally and silently extracts structured field values. Each dialogue turn sends the shared `chat.base` prompt below followed by the section-specific conversational prompt. Field extraction is communicated back via a `DATA_UPDATE:` marker followed by a fenced JSON block in the model's reply; the front-end parses this block out of the reply text before rendering. Conversational temperature 0.2, top-p 0.5.

Base Prompt (`chat.base`)

```
You are an expert GDPR compliance assistant helping users fill out their
↪ Records of Processing Activities (ROPA) documentation.

**Your role is to:**
• Answer questions about the current section in a helpful, conversational way
• Guide users through what information they need to provide
• Based on user responses, fill out the data structure appropriately
• **CRITICAL: Automatically detect and capture specific items mentioned by
↪ users**

**IMPORTANT RESPONSE FORMAT:**
• NEVER mention JSON, code blocks, structured data, "other field", or "other
↪ checkbox"
• NEVER use technical field names (e.g., don't say "microsoftOffice365", "
↪ affectedPersons", "DSGVOArt6Abs1a")
```

```
• ALWAYS use natural language (e.g., "Microsoft Office 365", "affected persons
↪ ", "GDPR Article 6(1)(a)")
• Instead of saying "I'll add it to the other field", say "I've included [
↪ specific item] in your data sources" (or categories, or person groups, etc.)
• Be conversational and describe changes in plain language
• Keep responses concise with NO MORE than one blank line between paragraphs

**CRITICAL BEHAVIOR - DETECTING SPECIFIC ITEMS:**

When users mention SPECIFIC items in conversation:
• **Data Sources:** If they mention "BeanCloud", "Salesforce", "SAP", custom
↪ platforms, or any specific tool → Add it to dataSources.other field
• **Data Categories:** If they mention "biometric data", "GPS tracking", "voice
↪  recordings", "genetic data" → Add to dataCategories.other.other field
• **Person Categories:** If they mention "contractors", "volunteers", "minors",
↪  "students" → Add to persons.other field
• **Legal Basis:** If they mention specific laws or regulations → Add to
↪ legalBasis.other field

**When you detect specific items:**
1. **In your message to the user:** List them naturally alongside predefined
↪ options (e.g., "I've marked Microsoft Office 365, email provider, and
↪ BeanCloud as your data sources")
2. **In the DATA_UPDATE:** Add specific items to the appropriate "other" field
↪ in the JSON

**EXAMPLE:**
User says: "We use Microsoft Office 365 and BeanCloud for our employee data"
Your response: "Perfect! I've marked Microsoft Office 365 and BeanCloud as your
↪  data sources for employee data processing."
Your DATA_UPDATE:
```json
{
  "dataSources": {
    "microsoftOffice365": true,
    "emailProvider": false,
    ...,
    "other": "BeanCloud"
  }
}
```
```

````
**Format your responses:**
I've updated [section name] to include [describe ALL changes, including both
↪ predefined and specific items]. [Any additional helpful context or questions
↪ ].

DATA_UPDATE:
```json
{json structure here}
```

**KEY PRINCIPLE:** Treat specific user-mentioned items (like "BeanCloud") with
↪ the SAME importance as predefined options. Present them naturally to the
↪ user, but store them in the "other" field in the background.

**CRUCIAL:** If the user talks about a section other than specified by the
↪ source, you must refuse to answer and redirect them to the correct section.
````

Purpose of Data Processing (chat.purposeOfDataProcessing)

````
You are helping with the "Purpose of Data Processing" section. This describes
↪ WHY the organization is processing personal data.

**CRITICAL: Analyze the Document Title first** - it usually reveals the core
↪ purpose (e.g., "Employee Payroll Management" = purpose is payroll processing
↪ ).

When you have enough information from the conversation, describe what you're
↪ adding naturally (e.g., "I've set the purpose to focus on employee
↪ management and payroll processing"), then include:

DATA_UPDATE:
```json
{
  "purposeOfDataProcessing": "the suggested text based on conversation"
}
```

The text should be professional, clear, and suitable for GDPR documentation (
↪ max 1000 characters).
````

```
**Be creative and comprehensive** - describe purposes that are specific to the
↪ user's situation, even if they're unique or uncommon. Use the title as your
↪ primary guide.
```

Technical and Organisational Measures (chat.technicalOrganizationalMeasures)

```
You are helping with the "Technical and Organizational Measures" section. This
↪ describes the security measures in place to protect the data.

**CRITICAL: Consider the Document Title** - it indicates the type of processing
↪  and helps determine appropriate security measures.

When you have enough information from the conversation, describe what security
↪ measures you're adding naturally (e.g., "I've added encryption, access
↪ controls, and regular backups to your security measures"), then include:

DATA_UPDATE:
```json
{
  "technicalOrganizationalMeasures": "the suggested security measures based on
↪ conversation"
}
```

The text should list specific, actionable security measures (max 1000
↪ characters).

Suggest specific, context-relevant security measures tailored to the user's
↪ specific processing activities and risk profile based on the title and
↪ conversation.
```

Legal Basis (chat.legalBasis)

```
You are helping with the "Legal Basis" section. This identifies the legal
↪ justification for processing personal data.
```

```
Predefined options: GDPR Art. 6(1)(a) Consent, Art. 6(1)(b) Contract, Art. 6(1)
↪ (c) Legal obligation, Art. 6(1)(f) Legitimate interests, and specific laws (
↪ IfSG §28b, SARSCoV, BDSG §26, HGB §257, HGB §239(1), AO §147).

**CRITICAL: Detect specific legal bases from user messages.** When users
↪ mention laws, regulations, or legal justifications NOT in predefined options
↪ :
• Specific national laws
• Industry-specific regulations (HIPAA, PCI-DSS, etc.)
• Sector directives
• International agreements

→ Add them to legalBasis.other field in DATA_UPDATE
→ In your message, describe them naturally

**EXAMPLE:**
User: "We're processing this under contractual necessity and also HIPAA
↪ requirements"
Your message: "I've enabled GDPR Article 6(1)(b) for contractual necessity, and
↪  I've also noted the HIPAA compliance requirements as an additional legal
↪ basis."
```

Data Sources (`chat.dataSources`)

```
You are helping with the "Data Sources" section. This identifies where the
↪ personal data comes from.

Predefined options: Microsoft Office 365, email provider, Microsoft Azure,
↪ Google Workspace, Google Cloud, video conferencing, Amazon AWS, e-commerce
↪ platform, LibreOffice, online banking, financial services, web/intranet,
↪ credit services, HR software, enterprise systems, security and compliance
↪ tools, document processing, other specialized software.

**CRITICAL: Detect specific data sources from user messages.** When users
↪ mention:
• Specific platform names ("BeanCloud", "Salesforce", "SAP", "Workday", etc.)
• Custom or proprietary systems
• Industry-specific tools
• Company-specific platforms
```

```
→ Add them to the "other" field in the DATA_UPDATE JSON
→ In your message, list them naturally alongside predefined options

**EXAMPLE:**
User: "We use Google Workspace and our company's custom BeanCloud platform"
Your message: "Great! I've marked Google Workspace and BeanCloud as your data
↪ sources. These platforms are commonly used for storing and managing employee
↪  information."
Your DATA_UPDATE:
```json
{
  "dataSources": {
    "microsoftOffice365": false,
    "emailProvider": false,
    "microsoftAzure": false,
    "googleWorkspace": true,
    "googleCloud": false,
    "videoConferencing": false,
    "amazonAWS": false,
    "ecommercePlatform": false,
    "libreOffice": false,
    "onlineBanking": false,
    "financialServices": false,
    "webAndIntranet": false,
    "creditServices": false,
    "hrSoftware": false,
    "enterpriseSystems": false,
    "securityAndCompliance": false,
    "documentProcessing": false,
    "otherSpecializedSoftware": false,
    "other": "BeanCloud custom platform"
  }
}
```
```

Data Categories (`chat.dataCategories`)

```
You are helping with the "Data Categories" section. This identifies what types
↪ of personal data are being processed.
```

```
Main categories: personal data (name, address, contact), employment, finance,
↪ health, legal, behavior, documentation/records, vehicle, and various
↪ subcategories.

**CRITICAL: Detect specific data types from user messages.** When users mention
↪  data types NOT in predefined categories:
• "biometric data", "fingerprints", "facial recognition"
• "GPS location", "geolocation data", "tracking data"
• "voice recordings", "call recordings"
• "genetic information", "DNA data"
• "social media profiles", "online activity"
• "sensor data", "IoT data"

→ Add them to dataCategories.other.other field in DATA_UPDATE
→ In your message, list them naturally with other categories

**EXAMPLE:**
User: "We collect names, emails, and biometric fingerprint data for access
↪ control"
Your message: "I've enabled personal data fields including names and email
↪ addresses, and I've also added biometric fingerprint data for your access
↪ control system."
```

Person Categories (`chat.personCategories`)

```
You are helping with the "Person Categories" section. This identifies who is
↪ affected by or has access to the data.

Predefined categories: internal groups, external customers, legal entities,
↪ healthcare institutions, recruitment agencies (affected persons); management
↪ , administration, technical, finance, legal/compliance, marketing/sales,
↪ logistics/operations (internal recipients); government authorities, software
↪ /cloud providers, health/insurance providers, financial institutions,
↪ business partners/agencies, service providers (external recipients); various
↪  role-based categories (authorized persons).

**CRITICAL: Detect specific person categories from user messages.** When users
↪ mention groups NOT in predefined categories:
• "contractors", "freelancers", "consultants"
• "volunteers", "interns", "trainees"
```

```
• "minors", "students", "pupils"
• "patients", "subscribers", "members"
• Specific department names or unique roles

→ Add them to persons.other field in DATA_UPDATE
→ In your message, list them naturally with other categories

**EXAMPLE:**
User: "This affects our employees and external contractors who work on-site"
Your message: "I've marked internal groups (employees) and external contractors
↪  as the affected person categories for this processing activity."
```

Retention Periods (`chat.retentionPeriods`)

```
You are helping with the "Retention Periods" section. This specifies how long
↪ data is kept and when it's deleted.

Fields: afterTerminationOfEmployment, afterContractTermination, uponRevocation,
↪  deletionTime, exceptForBeyondRetentionObligations, specialDeletionConcept.

**IMPORTANT: Be specific in the "deletionTime" field** - extract any mentioned
↪ timeframes from the context and provide concrete retention periods.

Return ONLY a JSON object:
{
  "retentionPeriods": {
    "afterTerminationOfEmployment": false,
    "afterContractTermination": true,
    "uponRevocation": false,
    "deletionTime": "[Specific timeframe based on context, e.g., '3 years after
↪  contract termination', '10 years for tax records']",
    "exceptForBeyondRetentionObligations": true,
    "specialDeletionConcept": false
  }
}
```

Additional Information (`chat.additionalInfo`)

```
You are helping with the "Additional Information" section. This is for any
↪ supplementary information relevant to the ROPA documentation.

When you have enough information, describe what supplementary information you'
↪ re adding naturally (e.g., "I've added a note about the data transfer
↪ mechanisms and third-party processors involved"), then include:

DATA_UPDATE:
```json
{
  "additionalInfo": "additional text based on conversation (max 1000 characters
↪ )"
}
```

You can include any relevant information that doesn't fit elsewhere - be
↪ comprehensive and include context-specific details that would be valuable
↪ for compliance.
```

A.2 ROPAgen Mode Screenshots

The three figures below show the ROPAgen user interface as it appeared to participants in each of the three interaction modes, captured on a comparable step of the standardized study scenario (Section B.1). They illustrate the user-facing surface of each mode rather than the backend behavior reproduced in Section A.1.

Screenshot placeholder — to be inserted.

Figure A.1: Form mode. The eight thematic groups of the `DocumentData` schema are exposed as a linear questionnaire of checkboxes and short text fields. Each section offers an optional *AI Suggest* button that calls the LLM with the section-specific Form-mode prompt (Section A.1.2).

Screenshot placeholder — to be inserted.

Figure A.2: Ask mode. The same structured form as Form mode is paired with a per-section chat panel that explains the section's purpose and answers follow-up questions. The model's reply may include a `DATA_UPDATE` block that the front-end uses to auto-populate the corresponding form fields.

Screenshot placeholder — to be inserted.

Figure A.3: Chat mode. The structured form is hidden; participants describe their processing activity in free-form text and the LLM populates the underlying `DocumentData` object via silent `DATA_UPDATE` blocks emitted on every turn.

A.2.1 Form Mode

A.2.2 Ask Mode

A.2.3 Chat Mode

A.3 ROPAgen Reference Document

Reproduced below is the expert-authored ROPA reference document. It was created by the study author using Form mode (Section 3.1), applying the standardized study scenario reproduced in Appendix B.1. The document is the comparison target used by every reference-based NLP metric in Section 4.3.1 and Section 4.3.2: every participant document is scored for similarity against this single ROPA. It is reproduced here in its original German (the language in which participants worked); the run-metadata footer emitted by ROPAgen (user identifier, mode, elapsed-time stamp, generation date) is omitted as it is administrative rather than substantive content.

```
Verzeichnis von Verarbeitungstätigkeiten (VVT)
Mitarbeiterverwaltung (Arbeitszeit/Urlaub) & Parkplatzberechtigung
```

```
Verantwortliche Stelle und Kontaktdaten

Wir, die APOR GmbH, fungieren als datenschutzrechtlich Verantwortliche gemäß
↪ Art. 4 Nr. 7 DSGVO für die im Folgenden beschriebenen Verarbeitungstä
↪ tigkeiten.
• Firmenname: APOR GmbH
• Anschrift: Mainstraße 67
• E-Mail-Adresse: mail@apor.eu
• Telefonnummer: 012345678
• Vertreter: A. I. (ai@apor.eu)
• Datenschutzbeauftragter: A. I. (ai@apor.eu)

Zwecke und Rechtsgrundlagen der Verarbeitung

2.1 Verarbeitungszwecke
Wir verarbeiten personenbezogene Daten für folgende Zwecke:
• Verwaltung von Mitarbeitenden vom Eintritt bis zum Austritt, einschließlich:
• Anlegen neuer Mitarbeitender
• Dokumentation von Arbeitszeiten
• Verwaltung von Urlaub und Abwesenheiten
• Verwaltung von Parkplatzberechtigungen für berechtigte Mitarbeitende

2.2 Rechtsgrundlagen der Verarbeitung
Die Verarbeitung erfolgt auf Grundlage folgender Rechtsnormen:
• Art. 6 Abs. 1 lit. b DSGVO (Erfüllung eines Vertrags oder vorvertraglicher Ma
↪ ßnahmen)
• § 26 BDSG (Datenverarbeitung für Zwecke des Beschäftigungsverhältnisses)

Kategorien personenbezogener Daten und Datenquellen

3.1 Verarbeitete Datenkategorien
Wir verarbeiten folgende personenbezogene Daten:

Personenbezogene Grunddaten:
• Nachname
• Vorname
• Geburtsdatum
• Anschrift
• E-Mail-Adresse

Beschäftigungsdaten:
```

```
• Arbeitszeiten
• Berufliche Position
• Personalnummer
• Unternehmenszugehörigkeit
• Urlaubsanspruch
• Urlaubstage
• Urlaubszeiten

Finanzdaten:
• Gehalt/Lohn
• Bankverbindung
• Steueridentifikationsnummer

Vertrags- und Rechtsdaten:
• Vertragsdaten

Dokumentations- und Aufzeichnungsdaten:
• Individuelle Dokumentation

Fahrzeugdaten:
• Kennzeichen

Sonstige Daten:
• Parkplatzberechtigungsdaten

3.2 Datenquellen
Die Daten stammen aus folgenden Quellen:
• Microsoft Office 365 (intern gehostet)
• Mitarbeiterverwaltungssystem (z. B. für Arbeitszeit, Urlaub und
↪ Parkplatzberechtigungen)
• HR-Software

Betroffene Personen und Empfänger

4.1 Kategorien betroffener Personen
Die Verarbeitung betrifft folgende Personengruppen:
• Interne Gruppen (Mitarbeitende)

4.2 Interne Empfänger
Die Daten werden innerhalb unserer Organisation an folgende interne Empfänger ü
↪ bermittelt:
• Verwaltung
```

```
• Personal- und Rekrutierungsverantwortliche (HR-Rollen)

4.3 Externe Empfänger
Es erfolgt keine Übermittlung personenbezogener Daten an externe Empfänger
↪ innerhalb oder außerhalb der EU.

Löschfristen und Aufbewahrungsdauern

Wir löschen die personenbezogenen Daten drei Monate nach Beendigung des Beschä
↪ ftigungsverhältnisses, sofern keine gesetzlichen Aufbewahrungspflichten
↪ entgegenstehen.

Technische und organisatorische Maßnahmen (TOM)

Zum Schutz der verarbeiteten personenbezogenen Daten haben wir folgende
↪ technische und organisatorische Maßnahmen implementiert:
• Zugriffsbeschränkung nach dem Need-to-know-Prinzip: Nur berechtigte
↪ Mitarbeitende erhalten Zugang zu den Daten.
• Passwortschutz und Multi-Faktor-Authentifizierung (MFA): Wo technisch möglich
↪ , setzen wir MFA ein.
• Regelmäßige Backups: Sicherungskopien werden in definierten Intervallen
↪ erstellt.
• Zutrittskontrolle zu Serverräumen und Archiven: Physische Zugangskontrollen
↪ schützen die IT-Infrastruktur.
• Vertraulichkeitsverpflichtung: Berechtigte Mitarbeitende sind zur
↪ Vertraulichkeit verpflichtet.

Dieses Verzeichnis wird regelmäßig überprüft und bei Bedarf aktualisiert.
```

A.4 LLM-as-a-Judge Configuration

This section documents the inputs to the LLM-as-a-Judge evaluation layer that scored the 113 retained ROPA documents (Section 4.4). It contains the verbatim system and user prompts, the five-metric rubric, and the model configuration needed to reproduce the evaluation. The prompts below mirror the formatting of the ROPAgen backend prompts above in Section A.1: each prompt is reproduced as a single listing block so the reader sees exactly what the model received.

A.4.1 System and User Prompt

The system prompt below is sent verbatim as the `system` role on every evaluation call. It is taken from `python/judge/prompt.json` (key `judge.system`).

```
You are an expert GDPR compliance reviewer evaluating Records of
Processing Activities (ROPA) documents under Article 30 GDPR. The
documents are written by novices (non-experts, no prior GDPR training)
who described the same processing scenario using different LLM-assisted
tools. Your job is to judge each ROPA document against (a) the provided
scenario and (b) GDPR Article 30 requirements. Be strict but fair. Do not
reward verbosity. Do not penalise a document for merely restating
scenario facts. Respond with valid JSON only - no prose, no Markdown
fences, no commentary before or after the JSON object.
```

The accompanying user-message template is sent on every call with `{scenario}` (the German original from Appendix B.1.1) and `{document}` (the participant ROPA document) substituted in:

```
You will evaluate one ROPA document against the scenario it is supposed
to cover.

=== SCENARIO (verbatim, German) ===
{scenario}
=== END SCENARIO ===

=== ROPA DOCUMENT TO EVALUATE ===
{document}
=== END DOCUMENT ===

Score the document on each of the following five metrics using a 1-5
Likert scale, and give a short rationale (max 2 sentences, English) for
each score. The scale for all metrics is:
1 = very poor, 2 = poor, 3 = adequate, 4 = good, 5 = excellent.

Metrics and what to assess:
- completeness: Coverage of Art. 30(1) GDPR required elements -
  controller identity and contact, DPO, purposes of processing, legal
  basis, categories of data subjects and personal data, categories of
  recipients, retention periods, and technical/organisational measures
  (TOMs). Missing elements lower the score.
- scenario_faithfulness: How accurately the document reflects the
```

```
  concrete facts of THIS scenario - APOR GmbH, Ulm (Mainstrasse 67), ~50
  employees, internally hosted Microsoft Office and internal servers (NO
  external cloud), no medical/sickness data, 3-month deletion after
  employee departure, parking lot authorisation using a distinguishing
  feature such as a licence plate. Generic ROPA boilerplate that ignores
  these specifics lowers the score.
- legal_correctness: Correct use of GDPR terminology and correct citation
  of legal basis (typically Art. 6(1)(b) performance of contract for
  employment data, Art. 6(1)(c) legal obligation where applicable, Art.
  6(1)(f) legitimate interest for parking authorisation). Wrong articles,
  invented legal bases, or confusing controller/processor/DPO roles lower
  the score.
- hallucination_inverse: Inverse score for invented facts not supported
  by the scenario (e.g. invented third-party processors, invented data
  categories like medical data, invented addresses, invented retention
  periods different from 3 months). 5 = no hallucinations, 1 = many
  invented facts.
- overall: Holistic judgement of how useful this document would be as a
  real Art. 30 ROPA record for this scenario, taking the four dimensions
  above into account.

Return a single JSON object with exactly this shape:
{
  "completeness":           {"score": <int 1-5>, "rationale": "<string>"},
  "scenario_faithfulness":  {"score": <int 1-5>, "rationale": "<string>"},
  "legal_correctness":      {"score": <int 1-5>, "rationale": "<string>"},
  "hallucination_inverse":  {"score": <int 1-5>, "rationale": "<string>"},
  "overall":                {"score": <int 1-5>, "rationale": "<string>"}
}
```

A.4.2 Rubric — Five Metrics

All five metrics share the same 1–5 Likert anchor: $1 =$ very poor, $2 =$ poor, $3 =$ adequate, $4 =$ good, $5 =$ excellent. The fourth metric, `hallucination_inverse`, is inverted internally so that for all five metrics higher equals better. For each metric the judge returns both an integer score and a short English rationale (maximum two sentences).

Metric	What is assessed
Completeness	Coverage of Art. 30(1) GDPR required elements: controller identity and contact, DPO, purposes of processing, legal basis, categories of data subjects and personal data, categories of recipients, retention periods, and technical/organizational measures (TOMs). Missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of this scenario: APOR GmbH, Ulm (Mainstraße 67), $\sim$50 employees, internally hosted Microsoft Office and internal servers (no external cloud), no medical/sickness data, 3-month deletion after employee departure, parking authorization using a distinguishing feature such as a license plate. Generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis (typically Art. 6(1)(b) for employment data, Art. 6(1)(c) where applicable, Art. 6(1)(f) for parking authorization). Wrong articles, invented legal bases, or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Inverse score for invented facts not supported by the scenario (e.g. invented third-party processors, invented categories such as medical data, invented addresses, retention periods other than 3 months). 5 = no hallucinations, 1 = many invented facts.
Overall	Holistic judgment of how useful the document would be as a real Art. 30 ROPA record for this scenario, taking the four metrics above into account.

Table A.1: The five judge metrics. All share the 1–5 Likert anchor; higher is better for every metric after the hallucination inversion.

A.4.3 Model Configuration and Limitations

The configuration in Table A.2 is taken from the `judge` block of `python/judge/prompt.json` and the corresponding cells of `ROPAgen_JUDGE.ipynb`. It is the minimal information required to reproduce the evaluation layer.

Setting	Value
Model	`google/gemini-3.1-pro-preview` (via OpenRouter)
Temperature	`0.0` (deterministic decoding)
Output format	JSON object, enforced via OpenRouter `response_format = json_schema` with `strict: true`
Schema name	`ropa_judge_result`
Required keys	`completeness`, `scenario_faithfulness`, `legal_correctness`, `hallucination_inverse`, `overall`
Per-key shape	`{"score": int in [1,5], "rationale": string}`
Additional properties	Disallowed at every object level
Documents evaluated	113 ROPA documents (Form 37, Ask 37, Chat 39)
Caching	Per-document JSON result cached on disk; reused on re-runs
Retry behavior	Schema-validation or transient API failures retried; persistent failures logged and skipped
Logging	Each call logs model, latency, prompt and completion token counts, and raw response payload

Table A.2: LLM-as-a-judge model configuration.

Known limitations. The following limitations of the judge layer are stated for full transparency and discussed further in the Discussion chapter:

- **Single-judge bias.** Only one judge model (Gemini 3.1 Pro Preview) operationalizes the rubric; inter-rater agreement with a human GDPR expert is not established.
- **Model-family bias.** Documents are produced by Mistral Large 3 inside ROPAgen, while the LLM-as-Judge belongs to a different model family (Google). This reduces but does not eliminate stylistic-preference bias.
- **Single-scenario bias.** All scores are conditional on the one APOR GmbH scenario; generalization to other ROPA contexts is not claimed.

- **Reference-free.** Unlike the NLP metric layer, the judge does not compare against an expert reference document; it scores against the scenario and Art. 30 directly.

B Study Materials

This appendix groups the two reproducible artifacts of the user study: the standardized scenario presented to all participants, and the verbatim items of the two standardized survey instruments (SUS and Raw NASA-TLX). Both are referenced repeatedly from the Methodology chapter and are reproduced here in their canonical form.

B.1 Study Scenario

The scenario reproduced below was presented to all participants in German, regardless of interaction mode (Form, Ask, Chat). The English translation is provided for the international reader; the German original is the version actually shown.

B.1.1 Original (German)

```
Szenario: Mitarbeiterverwaltung (Arbeitszeit/Urlaub) & Parkplatzberechtigung

Sie sind Mitarbeitender namens A.I. und arbeiten für die APOR GmbH, ein
Softwareunternehmen mit ca. 50 Mitarbeitenden in Ulm. Sie handeln im Namen
des Unternehmens und sind im Szenario sowohl Verantwortlicher (Data
Controller) als auch Datenschutzbeauftragte:r (DPO/DSB).

Zur Organisation:

Rolle: Data Controller, DPO
Adresse: Mainstraße 67, Ulm
```

E-Mail: mail@apor.eu
Telefon: 012345678
Vertreter: (Sie), A. I., ai@apor.eu
DPO: Das sind auch Sie

Verarbeitung
Das Unternehmen verarbeitet personenbezogene Daten, um Mitarbeitende vom Eintritt bis zum Austritt zu verwalten. Im Fokus stehen:

- Anlegen neuer Mitarbeitender
- Dokumentation von Arbeitszeiten
- Verwaltung von Urlaub und Abwesenheiten

Zusätzlich verwaltet APOR für berechtigte Mitarbeitende eine Parkplatzberechtigung für einen Firmenparkplatz mit Schranke. Damit berechtigte Mitarbeitende den Parkplatz nutzen können, wird ein geeignetes Merkmal zur Zuordnung der Berechtigung hinterlegt (z. B. ein Fahrzeugkennzeichen).

APOR arbeitet mit intern gehostetem Microsoft Office sowie internen Servern (keine externe Cloud). Krankentage oder medizinische Angaben sind für diese Verarbeitung nicht erforderlich.

Für die Verwaltung sind technische und organisatorische Maßnahmen vorgesehen, wie z.B. Zugriffsbeschränkung nach Rollen (Need-to-know), Passwortschutz (ggf. MFA), regelmäßige Backups, Zutrittskontrolle zu Serverraum/Archiv und Vertraulichkeitsverpflichtung der berechtigten Mitarbeitenden.

Alle in diesem Zusammenhang gespeicherten Informationen werden spätestens 3 Monate nach Austritt der jeweiligen Person gelöscht.

B.1.2 English Translation

```
Scenario: Employee Administration (Working Time / Leave) & Parking
Authorisation

You are an employee named A.I., working for APOR GmbH, a software company
with approximately 50 employees in Ulm. You act on behalf of the company
and, in this scenario, are both the Data Controller and the Data
Protection Officer (DPO).

Organisation details:

Role: Data Controller, DPO
Address: Mainstrasse 67, Ulm
E-mail: mail@apor.eu
Phone: 012345678
Representative: (You), A. I., ai@apor.eu
DPO: Also you

Processing
The company processes personal data in order to administer employees from
onboarding to exit. The focus is on:

- Onboarding new employees
- Documentation of working hours
- Administration of leave and absences

In addition, APOR manages a parking authorization for entitled employees
for a company car park with a barrier. To allow entitled employees to use
the car park, a suitable feature for assigning the authorization is
recorded (e.g. a vehicle license plate).

APOR uses internally hosted Microsoft Office and internal servers (no
external cloud). Sick days or medical information are not required for
this processing.
```

```
For administration, technical and organizational measures (TOMs) are
provided, such as role-based access restriction (need-to-know), password
protection (where appropriate, MFA), regular backups, physical access
control to the server room / archive, and a confidentiality obligation for
authorized employees.

All information stored in this context is deleted no later than 3 months
after the person's departure.
```

B.2 Survey Instruments

This section reproduces the verbatim items of the two standardized survey instruments referenced from the Methodology chapter: the System Usability Scale (SUS) and the Raw NASA Task Load Index (R-TLX). The six-item Perceived AI Support instrument is bespoke to this study and is listed in full in the Methodology chapter; it is not reproduced here.

B.2.1 System Usability Scale (SUS)

Five-point Likert scale, $1 =$ Strongly Disagree, $5 =$ Strongly Agree. Odd-numbered items are positively keyed; even-numbered items are negatively keyed. The canonical wording reproduced below is taken from Brooke [4].

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.

7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

B.2.2 NASA Raw Task Load Index Subscales

Six subscales, each rated on the standard NASA TLX 21-point scale. The Raw TLX variant used in this study takes the unweighted mean across the six subscales rather than applying the pairwise-comparison weighting of the original NASA TLX. The subscale definitions reproduced below follow Hart [10].

Mental Demand How mentally demanding was the task?

Physical Demand How physically demanding was the task?

Temporal Demand How hurried or rushed was the pace of the task?

Performance How successful were you in accomplishing what you were asked to do?

Effort How hard did you have to work to accomplish your level of performance?

Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?

C Inferential Statistics in Full

This appendix reproduces the complete inferential test output referenced in Chapter 4. The within-subjects survey tests use $n = 69$ (matched within-subjects, Friedman + Wilcoxon); the document-quality, timing, and confidence–quality analyses use $n = 113$ (between-groups, Kruskal–Wallis + Mann–Whitney U). The full sample-size derivation is reproduced first, followed by the survey inferential tests, the NLP inferential tests, the confidence–quality correlations, and the timing inferential tests.

C.1 Sample-Size Derivation

All sample sizes used in the thesis are derived from the raw data files committed to the repository: `python/survey/docs/data.csv` for the survey (one row per participant) and `python/metrics/output/all_scores_combined.csv` for the NLP and judge scores (one row per LLM-generated document).

Survey subset	**n**
Recruited participants (raw)	73
Participants with at least one valid session ($p\_0001 \geq 1$)	69
Form responses (mode-level)	69
Ask responses (mode-level)	69
Chat responses (mode-level)	69
Matched-triple participants (valid response in all three modes)	69

Table C.1: Survey sample sizes after consent and quality filtering.

NLP / judge subset	**n**
Total documents with parseable NLP scores and judge ratings	113
Form documents	37
Ask documents	37
Chat documents	39
Unique participants represented	47
Participants with at least one document in each mode	28
Strict matched triple (exactly one document per mode)	26

Table C.2: NLP and judge sample sizes. The full $n = 113$ sample is used for the document-quality inferential tests (Kruskal–Wallis, Mann–Whitney U).

Survey $\times$ NLP join	**n**
Joined rows (document with matching survey response)	106
Form joined rows	34
Ask joined rows	34
Chat joined rows	38
Unique participants in the joined sample	44
Participants with survey $\times$ NLP coverage in all three modes	27

Table C.3: Triangulated samples used for confidence–quality analyses. The joined $n = 106$ is the canonical sample for the rank-inversion table; the within-subjects $n = 27$ is used for paired confidence $\times$ quality correlations.

C.2 Survey Inferential Tests

C.2.1 Normality (Shapiro–Wilk, $n = 69$)

Table C.4: Shapiro–Wilk normality tests for survey measures by mode.

Measure	Mode	W	p	Normal?
SUS	Form	0.944	0.004	No
SUS	Ask	0.976	0.210	Yes
SUS	Chat	0.927	0.001	No
R-TLX composite	Form	0.940	0.003	No
R-TLX composite	Ask	0.926	0.001	No
R-TLX composite	Chat	0.945	0.005	No
AI Support (6-item)	Form	0.923	< 0.001	No
AI Support (6-item)	Ask	0.965	0.048	No
AI Support (6-item)	Chat	0.952	0.010	No
AI Confidence (5_{ai})	Form	0.846	< 0.001	No
AI Confidence (5_{ai})	Ask	0.856	< 0.001	No
AI Confidence (5_{ai})	Chat	0.864	< 0.001	No
AI Reuse (6_{ai})	Form	0.799	< 0.001	No
AI Reuse (6_{ai})	Ask	0.872	< 0.001	No
AI Reuse (6_{ai})	Chat	0.861	< 0.001	No

Non-normal distributions are detected for nearly every measure. Non-parametric tests (Friedman omnibus and Wilcoxon signed-rank post-hoc) are used throughout.

C.2.2 Friedman Omnibus ($n = 69$ matched triple)

C.2.3 Pairwise Wilcoxon Signed-Rank Post-hoc (Bonferroni $\alpha = 0.0167$)

Post-hoc tests are reported for measures with a significant Friedman result and, for completeness, for SUS and the R-TLX composite. Sign convention: positive r means the first-named mode ranks higher. Effect-size labels: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

Table C.5: Friedman omnibus tests across the three modes for survey measures. Kendall's W: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

Measure	χ^2	df	p	W	Significant
SUS (0–100)	5.51	2	0.064	0.040	No
R-TLX composite (1–21)	5.24	2	0.073	0.038	No
R-TLX Mental Demand	3.13	2	0.209	0.023	No
R-TLX Physical Demand	2.08	2	0.354	0.015	No
R-TLX Temporal Demand	12.77	2	0.002	0.093	**Yes**
R-TLX Effort	3.78	2	0.151	0.027	No
R-TLX Frustration	8.93	2	0.012	0.065	**Yes**
R-TLX Performance (pos.)	2.33	2	0.311	0.017	No
AI Confidence (5_{ai})	3.76	2	0.152	0.027	No
AI Support (6-item composite)	1.05	2	0.592	0.008	No
AI Reuse intention (6_{ai})	11.12	2	0.004	0.081	**Yes**

Table C.6: Pairwise Wilcoxon signed-rank post-hoc tests for survey measures.

Measure	Pair	n	W	p	r	Sig (Bonf)
SUS	Form vs Ask	60	578.5	0.013	$+0.368$ medium	**Yes**
SUS	Form vs Chat	58	549.5	0.018	$+0.358$ medium	No
SUS	Ask vs Chat	61	885.5	0.666	-0.063 negligible	No
R-TLX composite	Form vs Ask	65	684.0	0.011	-0.362 medium	**Yes**
R-TLX composite	Form vs Chat	65	636.0	0.004	-0.407 medium	**Yes**
R-TLX composite	Ask vs Chat	63	995.5	0.932	$+0.012$ negligible	No
Temporal Demand	Form vs Ask	43	237.0	0.004	-0.499 medium	**Yes**
Temporal Demand	Form vs Chat	46	233.0	< 0.001	-0.569 large	**Yes**
Temporal Demand	Ask vs Chat	46	427.0	0.211	-0.210 small	No
Frustration	Form vs Ask	52	403.5	0.009	-0.414 medium	**Yes**
Frustration	Form vs Chat	57	519.5	0.014	-0.371 medium	**Yes**
Frustration	Ask vs Chat	52	629.0	0.584	$+0.087$ negligible	No
AI Reuse (6_{ai})	Form vs Ask	49	402.5	0.034	$+0.343$ medium	No
AI Reuse (6_{ai})	Form vs Chat	43	216.0	0.002	$+0.543$ large	**Yes**
AI Reuse (6_{ai})	Ask vs Chat	45	454.5	0.468	$+0.122$ small	No

C.2.4 Mode Preference Ranking

A chi-square goodness-of-fit test on the rank-1 vote distribution against a uniform null gives $\chi^2(2, N = 69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$ (medium effect). The Friedman test on the full rank scores (1–3) is non-significant: $\chi^2(2) = 5.07$, $p = 0.079$, $W = 0.037$.

C.3 NLP Metric Inferential Tests

C.3.1 Kruskal–Wallis Omnibus ($n = 113$)

Table C.7: Kruskal–Wallis H tests on the unbalanced full document set ($n = 113$). Primary omnibus for document-quality inferential testing.

Metric	H	p	Sig ($\alpha = 0.05$)	ε^2
BLEU	4.48	0.107	No	—
ROUGE-1	5.57	0.062	No	—
ROUGE-2	9.56	0.008	Yes	0.069 (moderate)
ROUGE-L	7.79	0.020	Yes	—
METEOR	9.80	0.007	Yes	0.071 (moderate)
BERTScore Precision	9.11	0.011	Yes	—
BERTScore Recall	32.38	< 0.001	Yes	0.276 (large)
BERTScore F1	9.75	0.008	Yes	0.071 (moderate)
SBERT (ModernBERT)	17.43	< 0.001	Yes	0.140 (large)

C.3.2 Pairwise Mann–Whitney U ($n = 113$)

Pairwise Mann–Whitney U tests on the unbalanced full sample, serving as the post-hoc family where the Kruskal–Wallis omnibus is significant. Bonferroni-corrected $\alpha = 0.0167$. Only significant comparisons are listed; non-significant comparisons are omitted for brevity.

Ask is the significantly lowest-scoring mode (Ask $<$ Form, Ask $<$ Chat) on the metrics that reach significance — most clearly on BERTScore Recall, F1, ROUGE-2,

Table C.8: Pairwise Mann–Whitney U post-hoc tests on the unbalanced full sample of $n = 113$ documents.

Metric	Comparison	p	r	Sig
BERTScore Recall	Form > Ask	< 0.001	0.756 (large)	Yes
BERTScore Recall	Form > Chat	< 0.001	0.512 (large)	Yes
BERTScore F1	Form > Ask	0.004	0.386 (medium)	Yes
BERTScore F1	Chat > Ask	0.011	0.340 (medium)	Yes
SBERT	Chat > Form	< 0.001	0.505 (large)	Yes
SBERT	Chat > Ask	< 0.001	0.453 (large)	Yes
ROUGE-2	Form > Ask	0.006	0.369 (medium)	Yes
ROUGE-2	Chat > Ask	0.011	0.342 (medium)	Yes
METEOR	Form > Ask	0.003	0.407 (medium)	Yes

and METEOR — while Form and Chat trade leadership depending on whether the metric emphasizes lexical recall or document-level semantic alignment.

C.4 Confidence–Quality Correlations

Spearman rank correlation ρ between the primary confidence measure (item 5_{ai}, "I am confident the AI identified the correct legal basis") and document-quality measures, computed on the survey $\times$ NLP joined sample. Per-mode joined samples: Form $n = 34$, Ask $n = 34$, Chat $n = 38$; pooled $n = 106$.

Table C.9: Spearman rank correlations: primary confidence (5_{ai}) versus document-quality measures, by mode.

	5_{ai} vs BERTScore F1			5_{ai} vs Judge Overall		
Mode	n	ρ	p	n	ρ	p
Form	34	-0.22	0.22	34	$+0.07$	0.70
Ask	34	-0.23	0.19	34	-0.21	0.24
Chat	38	$+0.28$	0.09	38	$+0.41$	**0.012**
Pooled	106	-0.02	0.88	106	$+0.14$	0.14

For comparison, the secondary 6-item Perceived AI Support composite produces weaker mode-specific correlations with BERTScore F1 (Form $\rho = -0.22$, Ask $\rho =$

-0.04, Chat $\rho = +0.20$) and a small but significant pooled correlation with Judge Overall ($\rho = +0.21$, $p = 0.033$, $n = 106$).

C.5 Timing Inferential Tests

Inferential tests on the per-mode completion time recorded in the `Time elapsed` line emitted by ROPAgen at document submission (one value per participant–mode pairing). Sample sizes match the NLP document-level analysis: full $n = 113$ (Form 37, Ask 37, Chat 39); matched-triple within-subjects subset $n = 26$ participants. Non-parametric tests are used throughout, consistent with the NLP and survey pipelines, as Shapiro–Wilk rejects normality on every mode (Form $W = 0.780$, $p < 0.001$; Ask $W = 0.912$, $p = 0.030$; Chat $W = 0.601$, $p < 0.001$). Source: `python/metrics/output/statistical_tests.md` (*Per-Mode Completion Time* section).

C.5.1 Kruskal–Wallis Omnibus ($n = 113$)

Table C.10: Kruskal–Wallis H test on per-mode completion time across the unbalanced full sample of $n = 113$ documents.

Measure	H	p	Sig ($\alpha = 0.05$)	ε^2
Completion time (s)	16.455	< 0.001	Yes	0.147 (large)

C.5.2 Pairwise Mann–Whitney U ($n = 113$)

Pairwise Mann–Whitney U tests on the unbalanced full sample, serving as the post-hoc family where the Kruskal–Wallis omnibus is significant. Bonferroni-corrected $\alpha = 0.0167$.

Form is significantly faster than both Ask and Chat (large and medium effects); Ask and Chat are statistically indistinguishable from one another.

Table C.11: Pairwise Mann–Whitney U post-hoc tests on per-mode completion time on the unbalanced full sample of $n = 113$ documents. Effect-size r is positive when the first-named mode is *slower*.

Pair	U	p	r	Median A (s)	Median B (s)	Sig (Bonf)
Form vs Ask	334.0	< 0.001	$+0.512$ large	420.0	858.0	**Yes**
Form vs Chat	420.0	0.002	$+0.418$ medium	420.0	717.0	**Yes**
Ask vs Chat	793.5	0.457	-0.100 negligible	858.0	717.0	No

D Acronyms

This appendix lists the acronyms used throughout the thesis, together with their expansions, for ease of reference.

Acronym	Expansion
API	Application Programming Interface
BDSG	Bundesdatenschutzgesetz (German Federal Data Protection Act)
BERTScore	Bidirectional Encoder Representations from Transformers Score
BLEU	Bilingual Evaluation Understudy
DBIS	Institute of Databases and Information Systems (University of Ulm)
DPO	Data Protection Officer
DSGVO	Datenschutz-Grundverordnung (German term for GDPR)
EU	European Union
GDPR	General Data Protection Regulation
HGB	Handelsgesetzbuch (German Commercial Code)
IfSG	Infektionsschutzgesetz (German Infection Protection Act)
LCS	Longest Common Subsequence
LLM	Large Language Model
METEOR	Metric for Evaluation of Translation with Explicit ORdering
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ROPA	Record of Processing Activities
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
R-TLX	Raw NASA Task Load Index
RQ	Research Question
SBERT	Sentence-BERT
SD	Standard Deviation
SUS	System Usability Scale
UX	User Experience

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Declaration

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated.

Ulm, May 21, 2026 ....

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